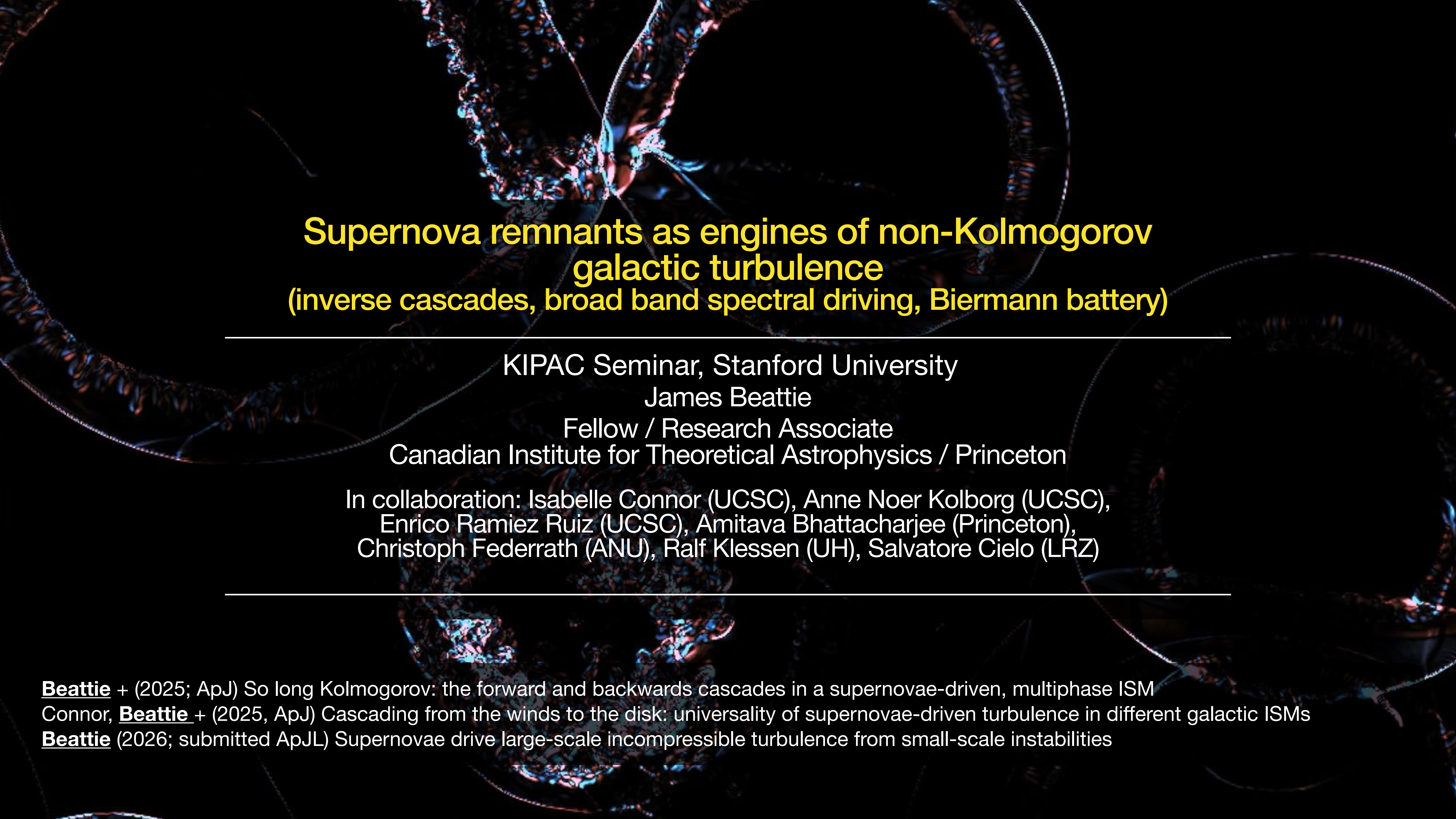




Supernova remnants as engines of non-Kolmogorov galactic turbulence (inverse cascades, broad band spectral driving, Biermann battery)

KIPAC Seminar, Stanford University

James Beattie

Fellow / Research Associate

Canadian Institute for Theoretical Astrophysics / Princeton

In collaboration: Isabelle Connor (UCSC), Anne Noer Kolborg (UCSC),
Enrico Ramirez Ruiz (UCSC), Amitava Bhattacharjee (Princeton),
Christoph Federrath (ANU), Ralf Klessen (UH), Salvatore Cielo (LRZ)

Beattie + (2025; ApJ) So long Kolmogorov: the forward and backwards cascades in a supernovae-driven, multiphase ISM

Connor, **Beattie** + (2025, ApJ) Cascading from the winds to the disk: universality of supernovae-driven turbulence in different galactic ISMs

Beattie (2026; submitted ApJL) Supernovae drive large-scale incompressible turbulence from small-scale instabilities

Roadmap for this seminar

1. Discuss vanilla K41 turbulence in a pedagogical manner & turbulence in our galaxy.
2. Supernova-driven turbulence — how to fit them into the cascade? Simulations.
3. Spectra, cascade directions.
4. Where does the incompressible turbulence come from — shell instabilities and spectra model.

The ISM: A laboratory for interesting nonlinear physics

- Multiphase trans-to-supersonic compressible plasma
- $Re = UL/\nu \gg 1 \implies$ highly-turbulent
- Turbulence driven by a zoo of sources
 - Gravitational instabilities, SN explosions, stellar feedback...
- Pervaded by ultra-relativistic particles with non-negligible energy densities (peaked number density in GeV – TeV range)
- Non-Gaussian gas-density structures related to star formation (multi-scale filaments, ribbons, striations, sheets)
- Dynamically important, maintained magnetic fields
- $Rm = UL/\eta \gg 1 \implies$ highly-turbulent magnetic fields
-

Krumholz & McKee (2005); Federrath & Klessen (2012); Tritsis+(2016); Xu & Lazarian (2016); Soler & Hennebelle (2017); Squire & Hopkins (2017); Mocz & Burkhart (2018), Burkhart & Mocz (2018); Heyer+(2012); Tritsis+(2018); Hu+(2019); Heyer+(2020); Krumholz+(2020); Beattie & Federrath (2020); Beattie+(2020); Körtgen & Soler (2020); Skolidis & Tassis (2020); Federrath+(2021); Burkhart (2021); Barreto-Mota+(2021); Hopkins+(2022); Beattie+(2022); Fielding+(2022); Sampson+ (2022); Kriel+(2022); Galishnikova+(2022); Beattie+(2023)

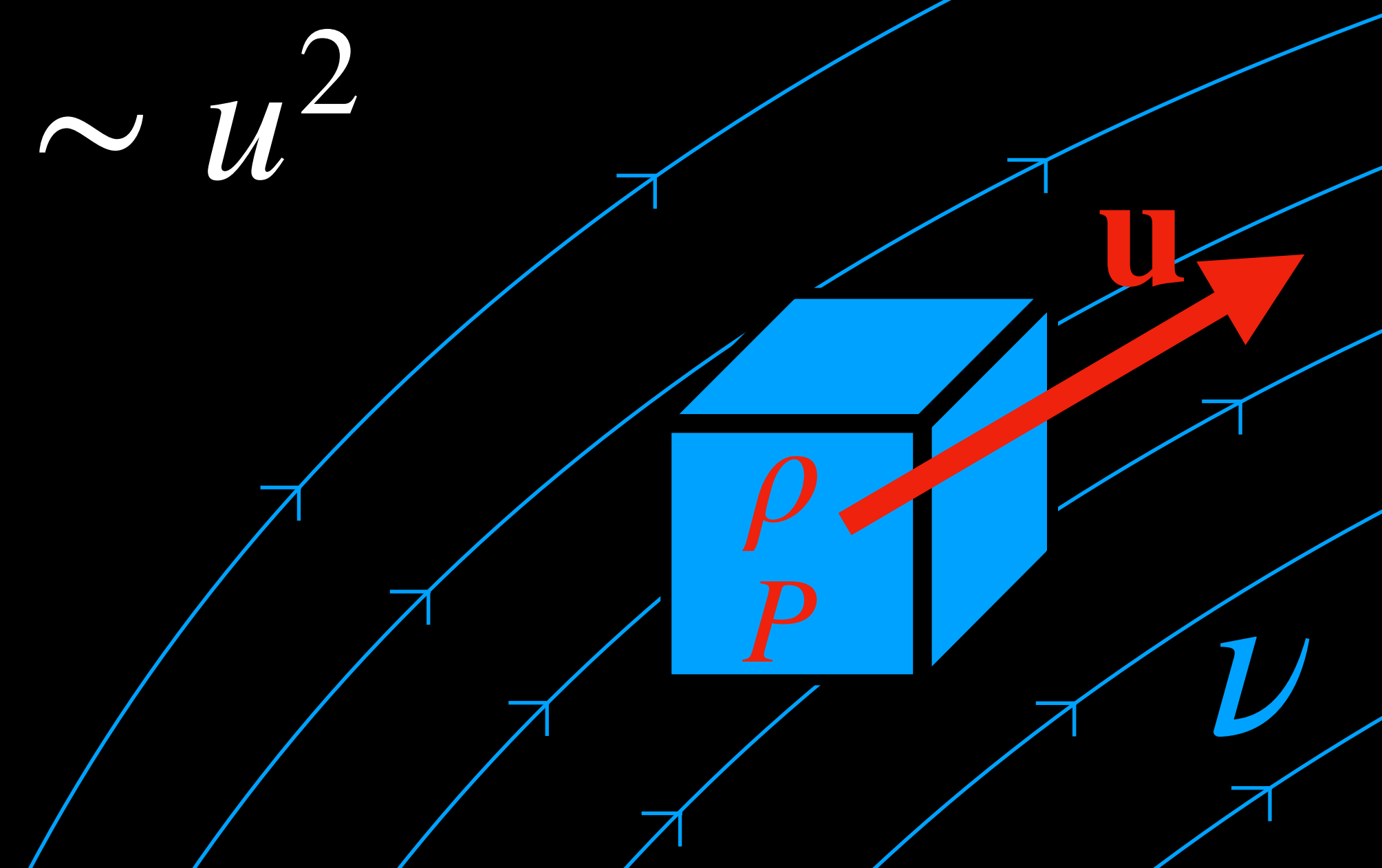
Turbulence

What is it (hand wavy)?

Momentum conservation for a hydrodynamical fluid element
(first moment of Boltzmann eq. for monoatomic plasma slightly out of LTE)

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = -\nabla \cdot P\mathbb{I} + 2\nu \nabla \cdot (\rho \mathcal{S})$$
$$\mathbf{u} \otimes \mathbf{u} = u_i u_j \sim u^2$$

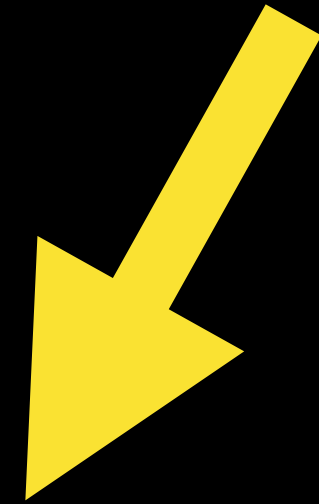
$$\lambda_{\text{mfp}}/L \ll 1$$



Turbulence

What is it (hand wavy)?

Reynolds stress



$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = -\nabla \cdot P\mathbb{I} + 2\nu \nabla \cdot (\rho \mathcal{S})$$

$$\mathbf{u} \otimes \mathbf{u} = u_i u_j \sim u^2$$

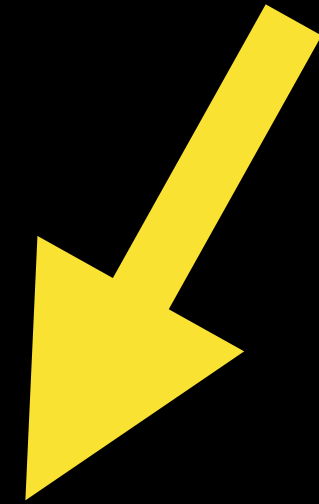


Viscous stress

Turbulence

What is it (hand wavey)?

creating nonlinearity



$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = -\nabla \cdot P \mathbb{I} + 2\nu \nabla \cdot (\rho \mathcal{S})$$
$$\mathbf{u} \otimes \mathbf{u} = u_i u_j \sim u^2$$



Smooths out nonlinear things in the fluid

Turbulence

What is it (hand wavy)?

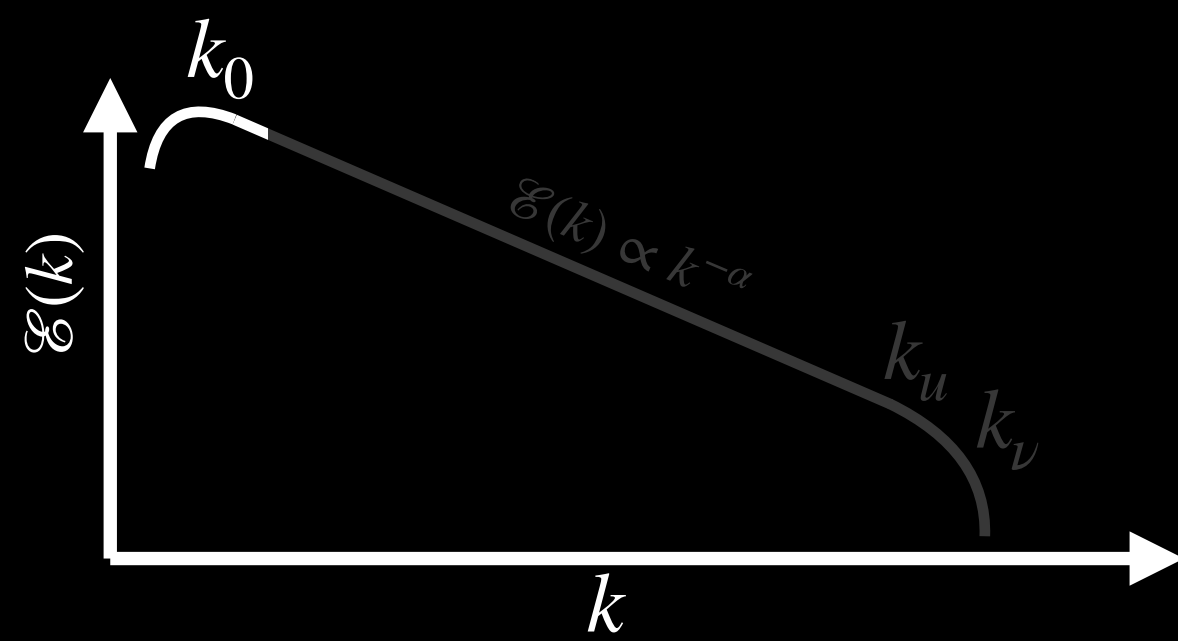
$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = -\nabla \cdot P\mathbb{I} + 2\nu \nabla \cdot (\rho \mathcal{S})$$

Creating nonlinear things in the fluid

$$\text{Re} = \frac{|\nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u})|}{|2\nu \nabla \cdot (\rho \mathcal{S})|} \sim \frac{UL}{\nu}$$

Smoothing out nonlinear things in the fluid

Turbulence Cascade



Credit: NASA, ESA, S. Beckwith (STScI) and the Hubble Heritage Team (STScI/AURA)

inspired by Shukurov, 2011

$$\begin{aligned} \text{WIM: } \text{Re} &\sim 10^7 & \lambda_{\text{mfp}} &\sim 5 R_{\oplus} \\ \text{WNM: } \text{Re} &\sim 10^7 & \lambda_{\text{mfp}} &\sim 1.3 \text{ AU} \\ \text{CNM: } \text{Re} &\sim 10^{10} & \lambda_{\text{mfp}} &\sim 0.3 \text{ pc} \end{aligned}$$

Ferrière, 2020;

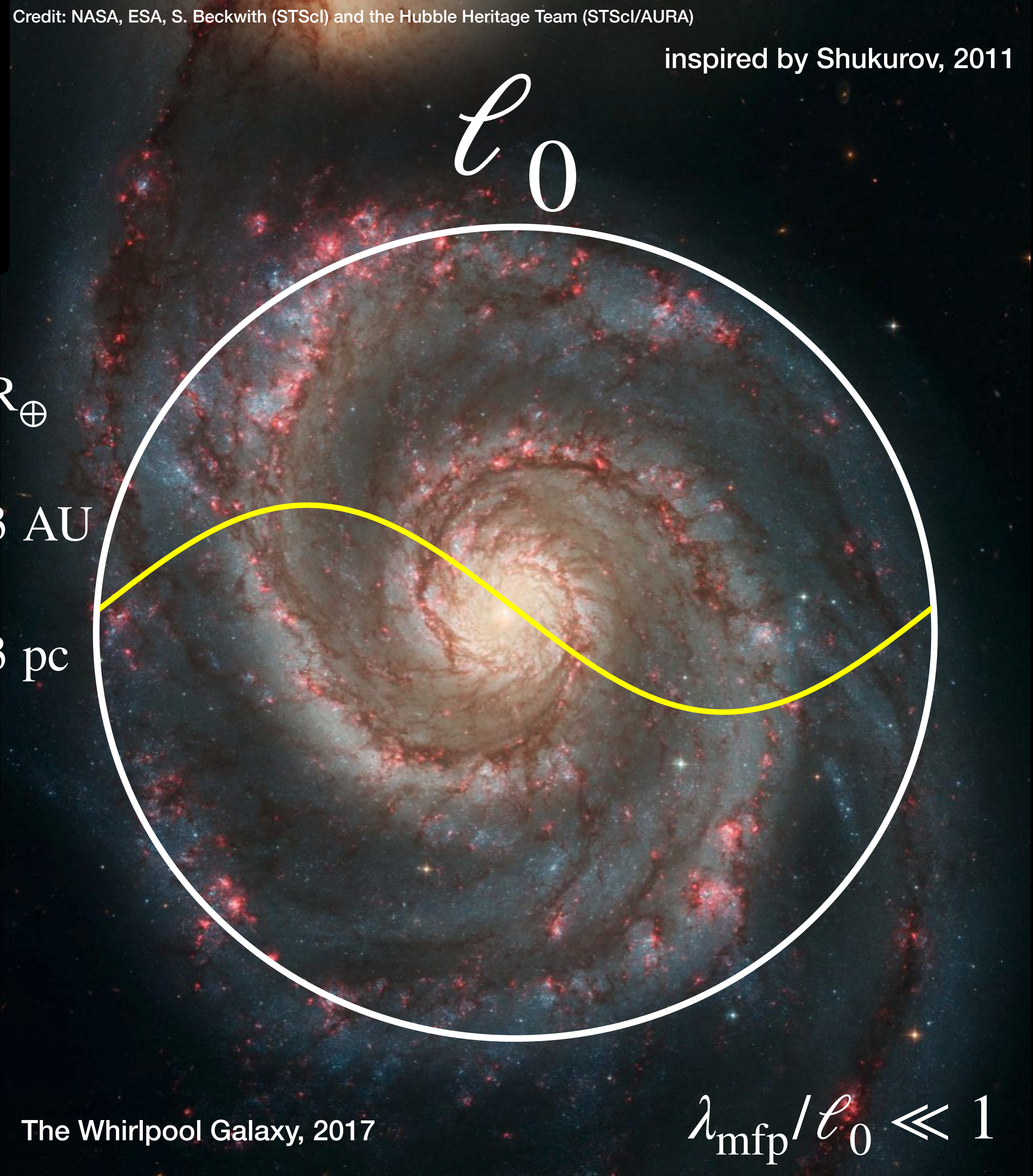
The quadratic nonlinear term

$$|\nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u})|$$

dominates on large scales, ℓ

The Whirlpool Galaxy, 2017

$$\lambda_{\text{mfp}} / \ell_0 \ll 1$$



Turbulence

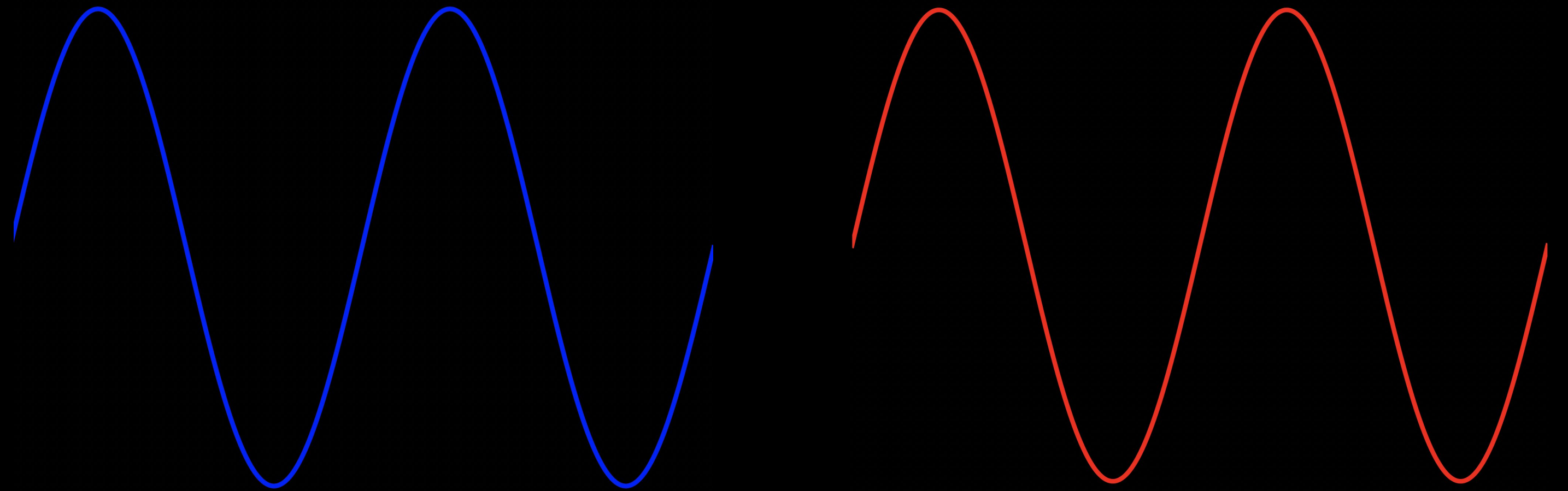
What does $\nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u})$ want to do?

$$k_1 = k_2$$

Consider two waves

$$u_1(x) = \sin(k_1 x)$$

$$u_2(x) = \sin(k_2 x)$$



The nonlinear term creates a new wave

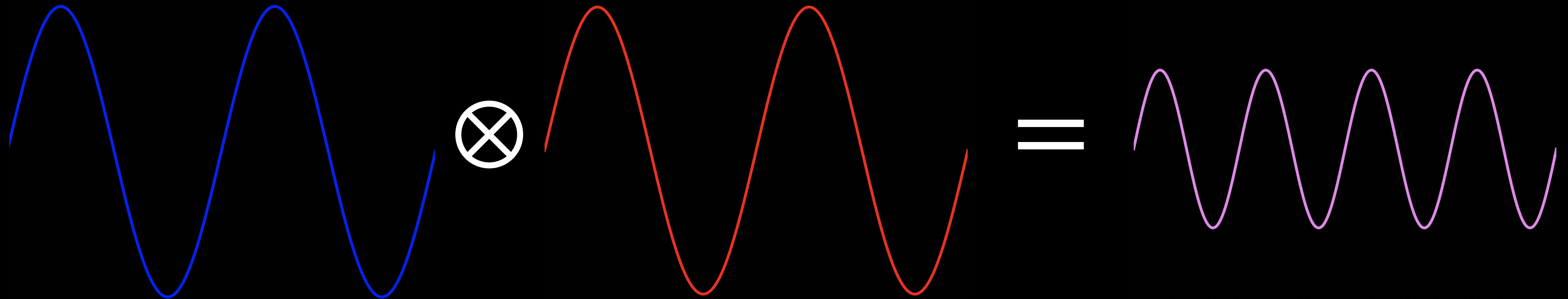
$$\nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) \sim u_1 \partial_x u_2 = k_2 \sin(k_1 x) \cos(k_2 x) \propto \sin(k_3 x)$$

$$k_1 + k_2 = k_3 \quad (\text{momentum conservation})$$

Turbulence

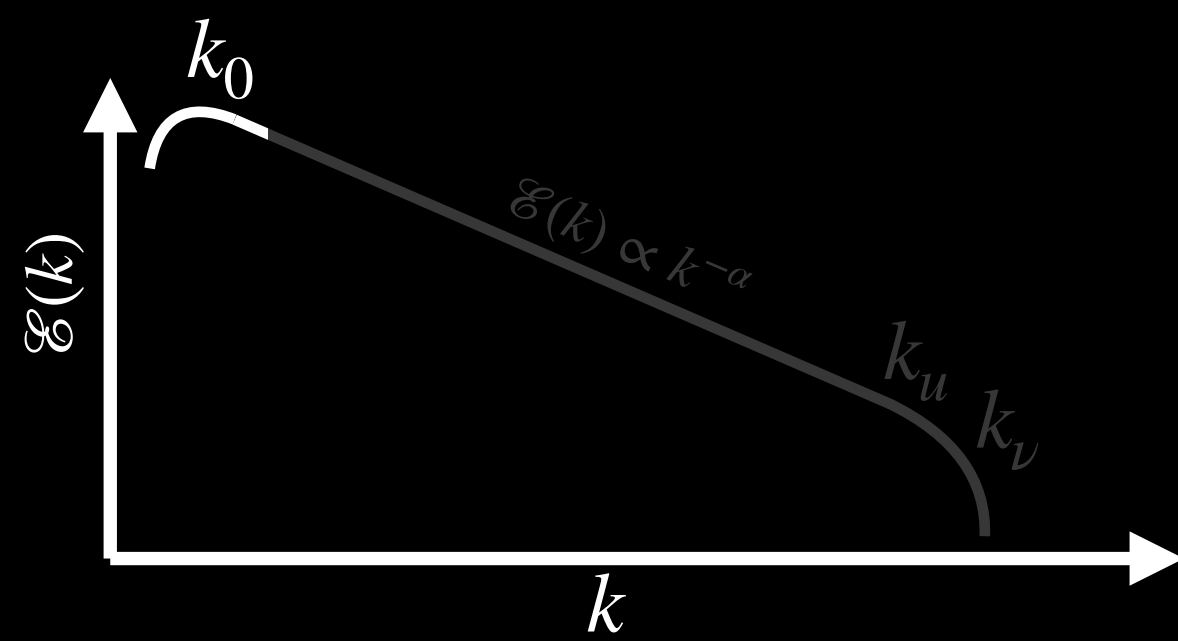
What does $\nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u})$ want to do — cascade!!!

$$\nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u})$$



$$k_1 + k_2 = k_3$$

Turbulence Cascade



Credit: NASA, ESA, S. Beckwith (STScI) and the Hubble Heritage Team (STScI/AURA)

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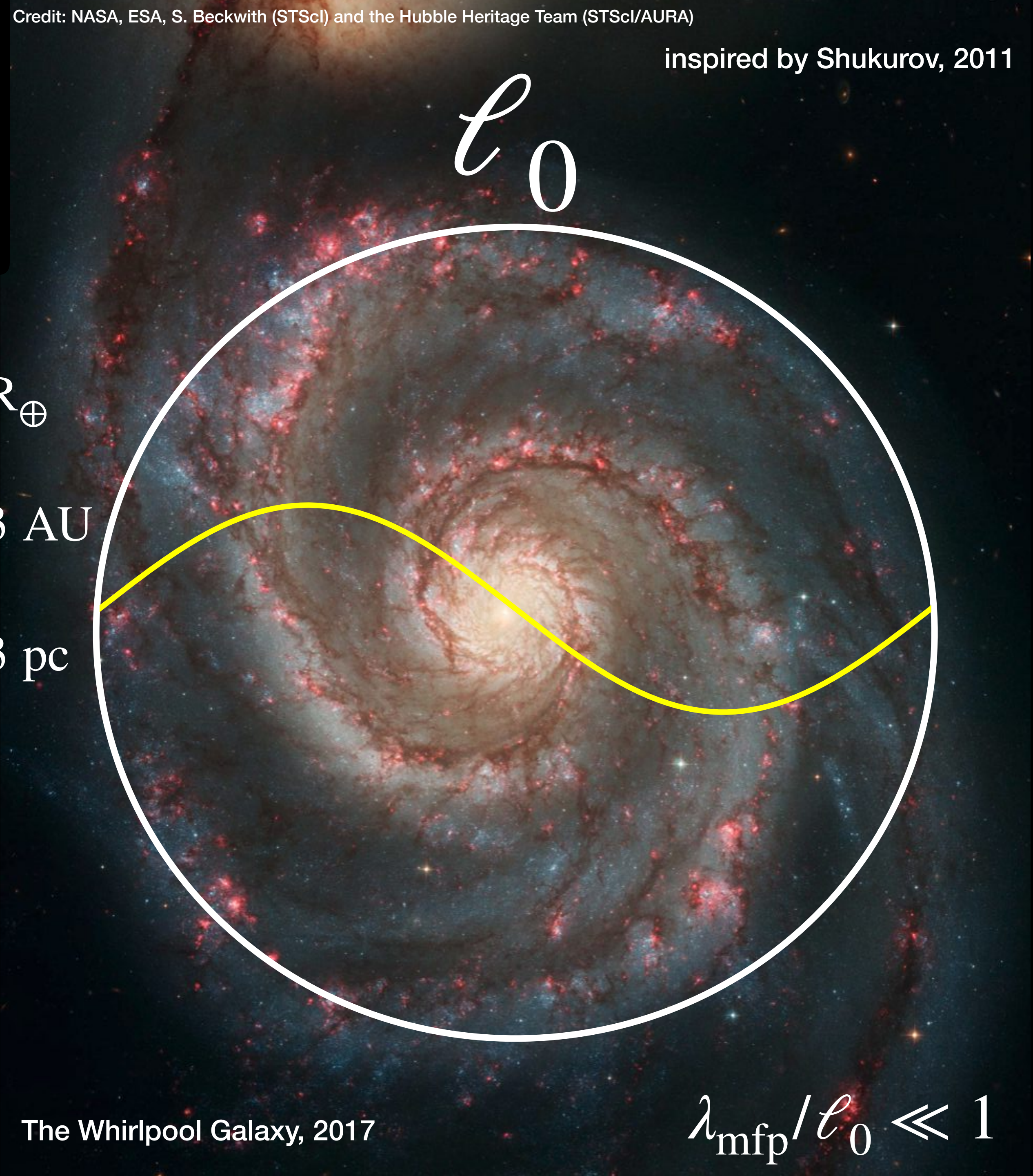
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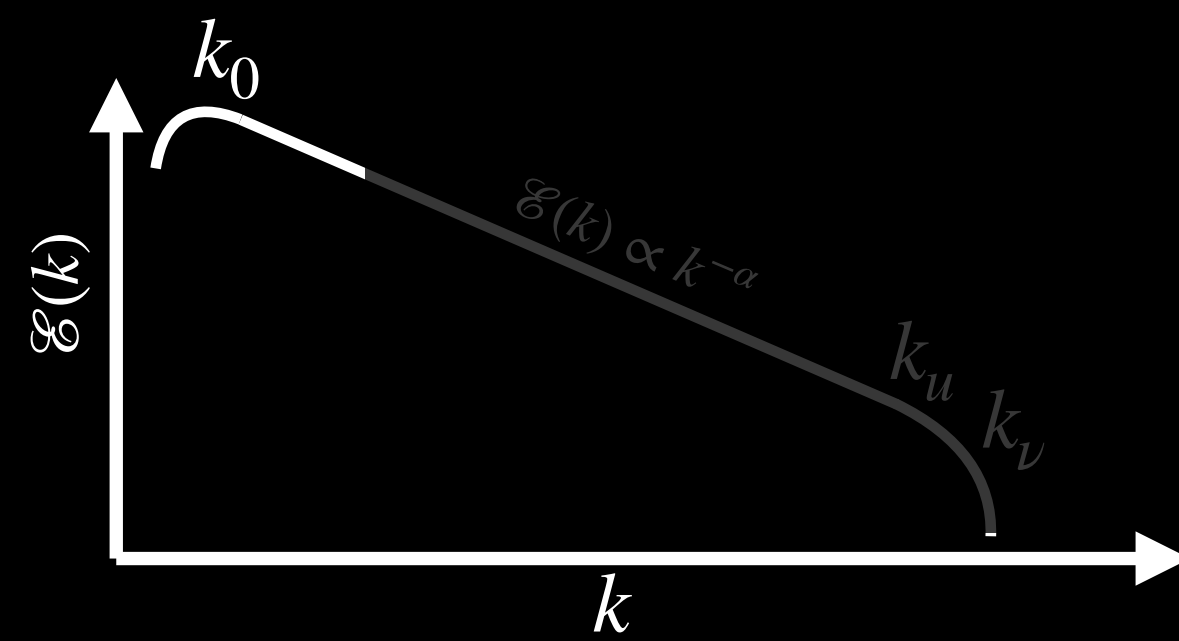
The Whirlpool Galaxy, 2017

$$\lambda_{\text{mfp}} / \ell_0 \ll 1$$



Turbulence

Cascade



The quadratic nonlinear term

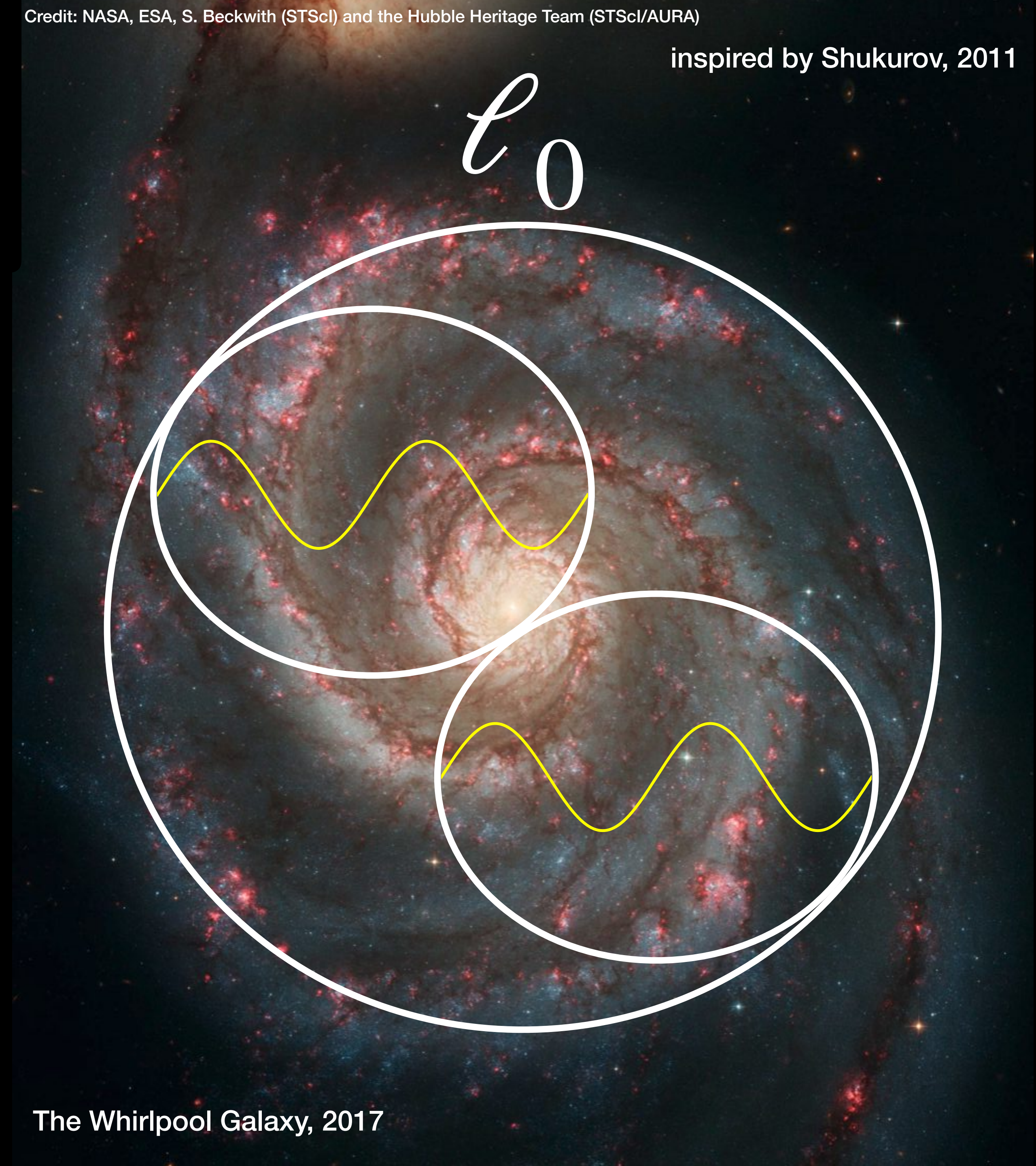
$$|\nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u})|$$

dominates on large scales, ℓ

Creates new modes on

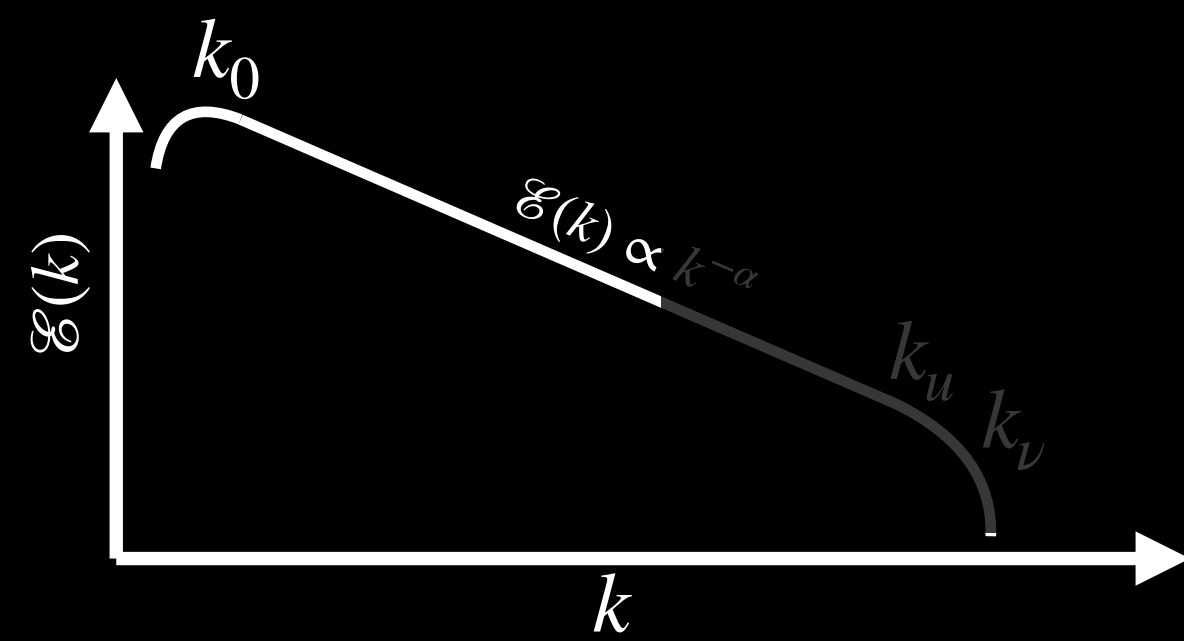
$$t_{\text{nl}} \sim L/U \sim \mathcal{O}(\text{Gyr})$$

$$\dot{\varepsilon} \sim u_0^3 / \ell_0$$



The Whirlpool Galaxy, 2017

Turbulence Cascade



inside of the cascade

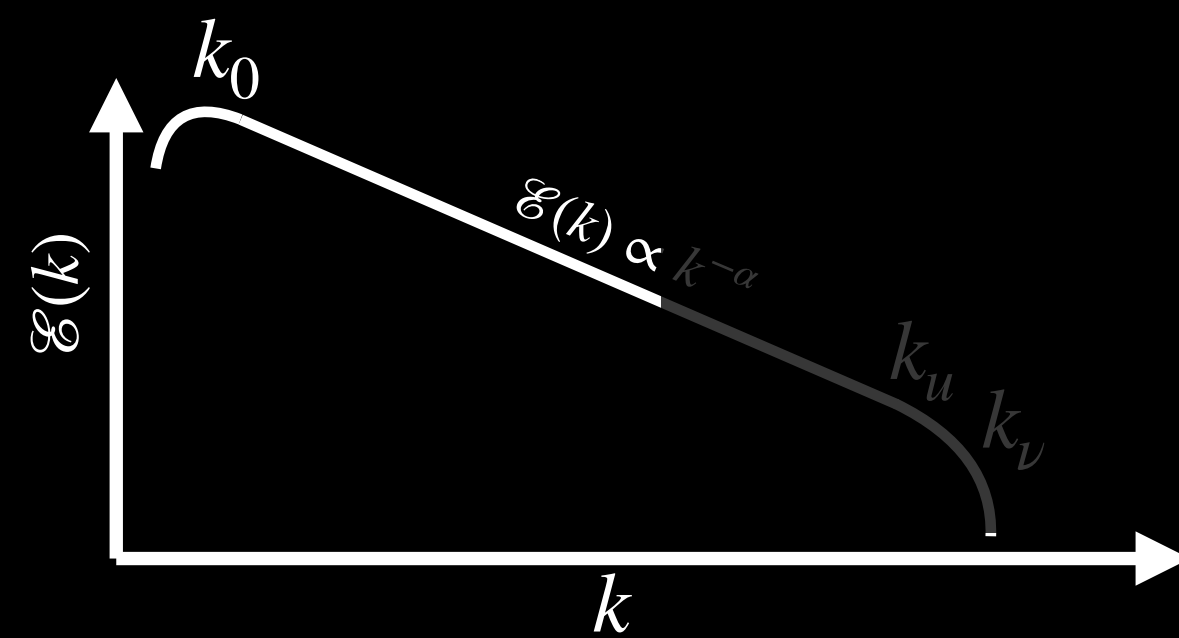
Credit: NASA, ESA, S. Beckwith (STScI) and the Hubble Heritage Team (STScI/AURA)

inspired by Shukurov, 2011



The Whirlpool Galaxy, 2017

Turbulence Cascade

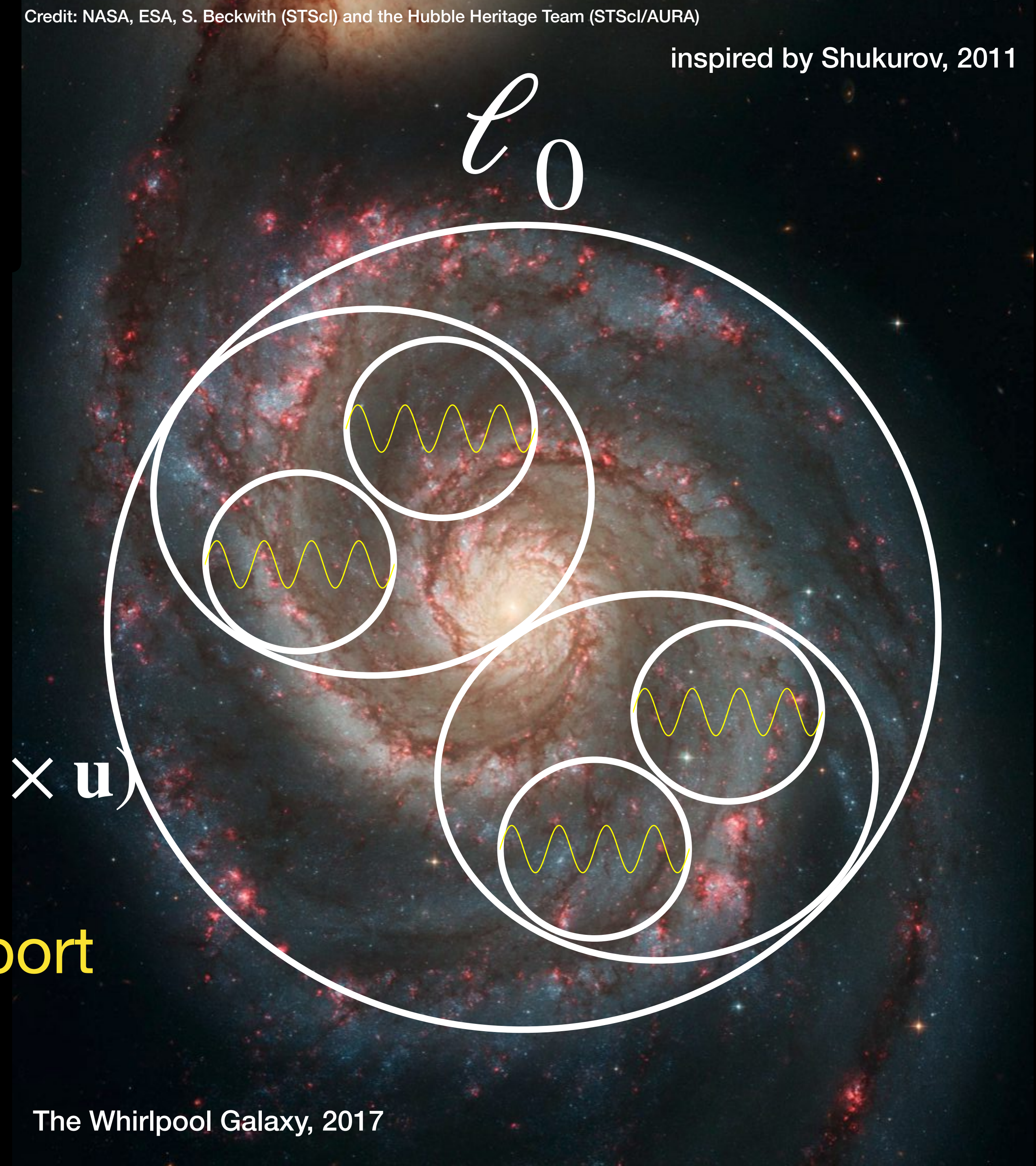


inside of the cascade

The quadratic nonlinear term

$$\nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = \underbrace{1/2 \nabla u^2}_{\text{turbulent pressure}} - \mathbf{u} \times (\nabla \times \mathbf{u})$$

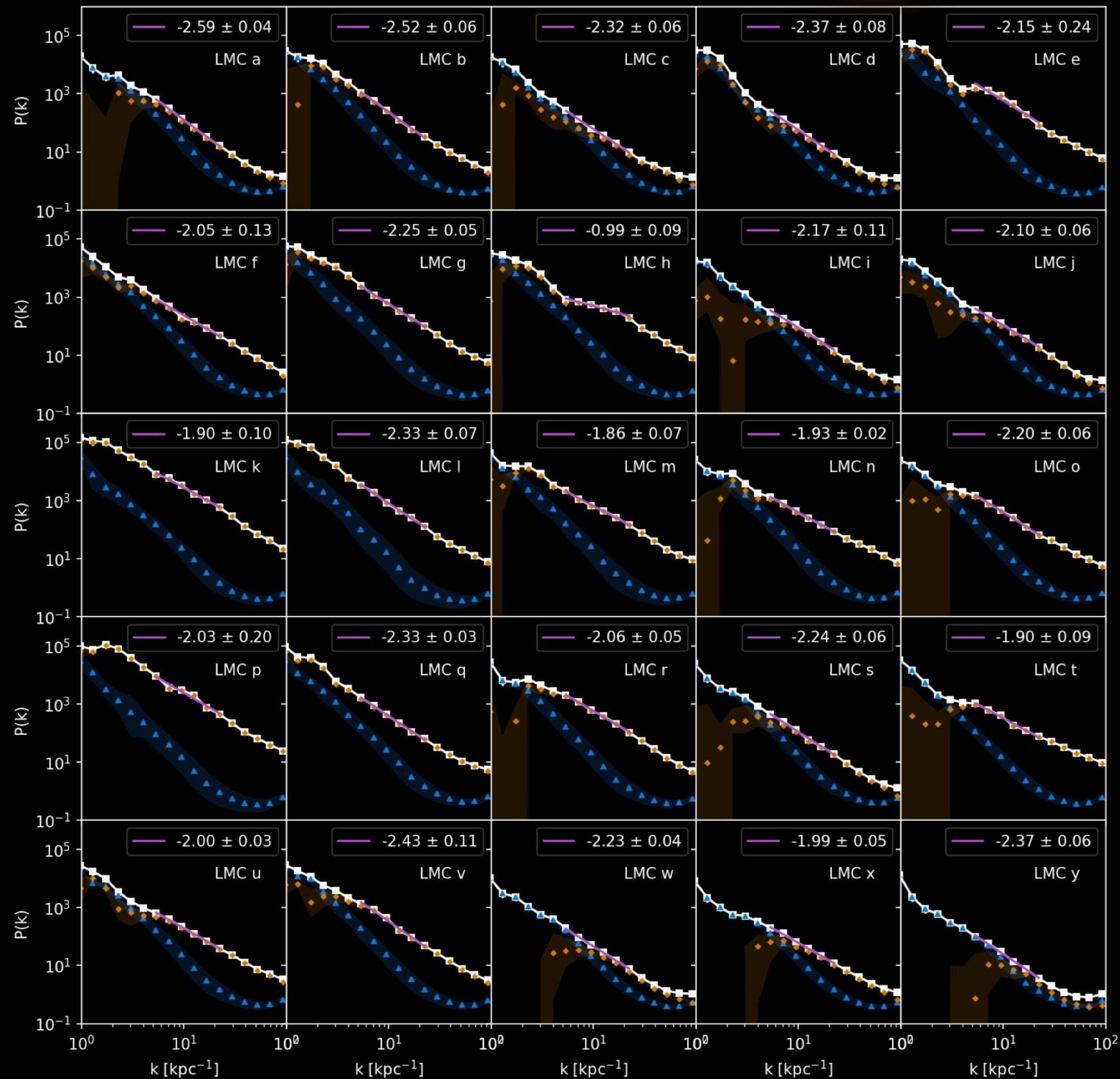
turbulent pressure = turbulent support



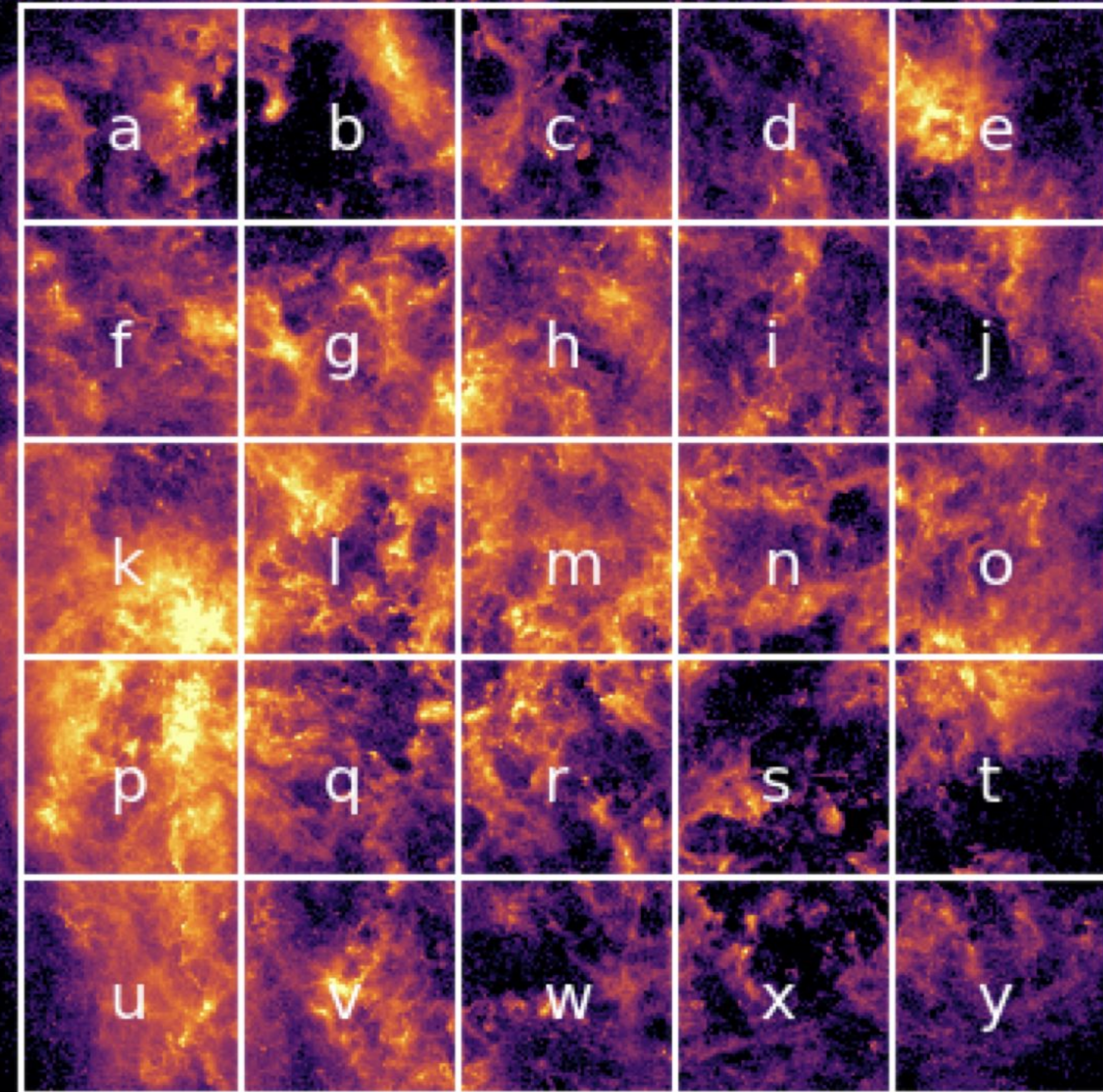
Turbulence



LMC: $500\mu\text{m}$, *Herschel* (processing Gordon et al. 2014)

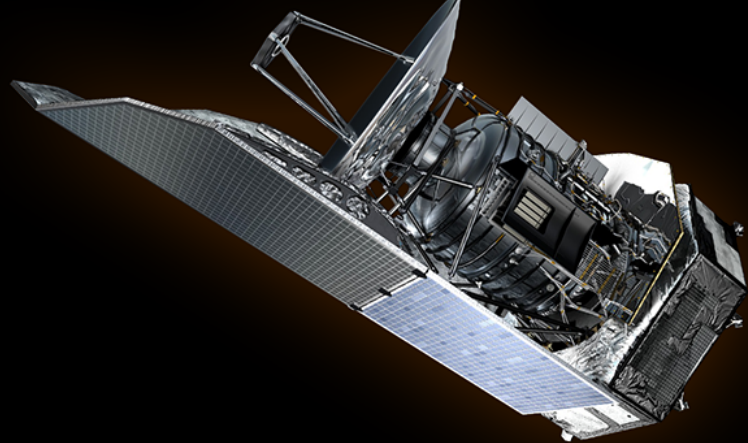


1 kpc

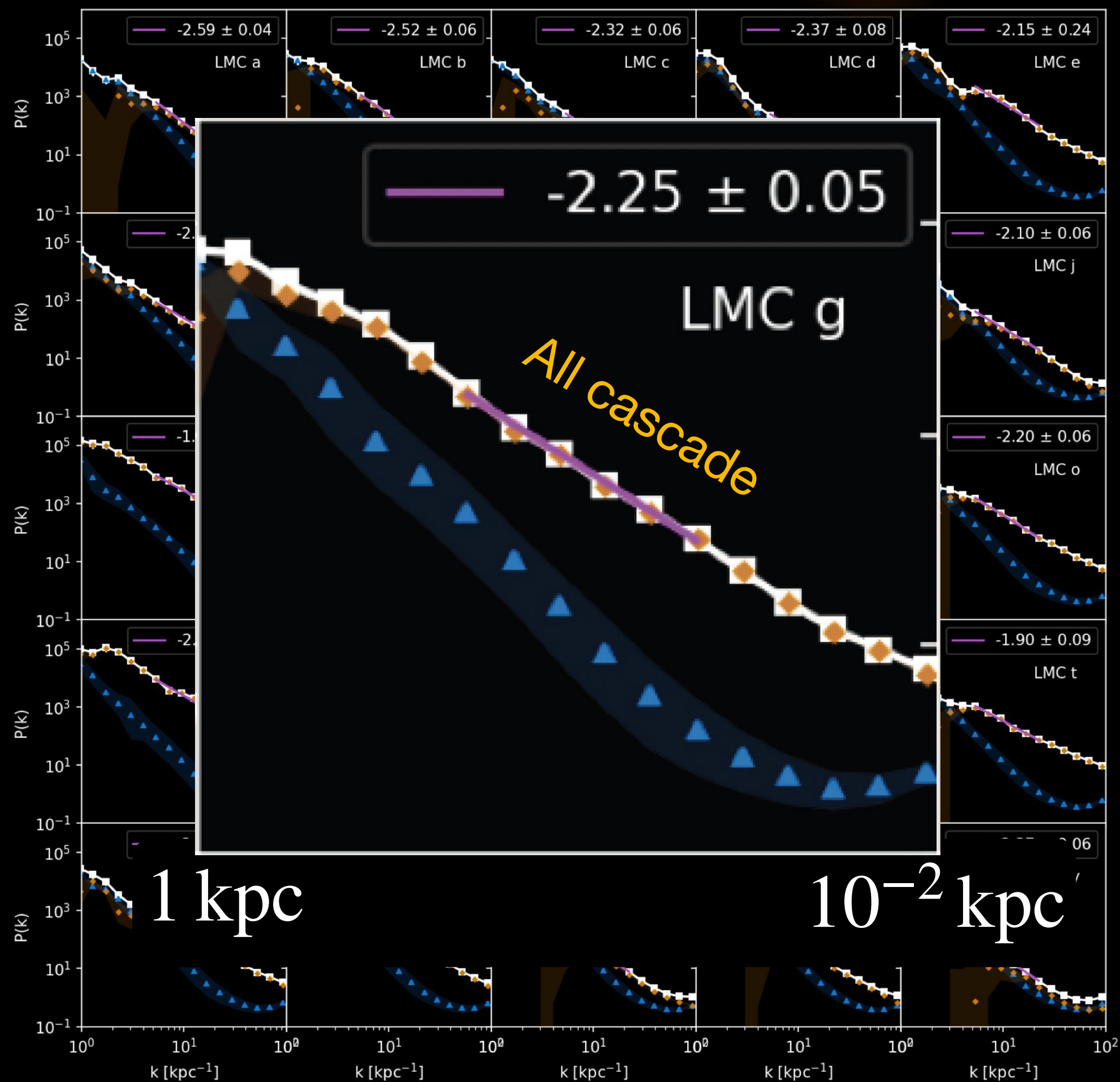


Colman, et al., 2022, MNRAS

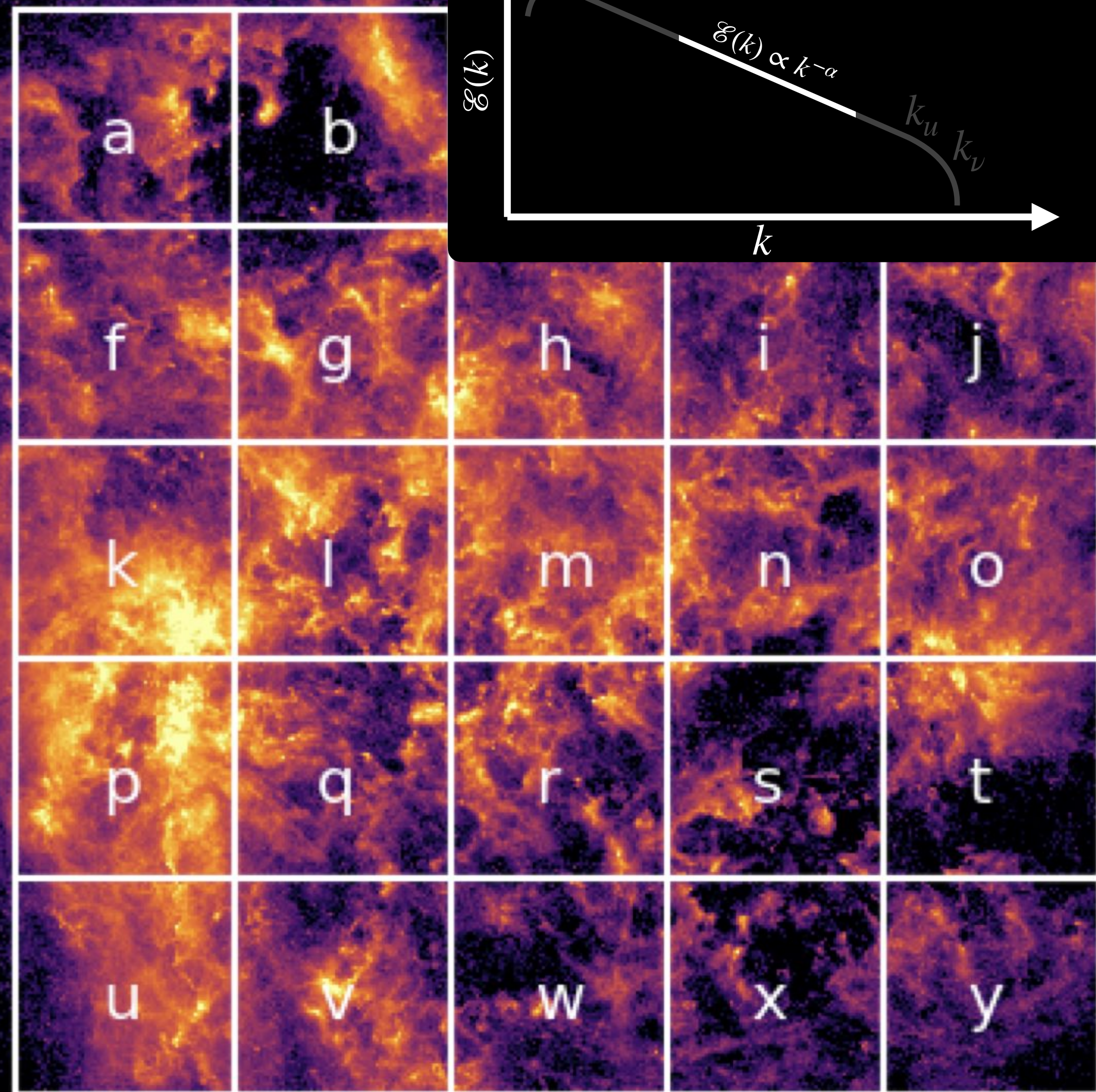
Turbulence



LMC: $500\mu\text{m}$, *Herschel* (processing Gordon et al. 2014)

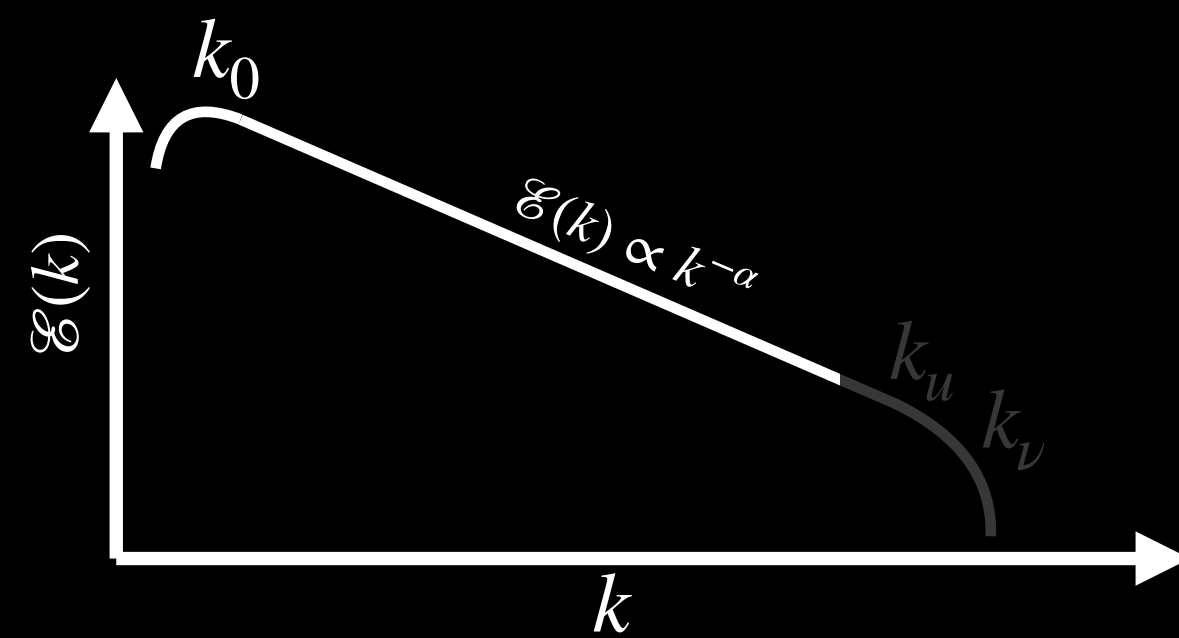


1 kpc



Colman, et al., 2022, MNRAS

Turbulence Cascade

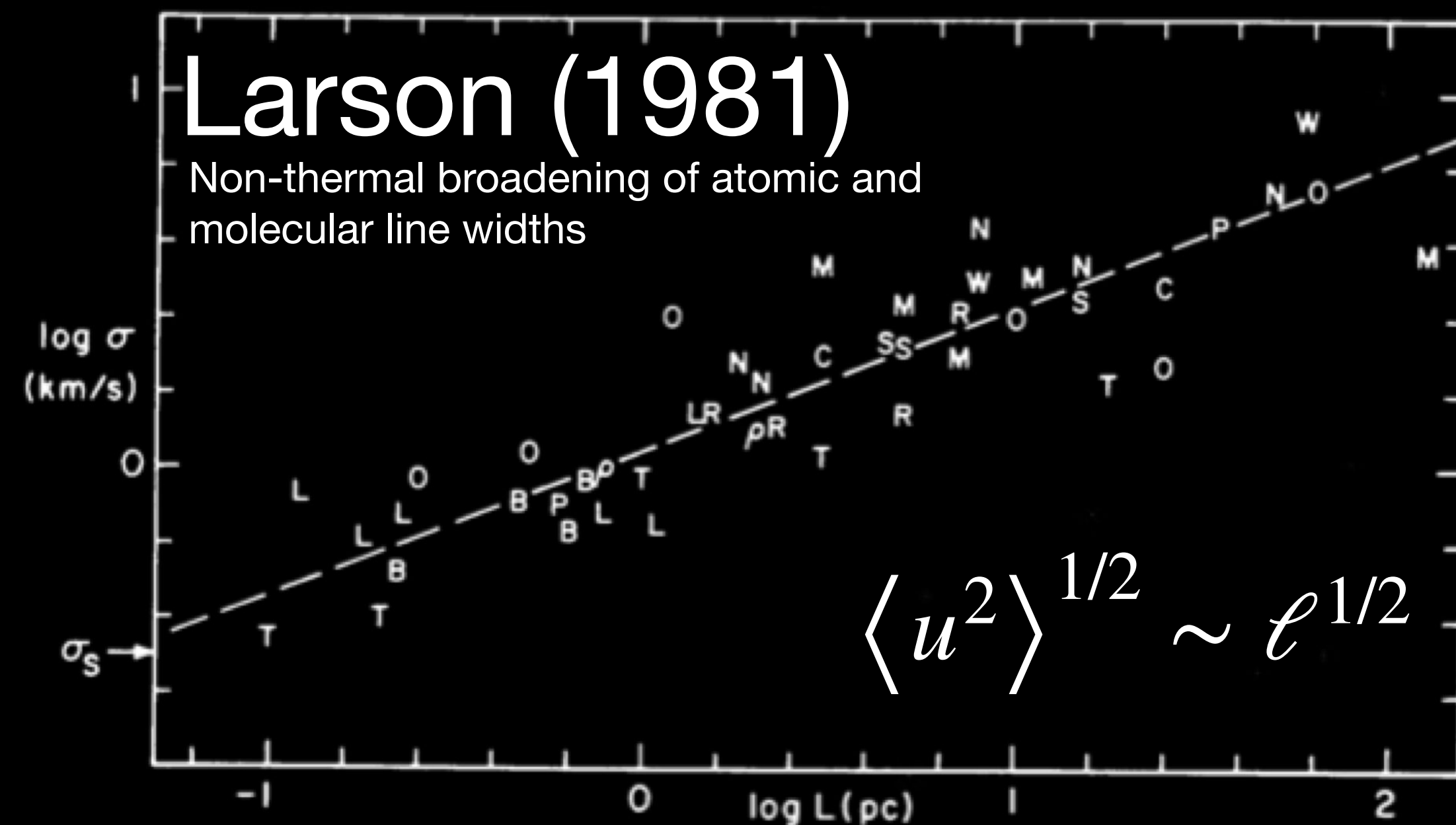


Credit: NASA, ESA, S. Beckwith (STScI) and the Hubble Heritage Team (STScI/AURA)

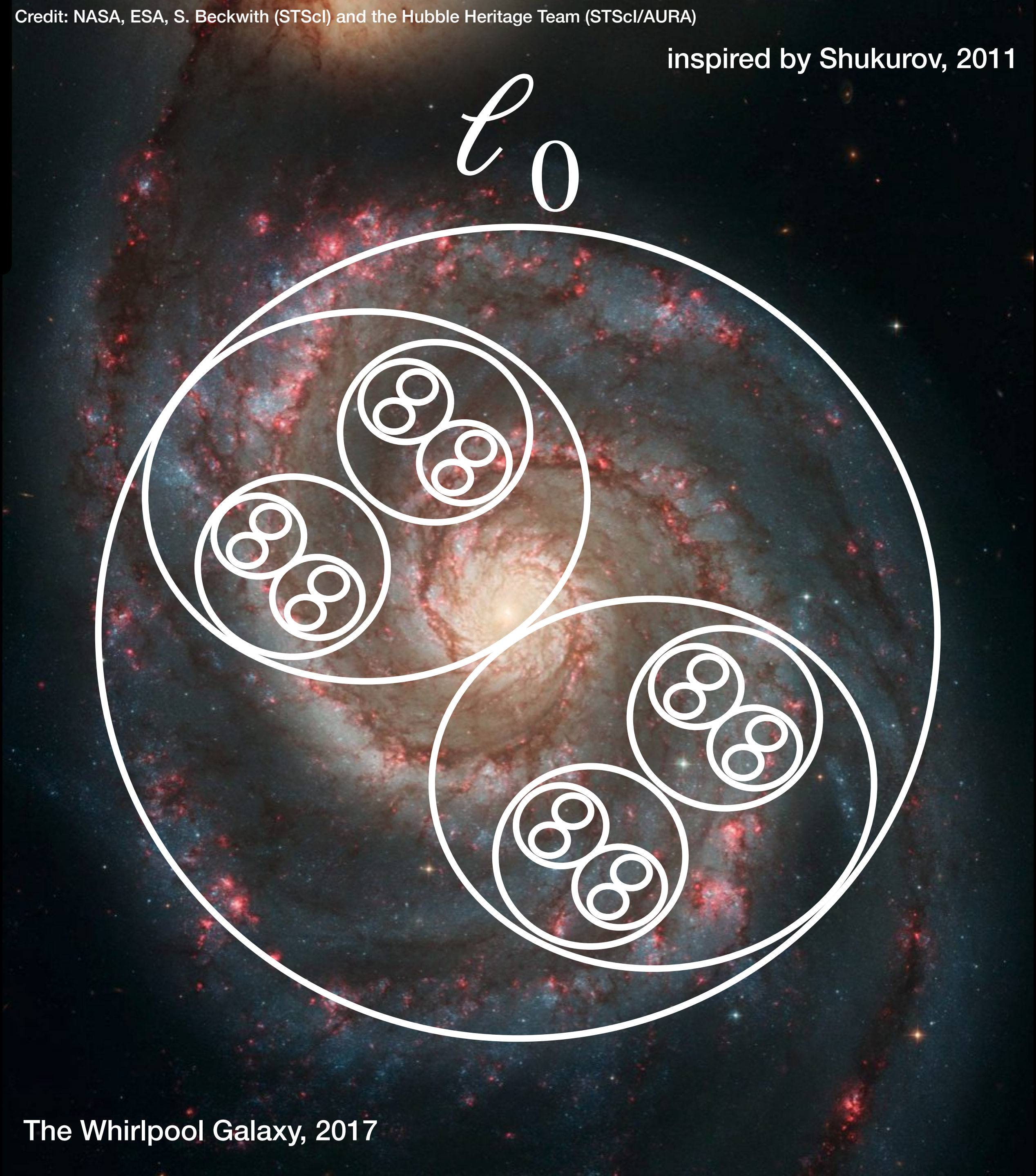
inspired by Shukurov, 2011

Reynolds number is shrinking

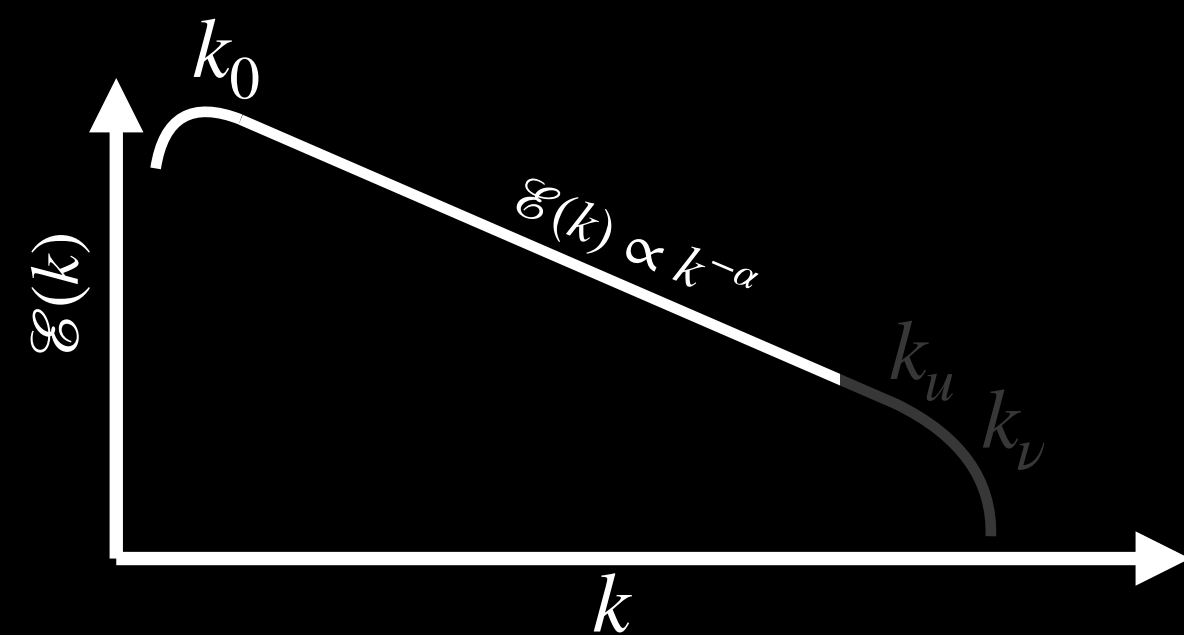
$$\text{Re} \sim \langle u^2 \rangle^{1/2} \ell / \nu \sim \ell^{3/2} / \nu$$



The Whirlpool Galaxy, 2017

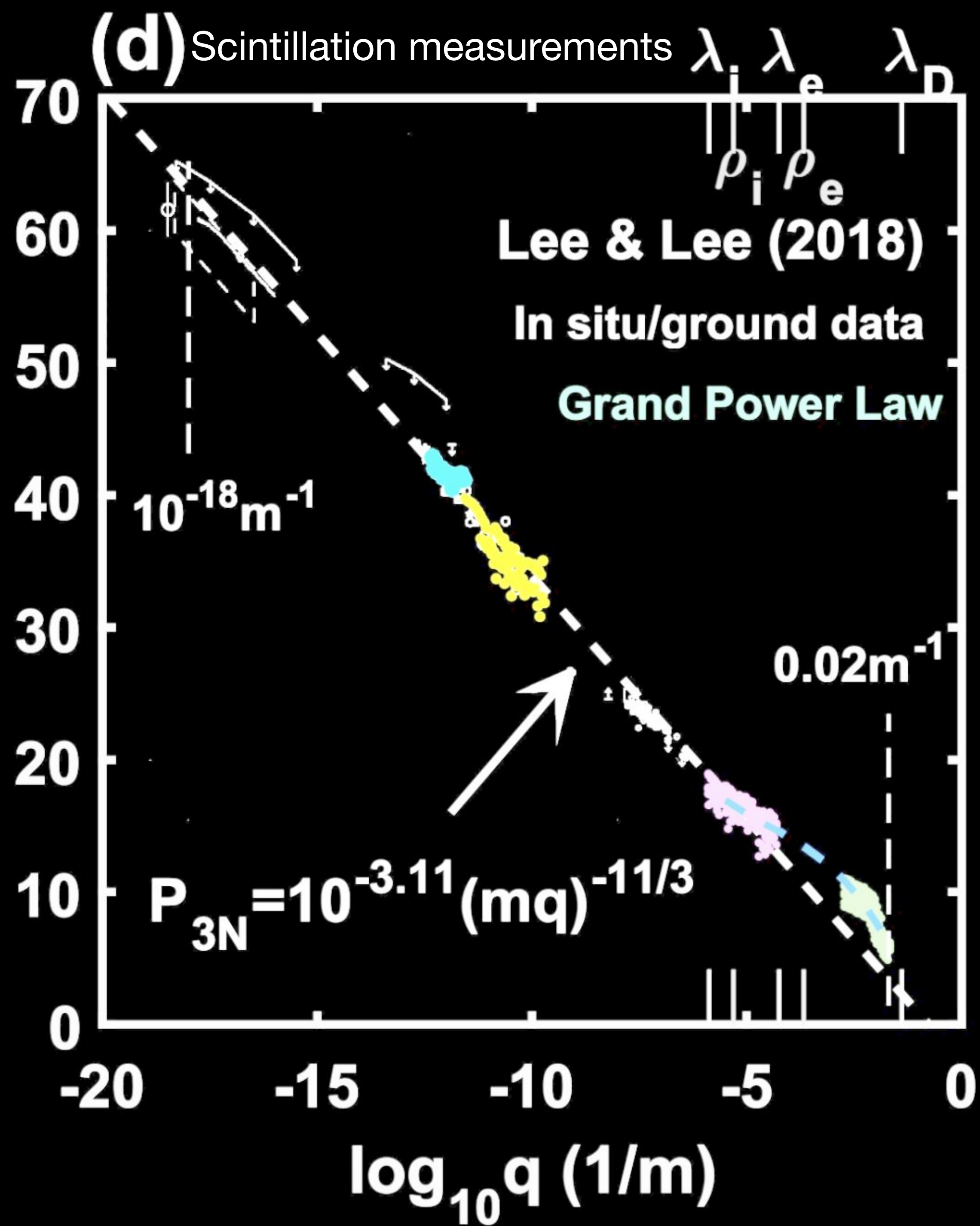
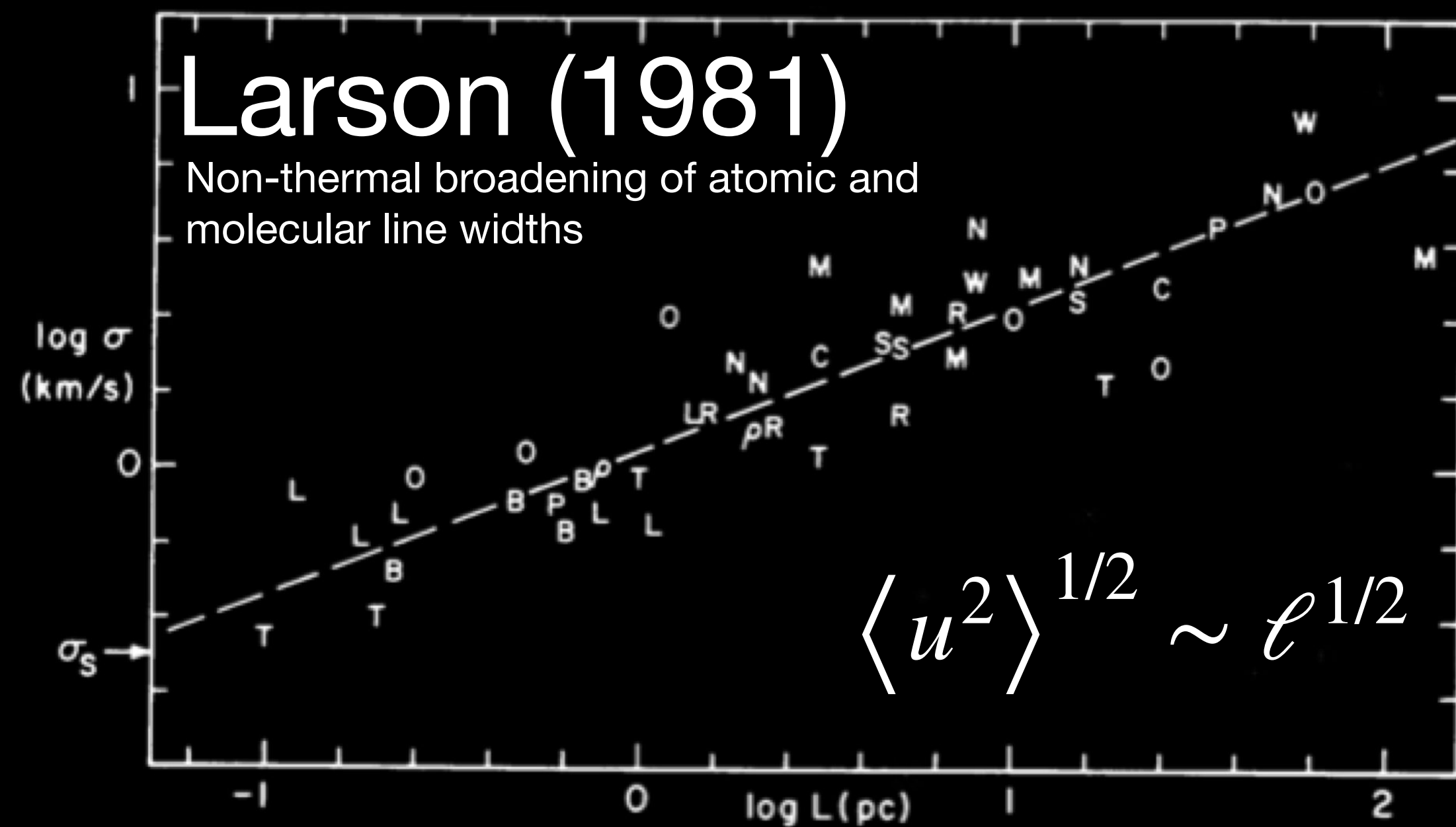


Turbulence Cascade



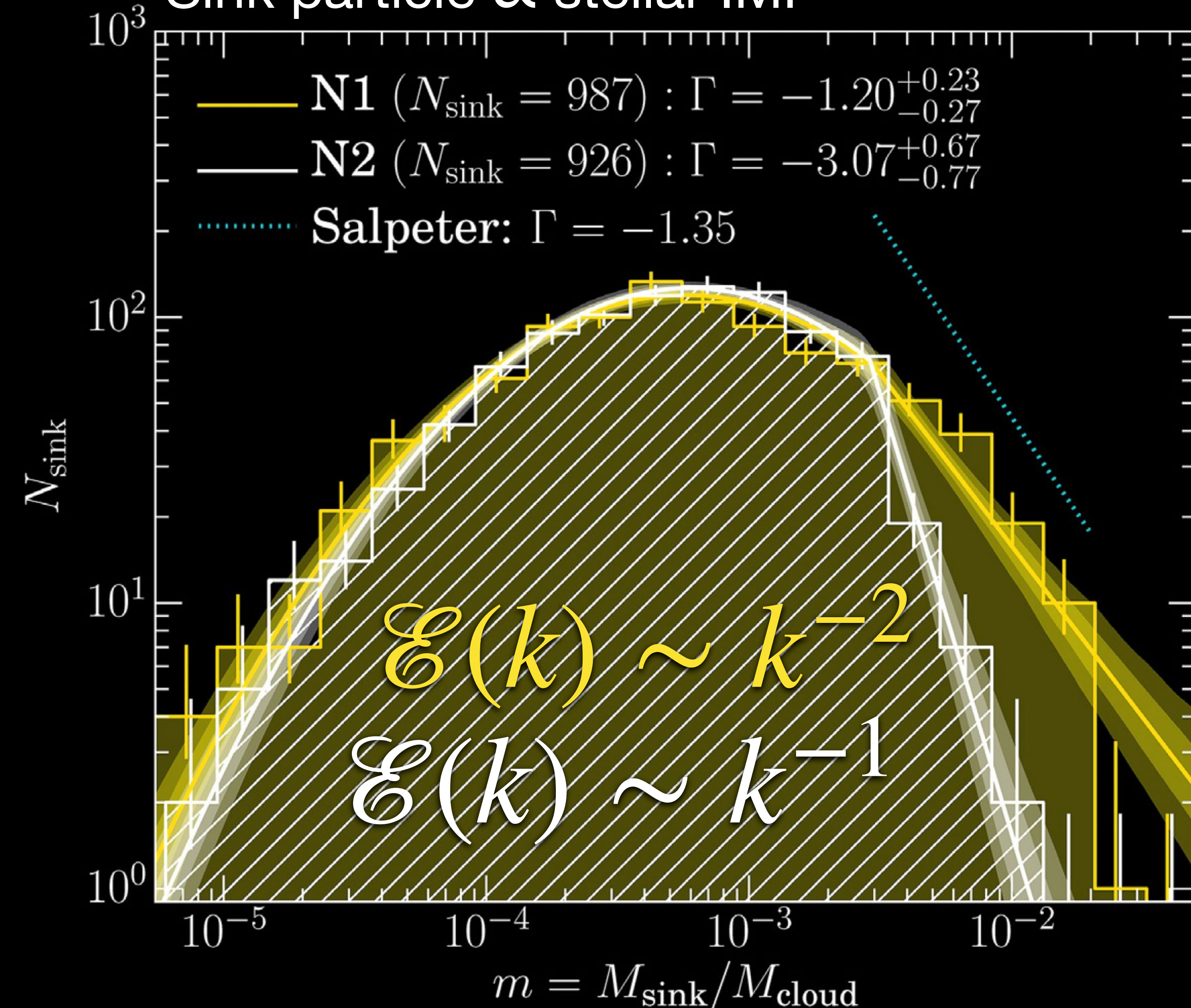
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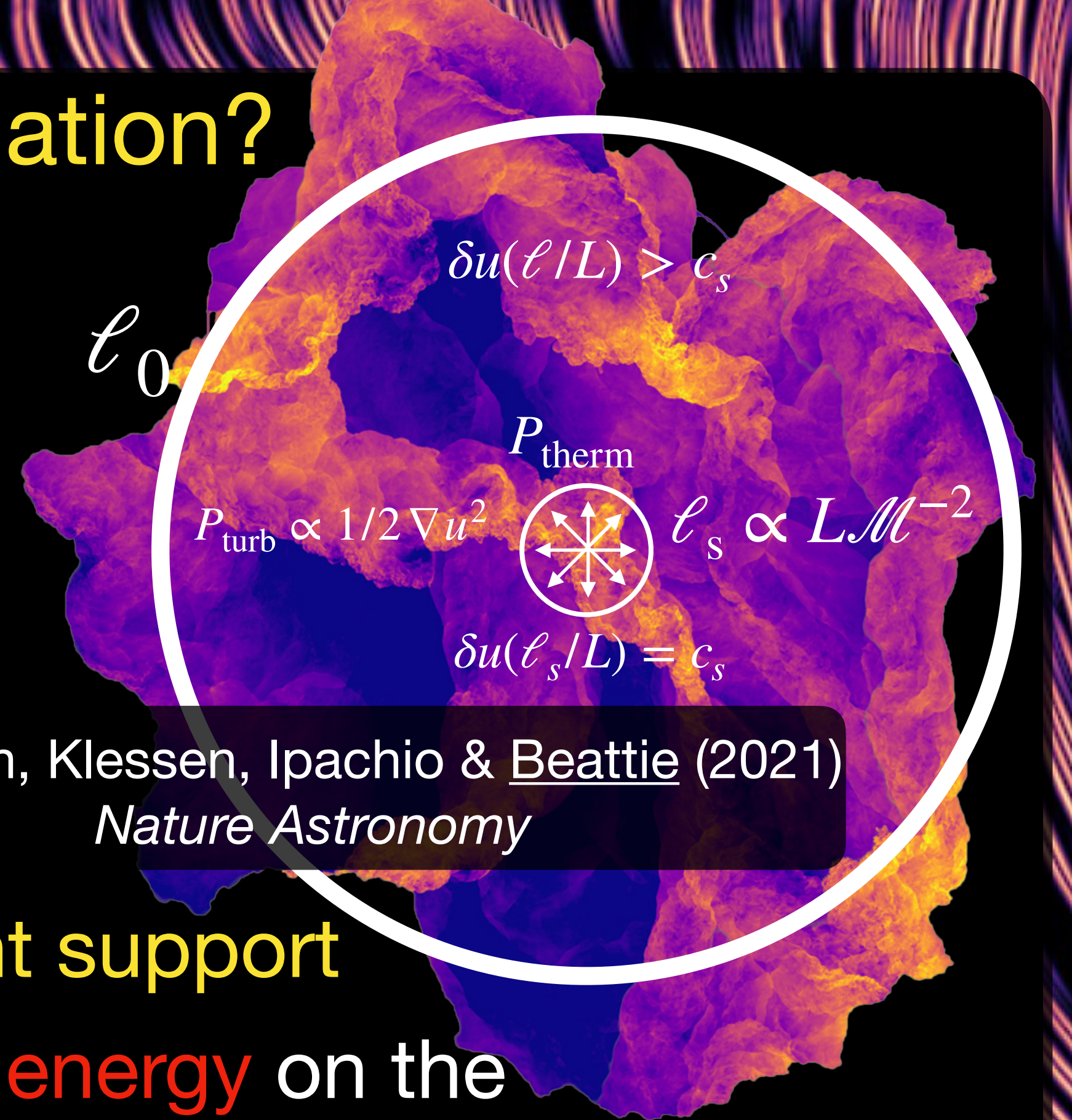


Why should astrophysicists care — star formation?

Sink particle $\propto$ stellar IMF



Nam, Federrath & Krumholz (2021)



Federrath, Klessen, Ipachio & Beattie (2021)
Nature Astronomy

Turbulent support

More energy on the small-scales (more small-scale support) implies less high-mass stars formation

$$\mathbf{u} \cdot \nabla \otimes \mathbf{u} = \frac{1}{2} \nabla u^2 - \mathbf{u} \times \nabla \times \mathbf{u}$$

Magnetic reconnection in a turbulent plasma (visualisation by J. Beattie).

Why should astrophysicists care — cosmic ray transport?

Testing Physical Models for Cosmic Ray Transport Coefficients on Galactic Scales: **Self-Confinement** and **Extrinsic Turbulence** at $\sim$ GeV Energies

Philip F. Hopkins¹, Jonathan Squire², T. K. Chan^{3,4}, Eliot Quataert⁵,
Suoqing Ji¹, Dušan Kereš², Claude-André Faucher-Giguère⁶

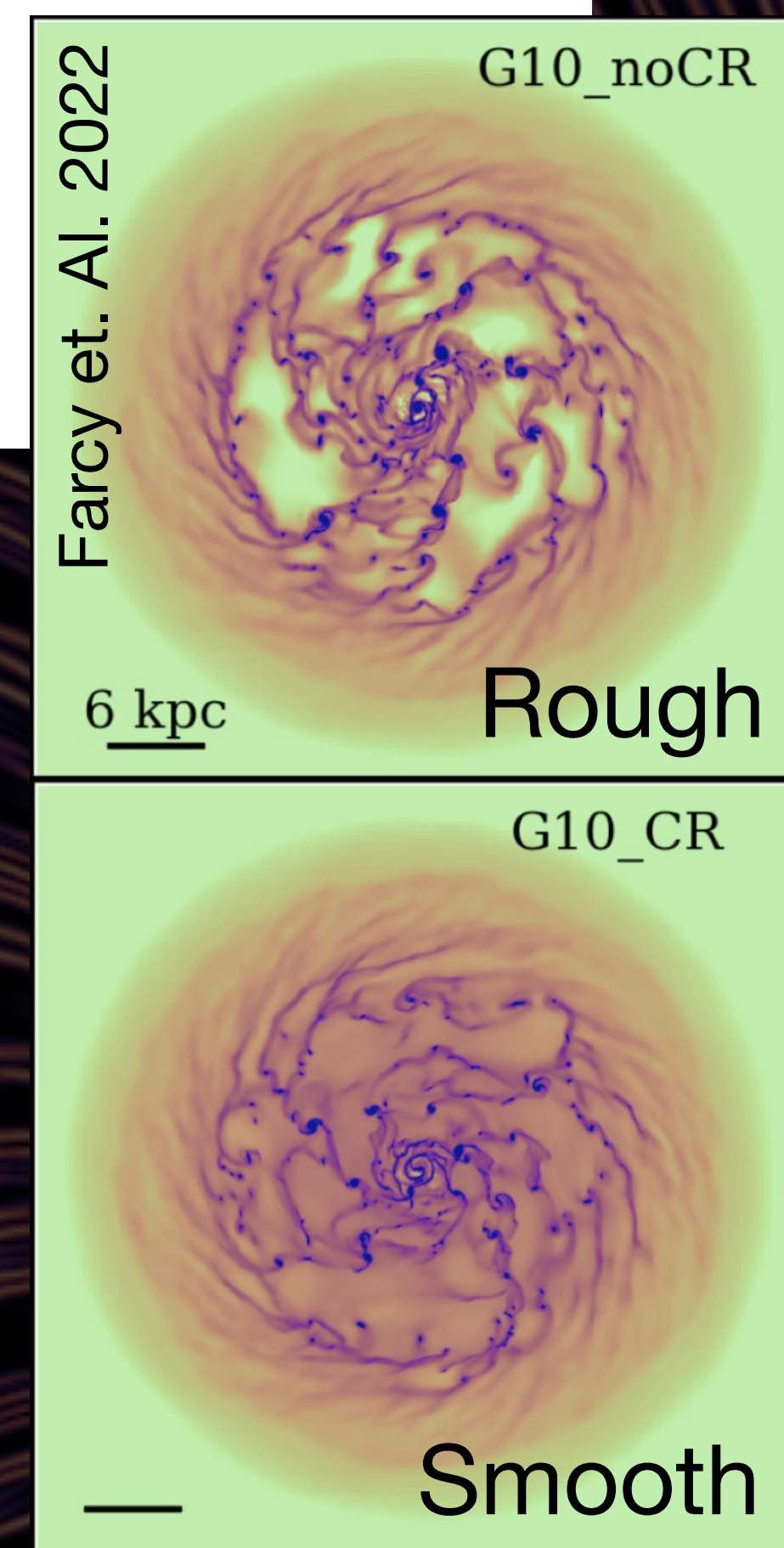
Sampson, [Beattie et al. \(2023, MNRAS\)](#). *Turbulent diffusion of streaming cosmic rays in compressible, partially ionised plasma*

Provides source of
diffusion on large-
scales

[Beattie et al. \(2023, Frontiers\)](#). *Ion Alfvén velocity fluctuations and implications for the diffusion of streaming cosmic rays*

Determines the
properties of
turbulence

CRs add
extra
pressure
support +
smooth
plasma
gradients



The energy cascade — fundamental for CR transport

Two great insights from Kolmogorov

- Constant energy flux between modes
- No magnetic field
- No inhomogeneities
- No density fluctuations
- Isotropic

$$\varepsilon \sim u_\ell^3 / \ell \sim \text{const.}$$

Two great insights from Kolmogorov

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$$\varepsilon \sim u_\ell^3 / \ell \sim \text{const.}$$

$$u_\ell \sim (\varepsilon \ell)^{1/3}$$

Two great insights from Kolmogorov

- Constant energy flux between modes
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- Isotropic

$$\varepsilon \sim u_\ell^3 / \ell \sim \text{const.}$$

$$u_\ell \sim (\varepsilon \ell)^{1/3}$$

$$u_\ell \sim (\varepsilon \ell)^{1/3} \iff u^2(k) \sim k^{-5/3} dk$$

Two great insights from Kolmogorov

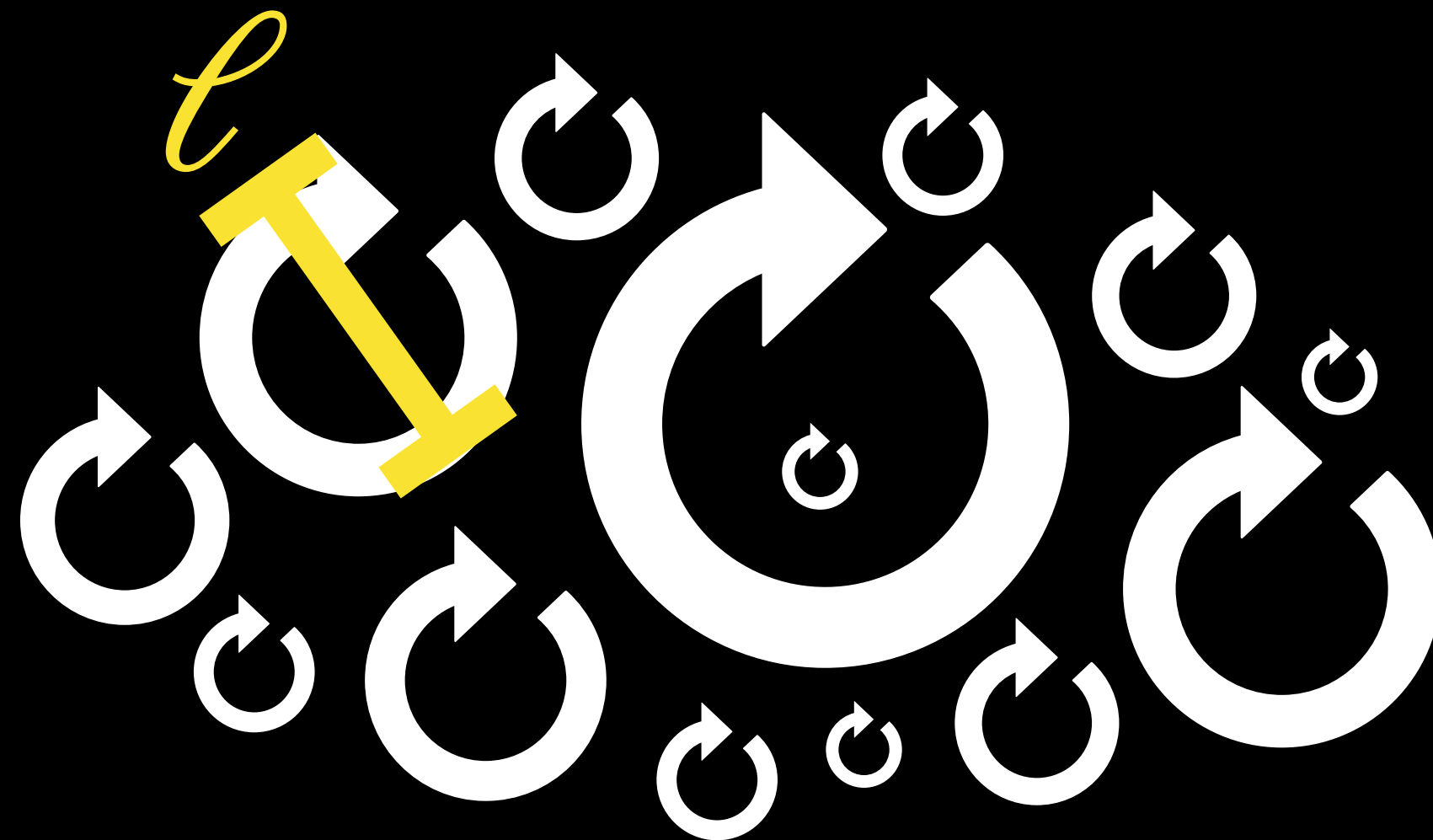
- Constant energy flux between modes
- No magnetic field
- No inhomogeneities
- No density fluctuations
- Isotropic

$$u_\ell \sim (\varepsilon \ell)^{1/3}$$

“structure function”

$$u(\ell) = \langle |u(\mathbf{x}) - u(\mathbf{x} + \ell) \cdot \ell| \rangle_{\mathbf{x}} \sim \ell^{1/3}$$

ℓ picks out an eddy



Two great insights from Kolmogorov

- Constant energy flux between modes
- No magnetic field
- No inhomogeneities
- No density fluctuations
- Isotropic

$$u_\ell \sim (\varepsilon \ell)^{1/3}$$

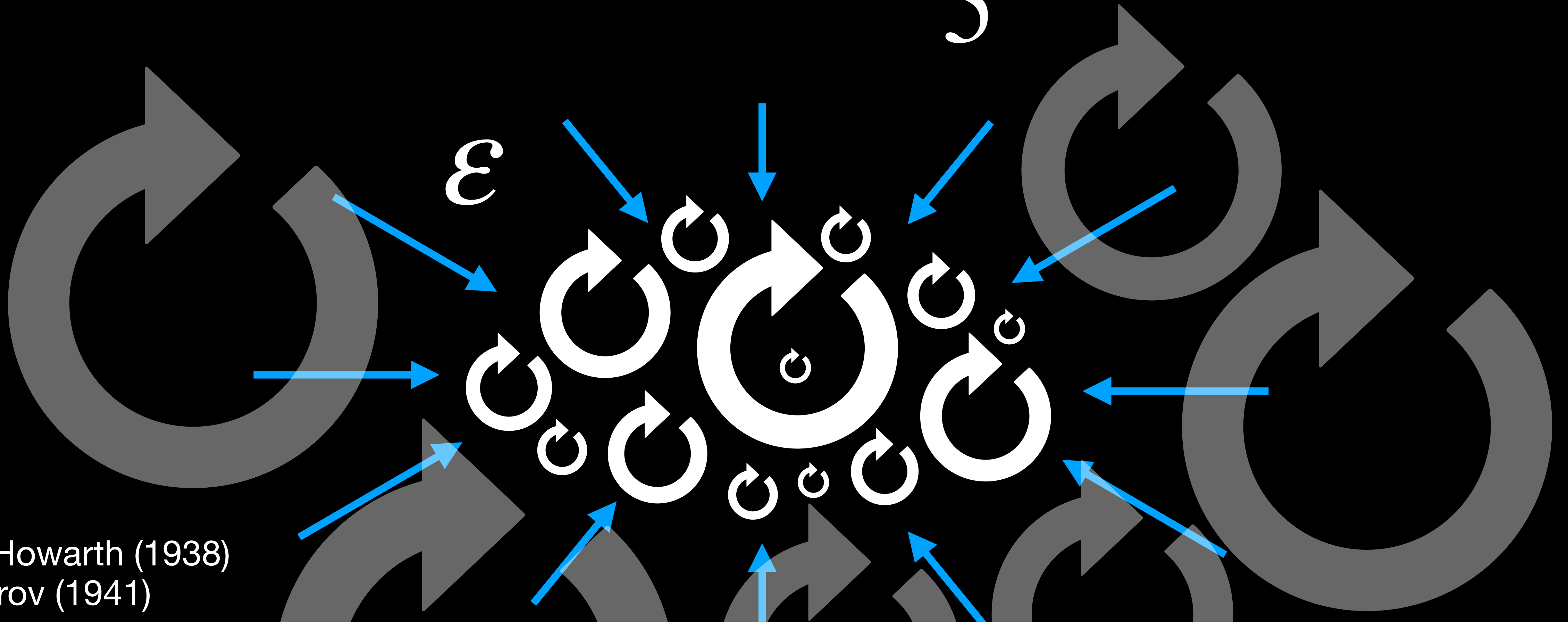
$$u(\ell) = \langle |u(\mathbf{x}) - u(\mathbf{x} + \ell) \cdot \ell| \rangle_{\mathbf{x}} \sim \ell^{1/3}$$

$$u(\ell) = \langle |\delta u_L| \rangle_{\mathbf{x}} \sim \ell^{1/3}$$

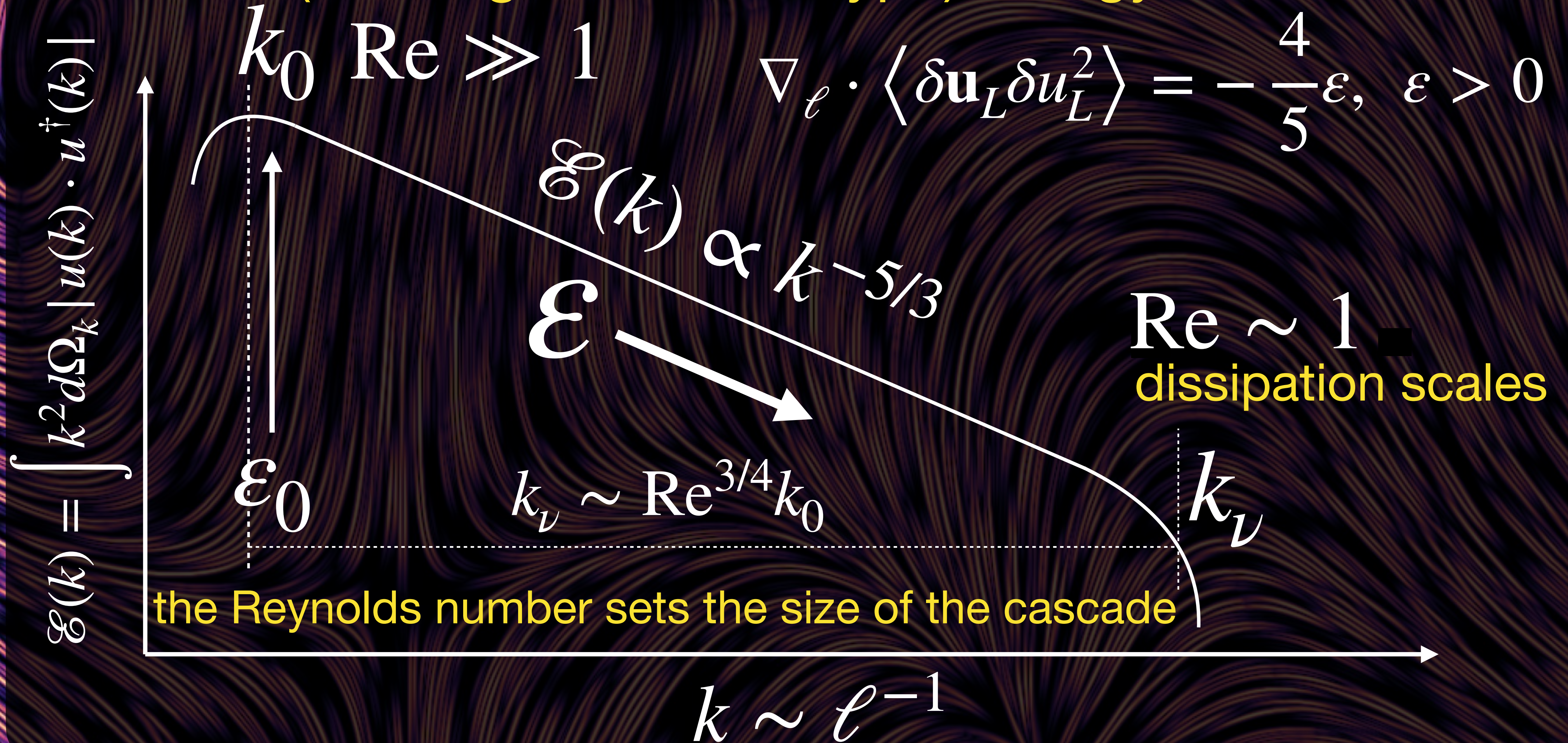
An exact relation for turbulence

Kolmogorov's 4/5 law

$$\nabla_{\ell} \cdot \langle \delta \mathbf{u}_L \delta u_L^2 \rangle = -\frac{4}{5} \varepsilon, \quad \varepsilon > 0$$



The (Kolmogorov, 1941-type) energy cascade



Fact: the galaxy that we reside in is in a state of highly-compressible fluid turbulence

$$E_{\text{SNe}} \approx 10^{51} \text{ erg}$$

$$\gamma_{\text{SNe}} \approx 0.01 - 0.03 \text{ yr}^{-1} \quad \text{Diehl + (2016)}$$

$$\dot{E}_{\text{SNe}} \approx E_{\text{SNe}} \gamma_{\text{SNe}} \approx 10^{41} \text{ erg s}^{-1}$$

$$E_{\text{turb}} \approx 10^{54} \text{ erg}$$

$$t_{\text{turb}} \approx \ell_0 / \langle u^2 \rangle^{1/2} \approx 10^7 \text{ yr} \quad \text{Ferriere + (2020)}$$

$$\dot{E}_{\text{turb}} \approx E_{\text{turb}} / t_{\text{turb}} \approx 10^{39} \text{ erg s}^{-1}$$

$$\dot{E}_{\text{turb}} / \dot{E}_{\text{SNe}} \approx 10^{-2}$$

CC-supernovae alone provide enough energy flux to fuel a continuously driven cascade

Connor, Beattie + (2025, ApJ) Cascading from the winds to the disk: universality of supernovae-driven turbulence in different galactic ISMs

The (Kolmogorov, 1941-type) energy cascade

$$k_0 \text{ Re} \gg 1$$

$$\nabla_\ell \cdot \langle \delta \mathbf{u}_L \delta u_L^2 \rangle = -\frac{4}{5} \varepsilon, \quad \varepsilon > 0$$

Does it resemble this
idea at all?

ales

$$\mathcal{E}(k) = \int k^2 d\Omega_k |u(k) \cdot u^\dagger(k)|$$

$$k_0$$

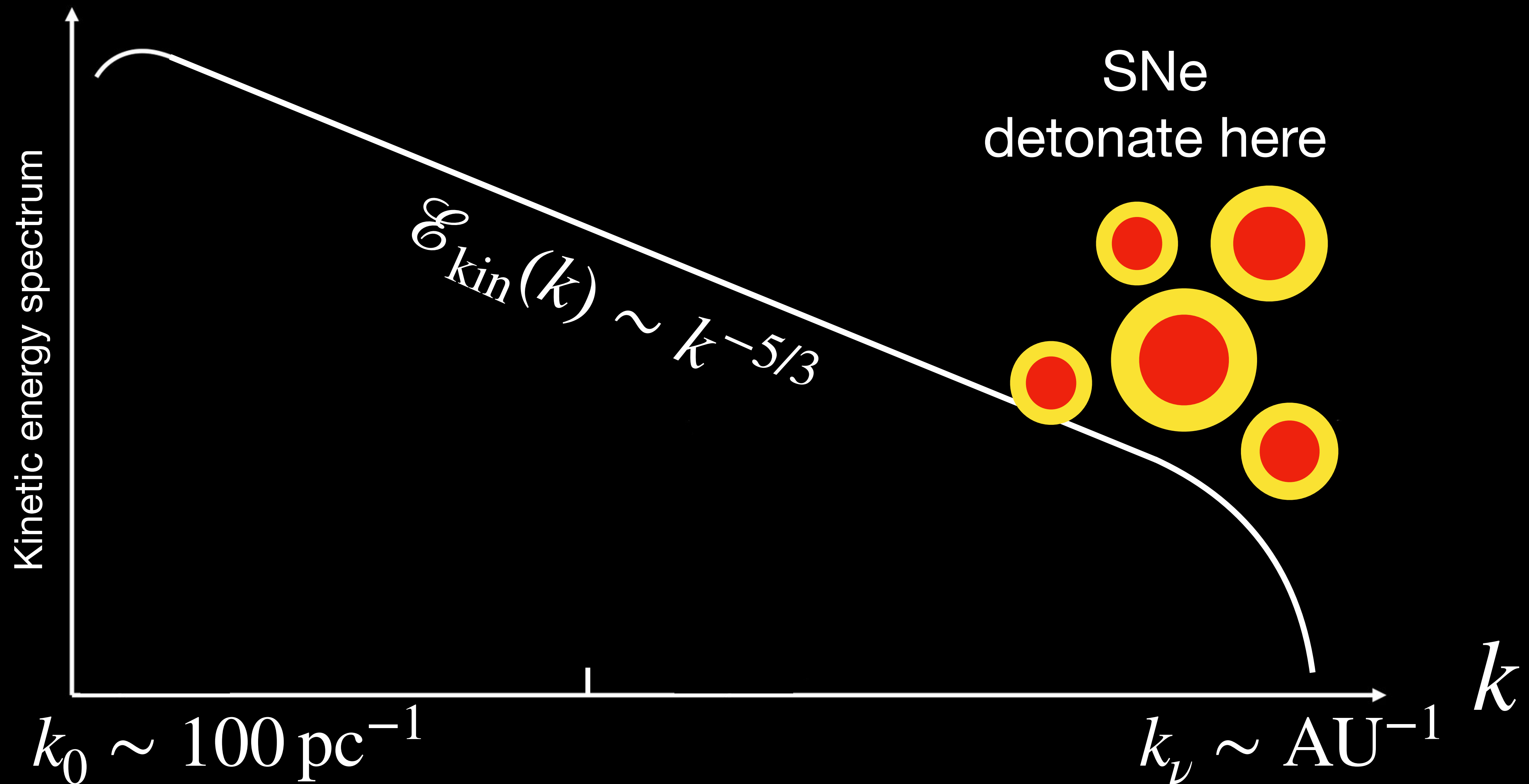
$$k_\nu \sim \text{Re}^{3/4} k_0$$

$$K_\nu$$

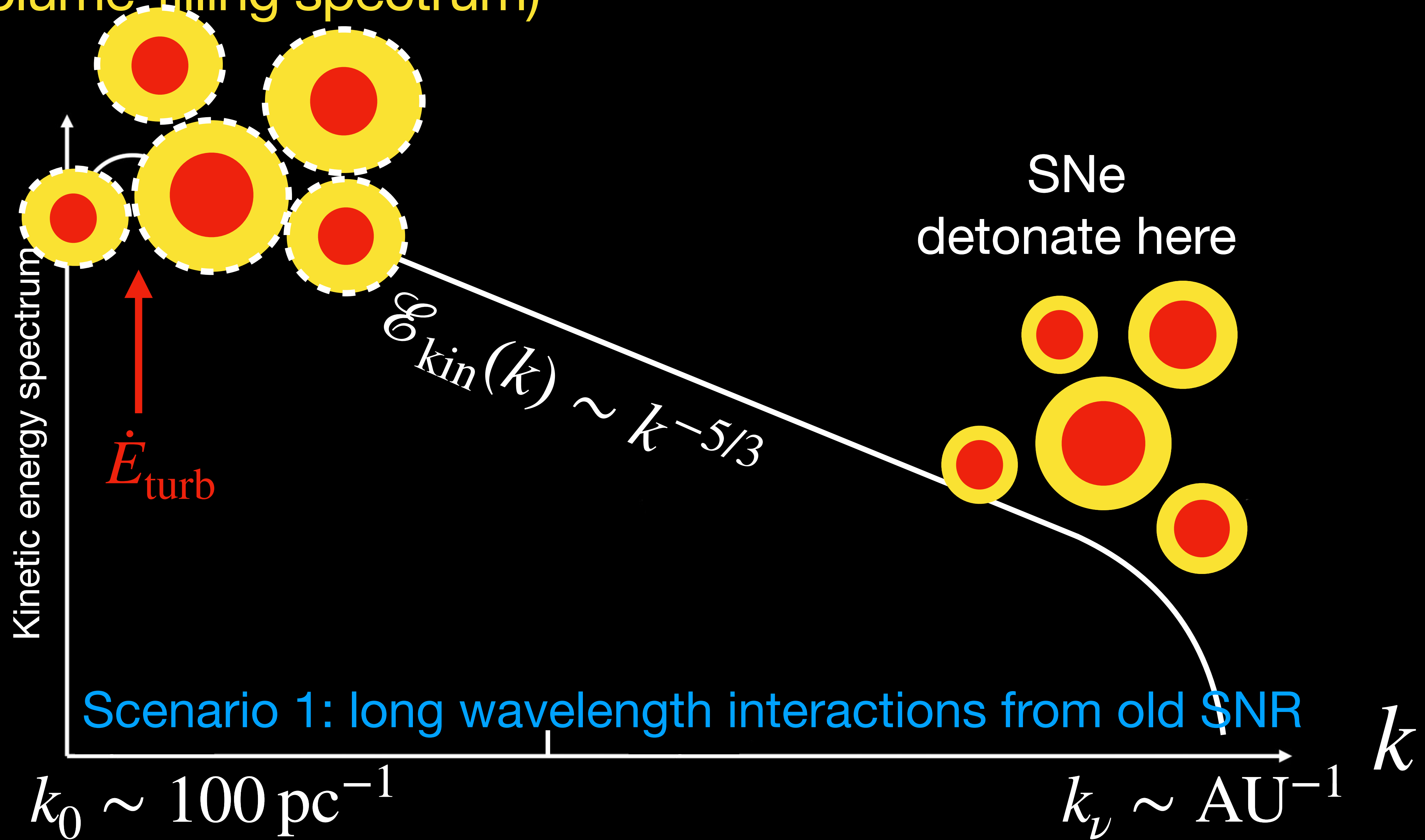
the Reynolds number sets the size of the cascade

$$k \sim \ell^{-1}$$

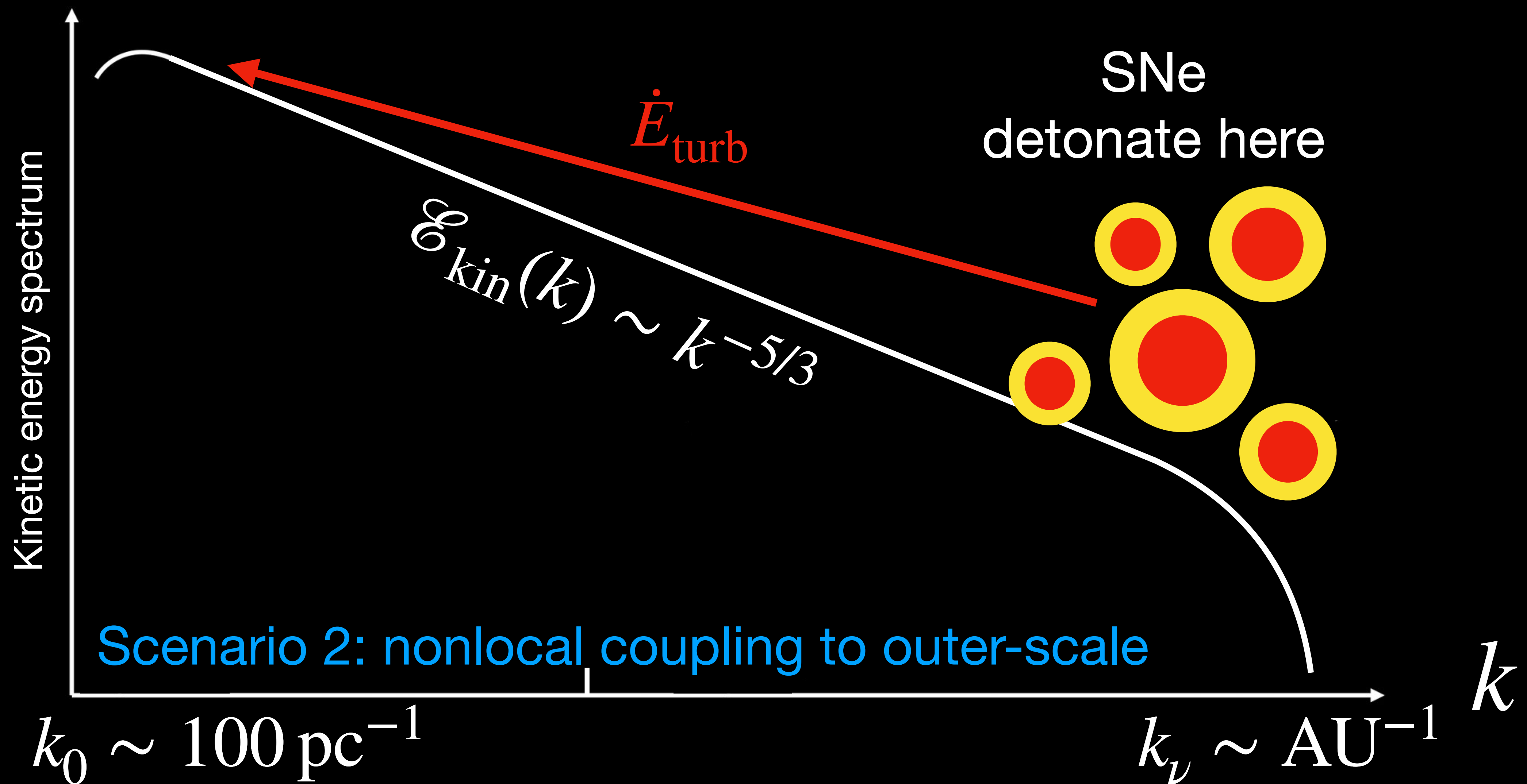
The naive ($\mathcal{M} \approx 2$) warm ionized medium spectrum
(the volume-filling spectrum)



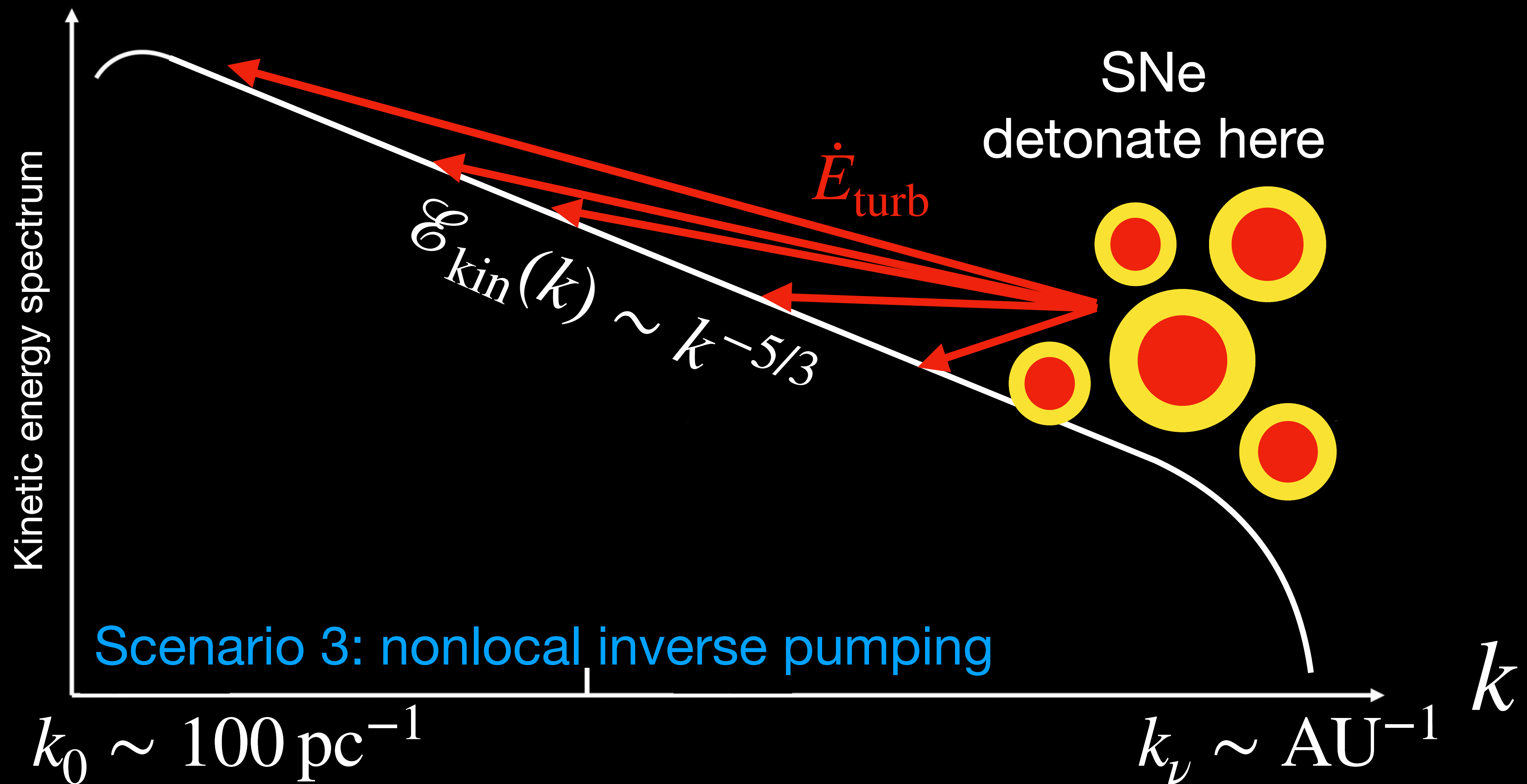
The naive ($\mathcal{M} \approx 2$) warm ionized medium spectrum
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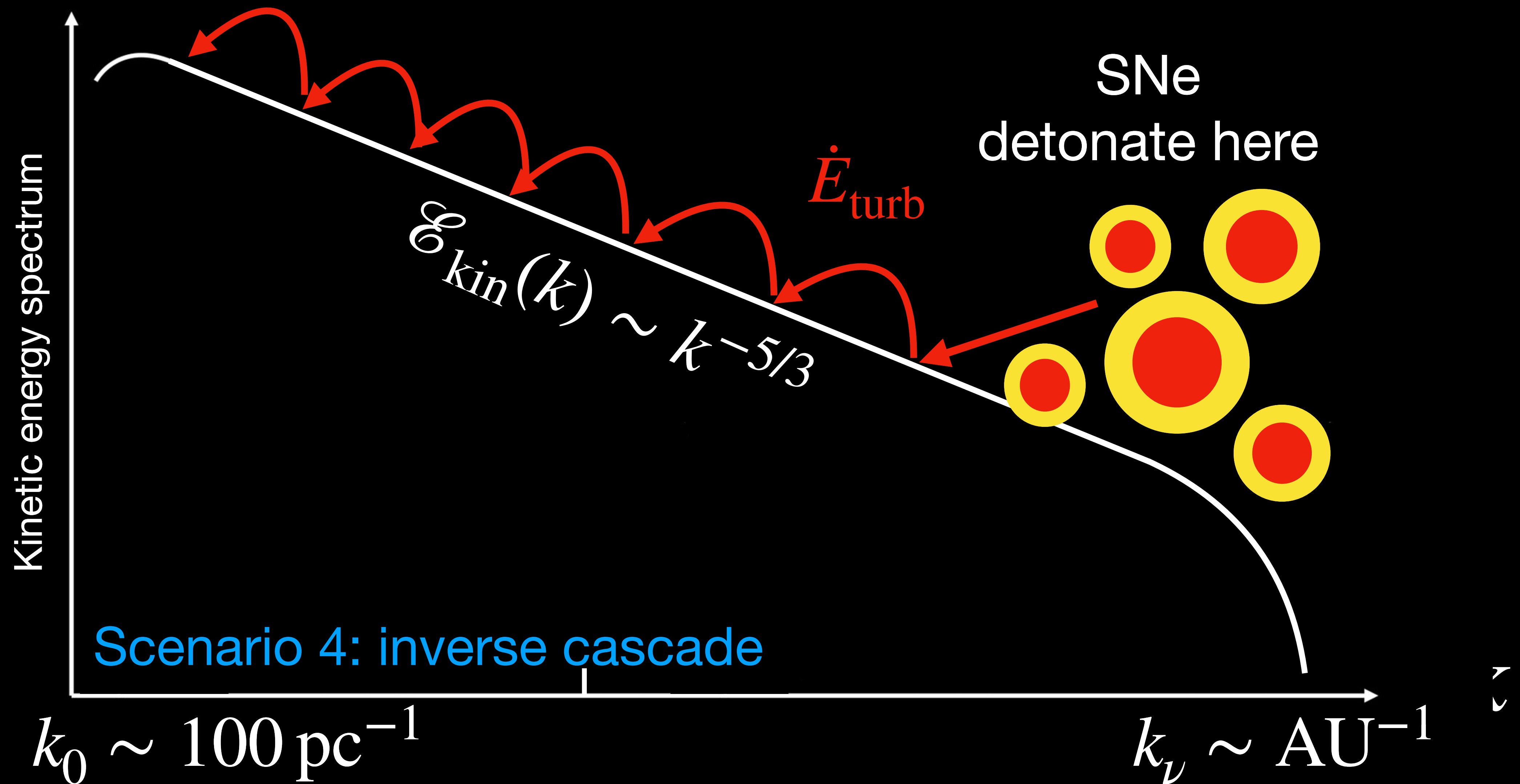
The naive ($\mathcal{M} \approx 2$) warm ionized medium spectrum (the volume-filling spectrum)



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The naive ($\mathcal{M} \approx 2$) warm ionized medium spectrum
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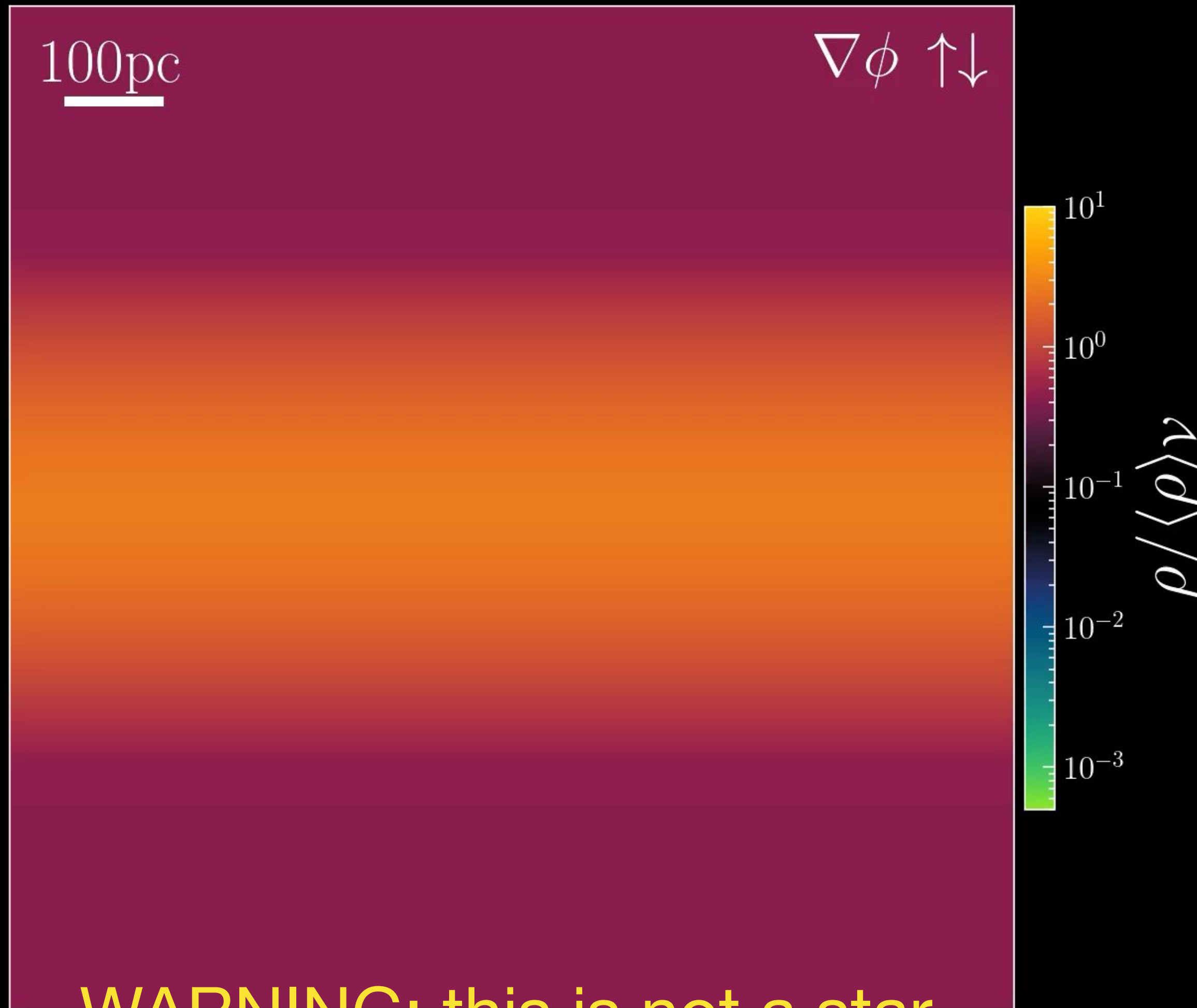
The simplest SNe-driven simulations possible

$L = 1 \text{ kpc}$

Martizzi+2016

RAMSES (Teyssier 2002)

Supernova driven
gravito-hydro dynamical model



WARNING: this is not a star
formation simulation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = \dot{n}_{\text{SNe}} M_{\text{ej}}, \quad (1)$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u} + P \mathbb{I}) = -\rho \nabla \phi + \dot{n}_{\text{SNe}} \mathbf{p}_{\text{SNe}}(Z, n_{\text{H}}), \quad (2)$$

$$\frac{\partial \rho e}{\partial t} + \nabla \cdot [\rho (e + P) \mathbf{u}] = -n_{\text{H}}^2 \Lambda - \quad (3)$$

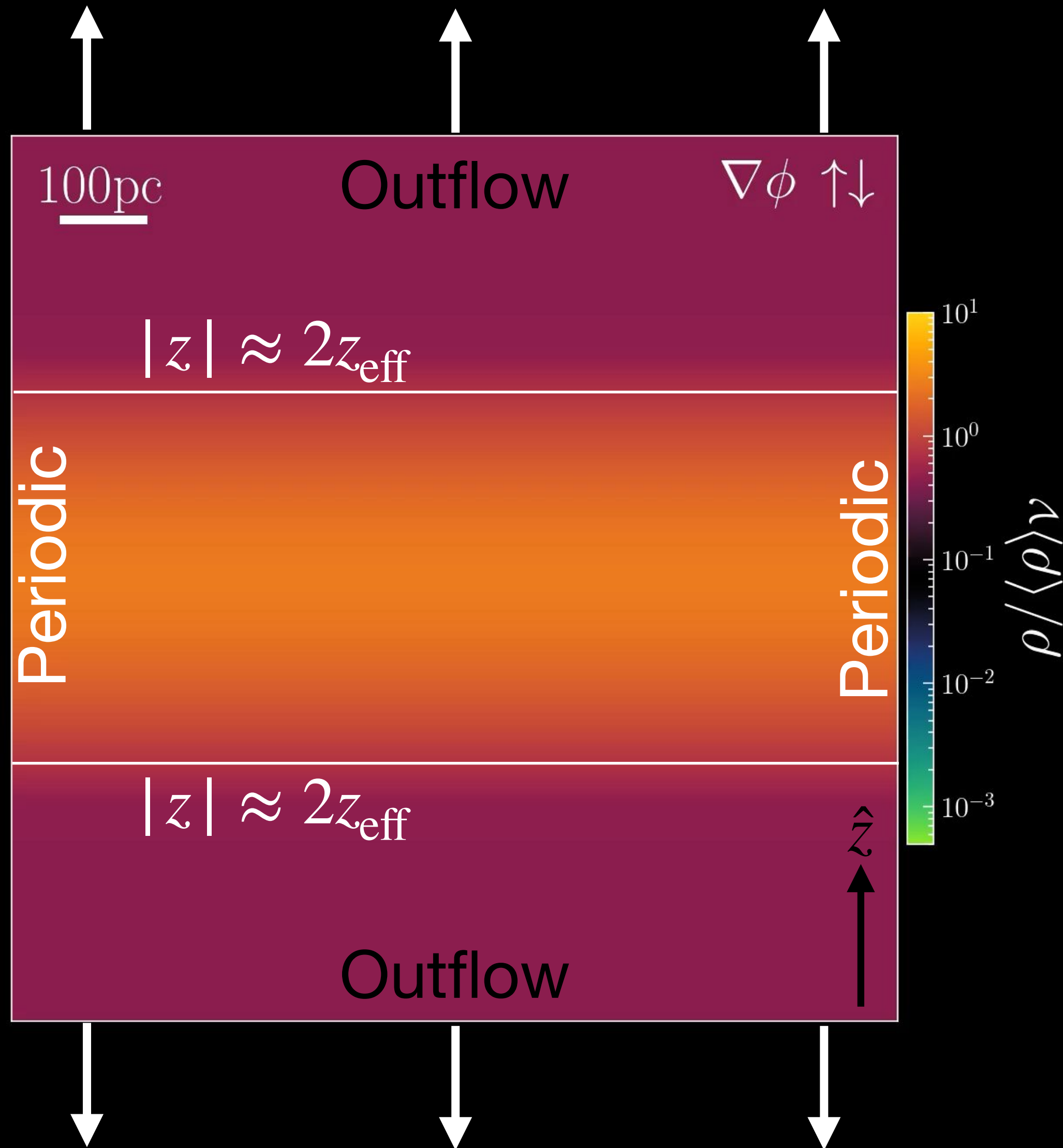
$$\rho \mathbf{u} \cdot \nabla \phi + \dot{n}_{\text{SNe}} \left[E_{\text{th,SNe}}(Z, n_{\text{H}}) + \frac{\mathbf{p}_{\text{SNe}}(Z, n_{\text{H}})^2}{2(M_{\text{ej}} + M_{\text{swept}})} \right], \quad (4)$$

$$e = \epsilon + \frac{u^2}{2}, \quad P = \frac{2}{3} \rho e, \quad (5)$$

$$N_{\text{grid}}^3 = 1024^3 \implies \Delta x \sim 1 \text{ pc}$$

$$\text{numerical diss.} \implies \Delta x \sim 10 \text{ pc}$$

The simplest SNe-driven simulations possible



Static gravitational potential

$$\phi(z) = \underbrace{2\pi G \Sigma_* \left(\sqrt{z^2 - z_0^2} - z_0 \right)}_{\text{stratified disk}} + \underbrace{\frac{2\pi G \rho_{\text{halo}}}{3} z^2}_{\text{spherical halo}}$$

Supernova driving prescription

(following 1D evolution models for momentum and energy deposition in Martizzi + 2016)

$$\dot{n}_{\text{SNe}} = \frac{\dot{\Sigma}_*}{2z_{\text{eff}} 100 M_{\odot}} \quad \text{1 SN per } 100 M_{\odot} \text{ of SF}$$

$$\dot{\Sigma}_* \propto \Sigma^{1.4}$$

KS relation

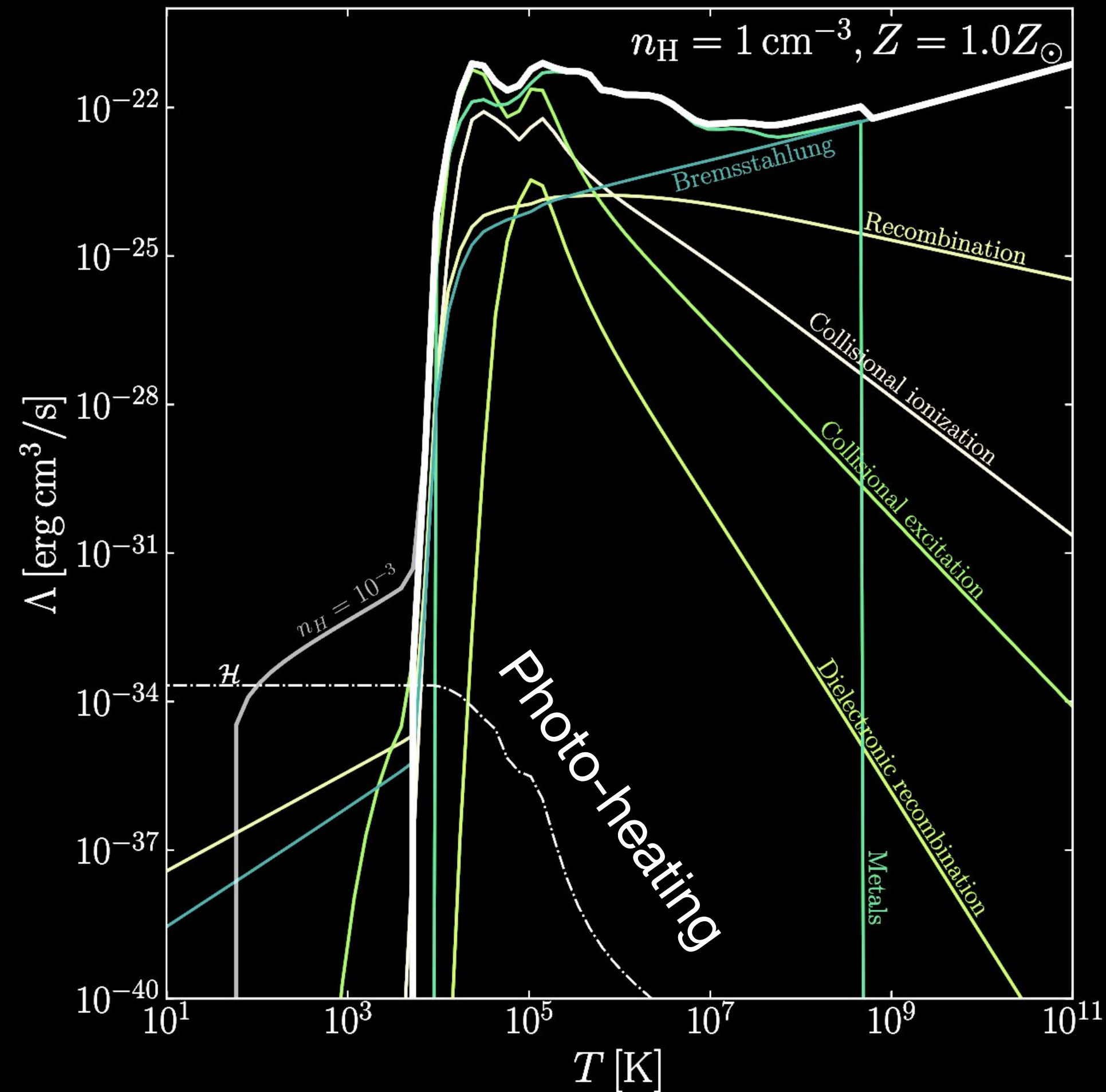
$$p(z) = \begin{cases} \frac{1}{4z_{\text{eff}}}, & |z| \leq 2z_{\text{eff}} \\ 0, & |z| > 2z_{\text{eff}} \end{cases}$$

$$p(n | \lambda) = \lambda^n e^{-\lambda} / n!$$

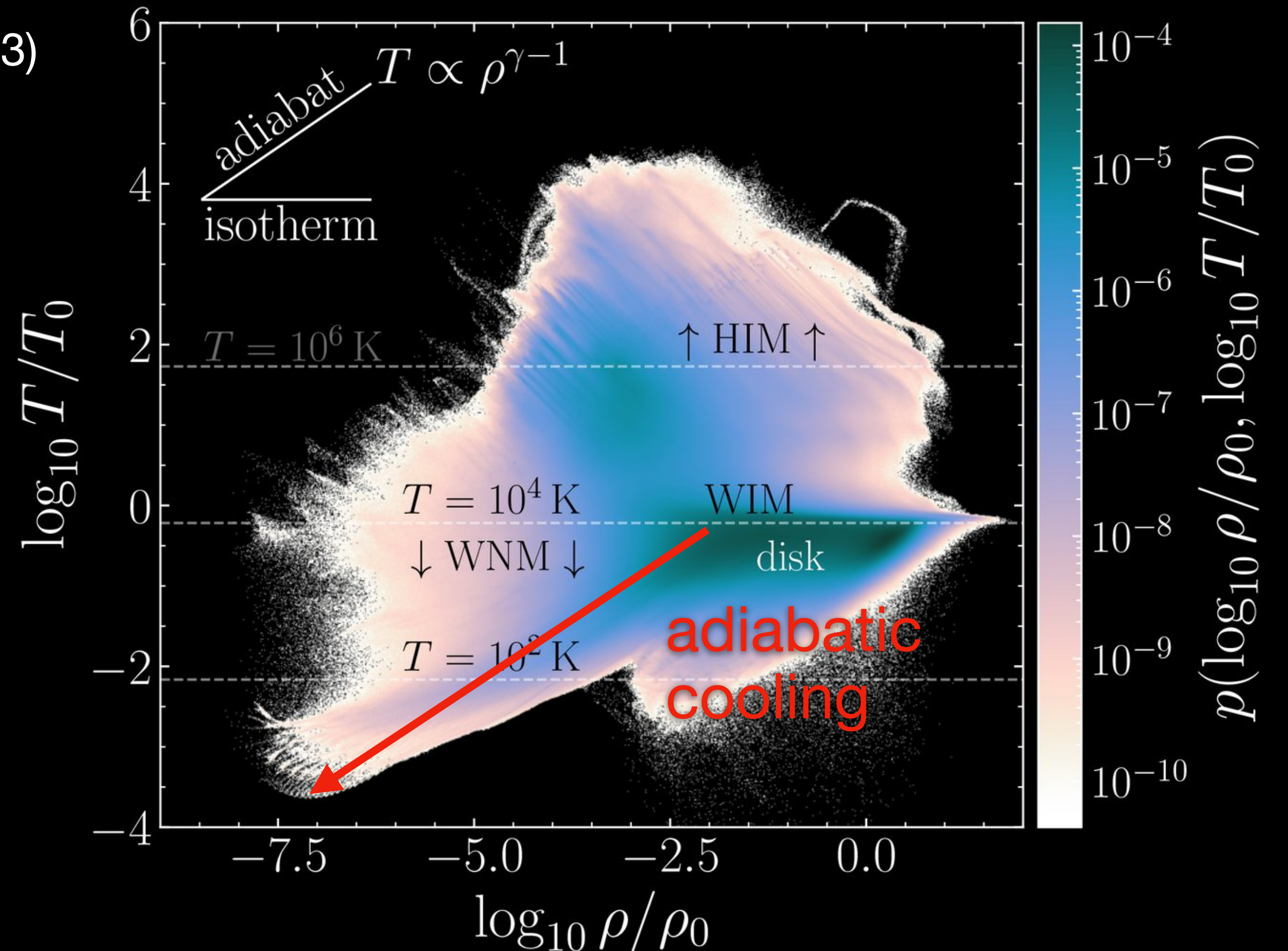
$$\lambda(x, y, z) = \dot{n}_{\text{SNe}} \mathcal{V}_{\text{cell}} \Delta t$$

The simplest SNe-driven simulations possible

Cooling function Theuns+(1998)
Sutherland & Dopita(1993)



HI, HII, HeI, HeII, HeIII and free electrons



$$\rho_0 = 2.1 \times 10^{-24} \text{ g cm}^{-3}$$

$$P_0 = 2.2 \times 10^{-12} \text{ Ba}$$

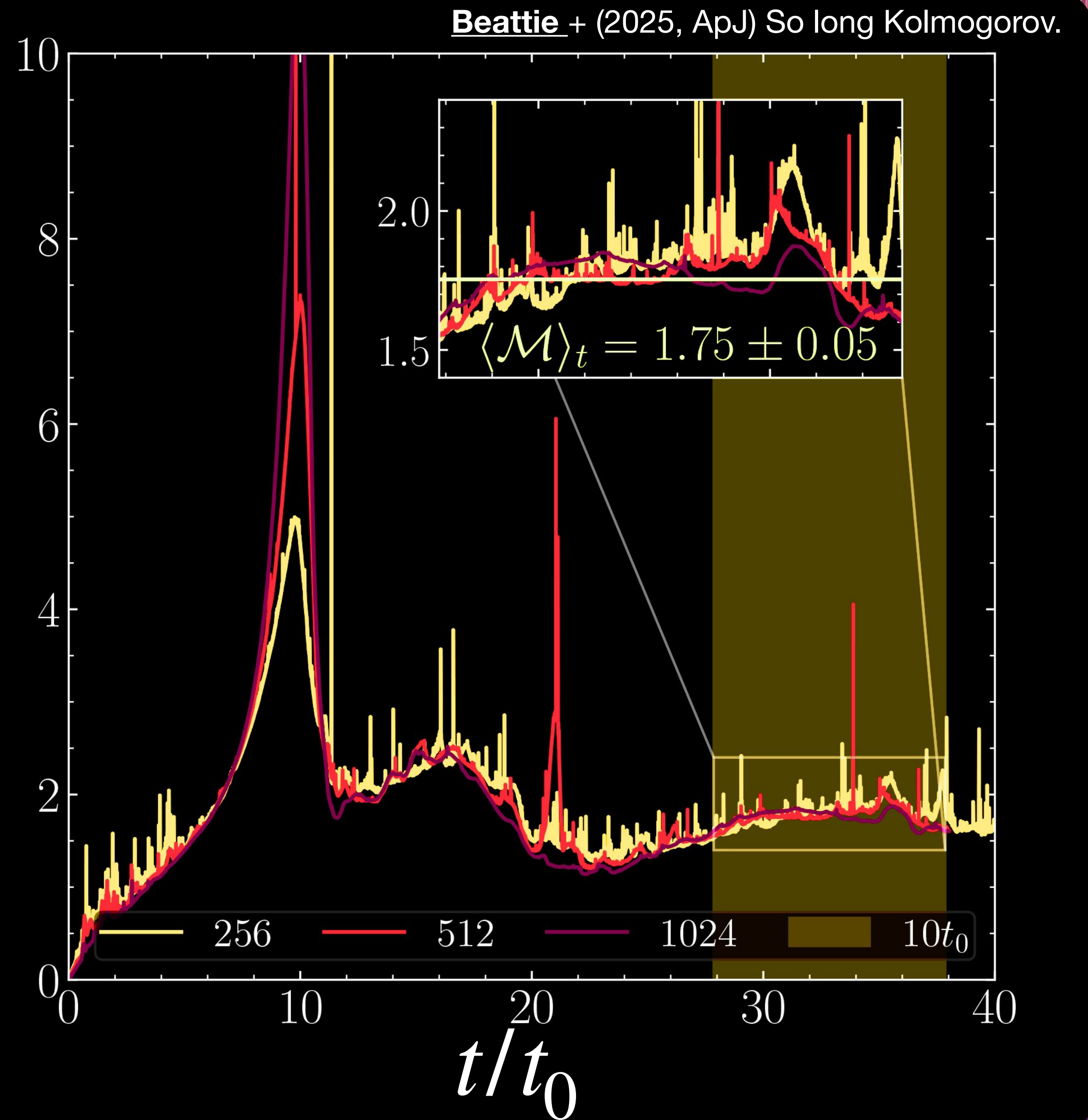
Stationarity and Mach number

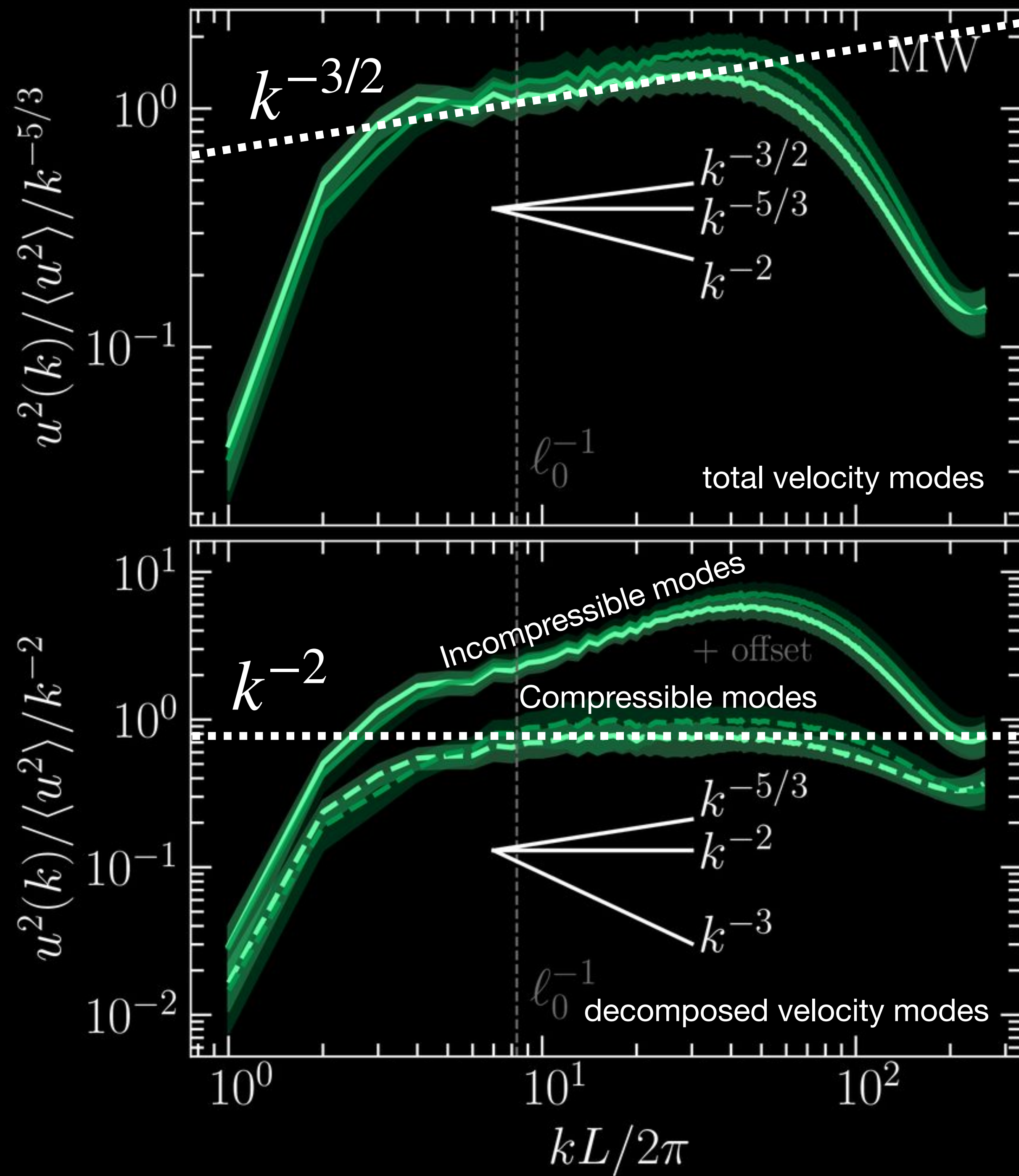
$$\mathcal{M} = \left\langle \left(\frac{u}{c_s} \right)^2 \right\rangle^{1/2}$$

$\mathcal{M}$

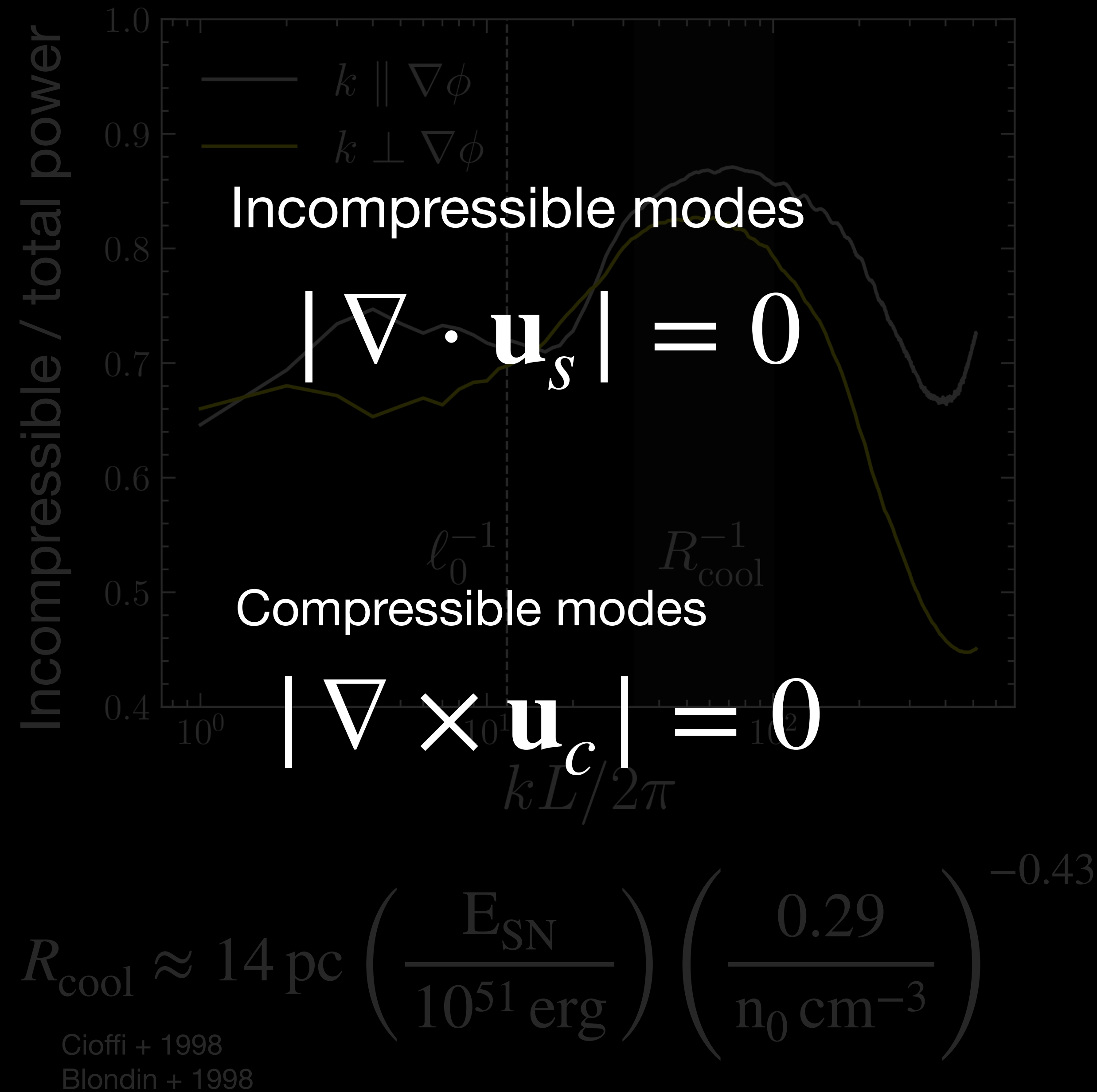
$$t_0 \sim \frac{\ell_0}{\langle u^2 \rangle^{1/2}} \sim 80 \text{ Myr}$$

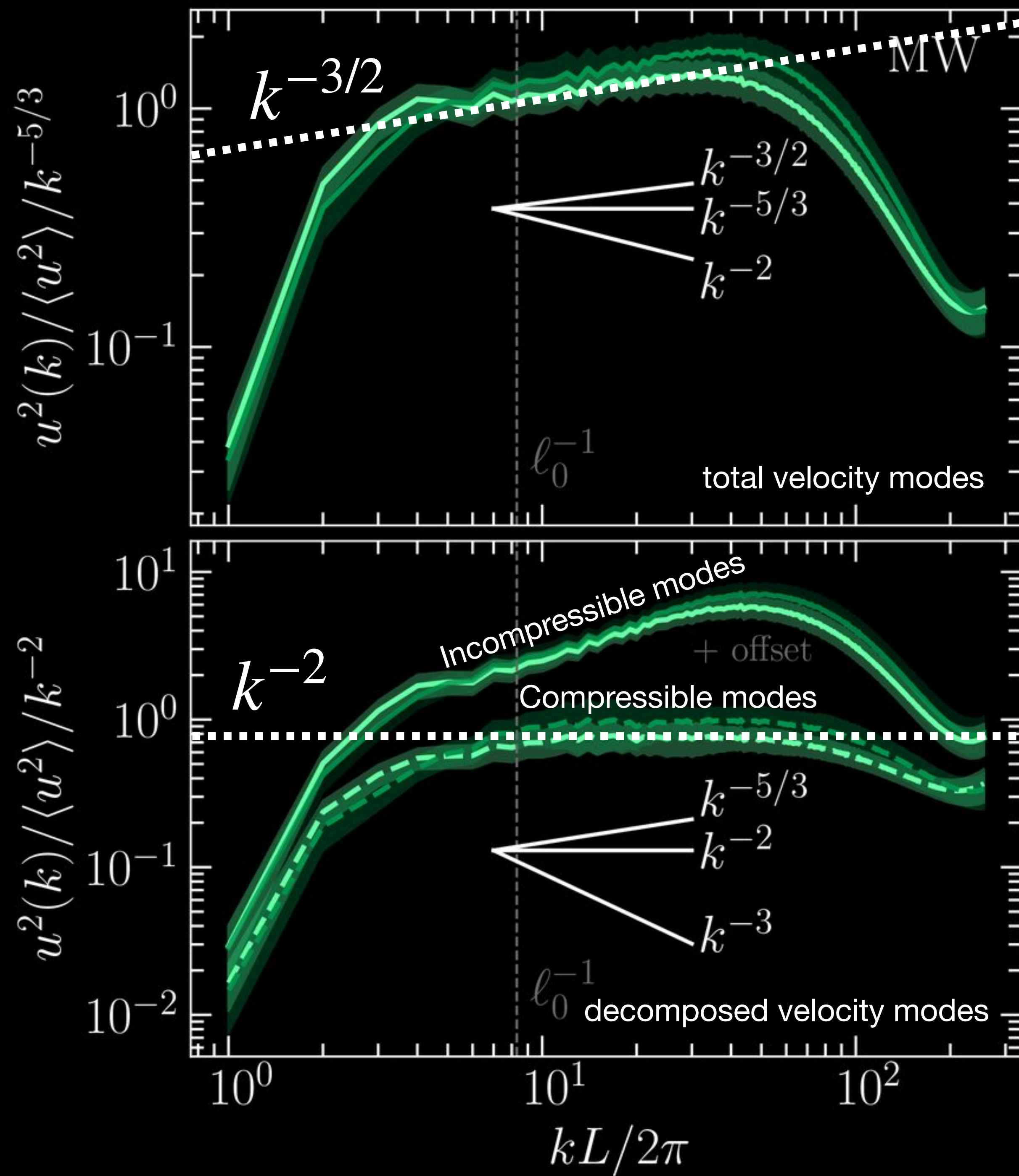
reaches a quasi stationary state





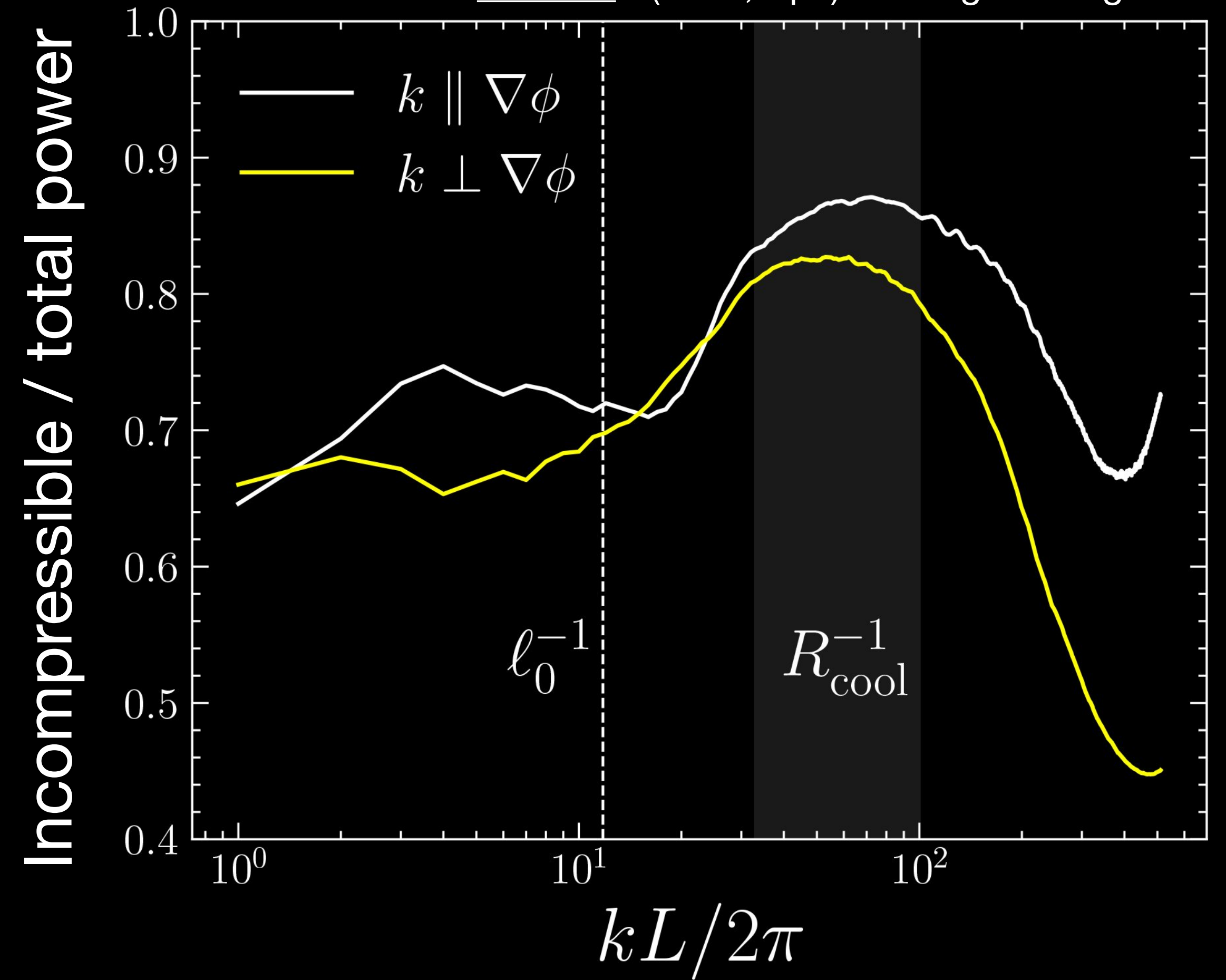
Basic spectral properties





Basic spectral properties

Beattie + (2025, ApJ) So long Kolmogorov.



$$R_{cool} \approx 14 \text{ pc} \left(\frac{E_{SN}}{10^{51} \text{ erg}} \right) \left(\frac{0.29}{n_0 \text{ cm}^{-3}} \right)^{-0.43}$$

Cioffi + 1998
Blondin + 1998

Consider the adiabatic limit

$$t_{\text{turb}} \ll t_{\text{cool}}$$

via linearized continuity equation

$$\delta \tilde{c}_s \sim c_{s,0} \frac{\gamma - 1}{2\omega} \mathbf{k} \cdot \tilde{\mathbf{u}}_c$$

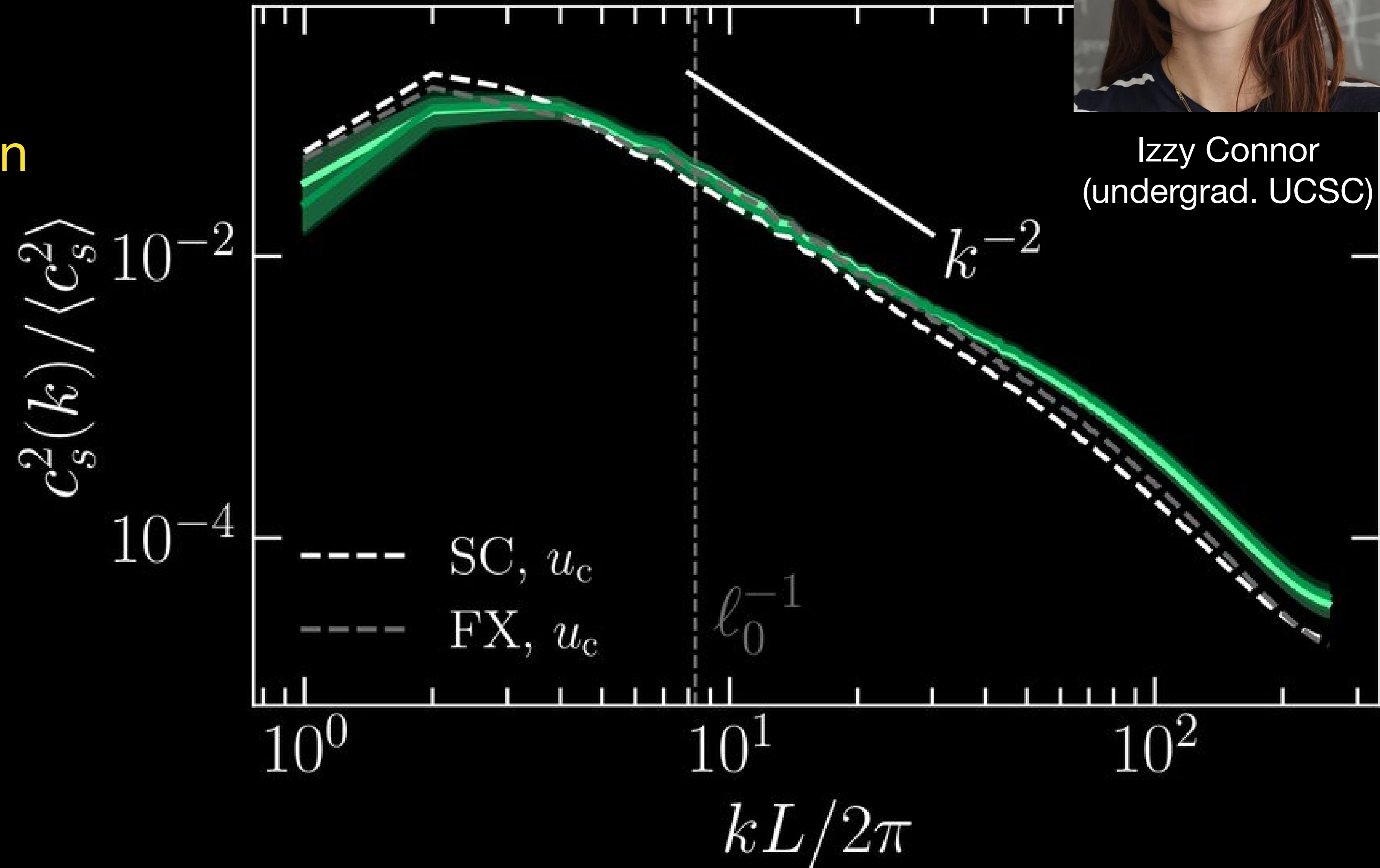
$$\omega \sim c_{s,0} k$$

$$\delta \tilde{c}_s \sim \frac{\gamma - 1}{2} \tilde{u}_c$$

Phase fluctuations and thermal response is mostly adiabatic



Izzy Connor
(undergrad. UCSC)

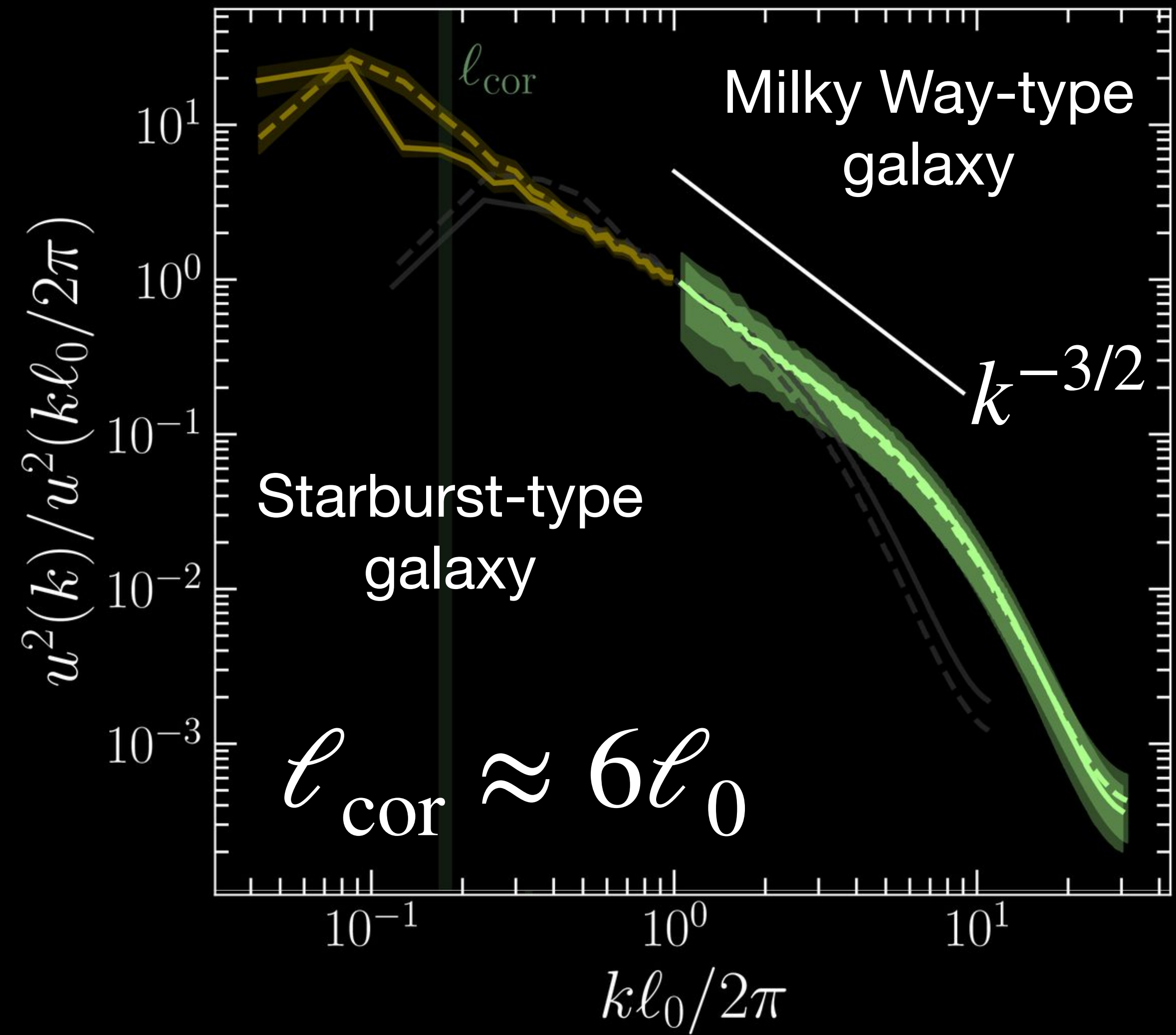


thermal response / phase fluctuations controlled by compressible modes

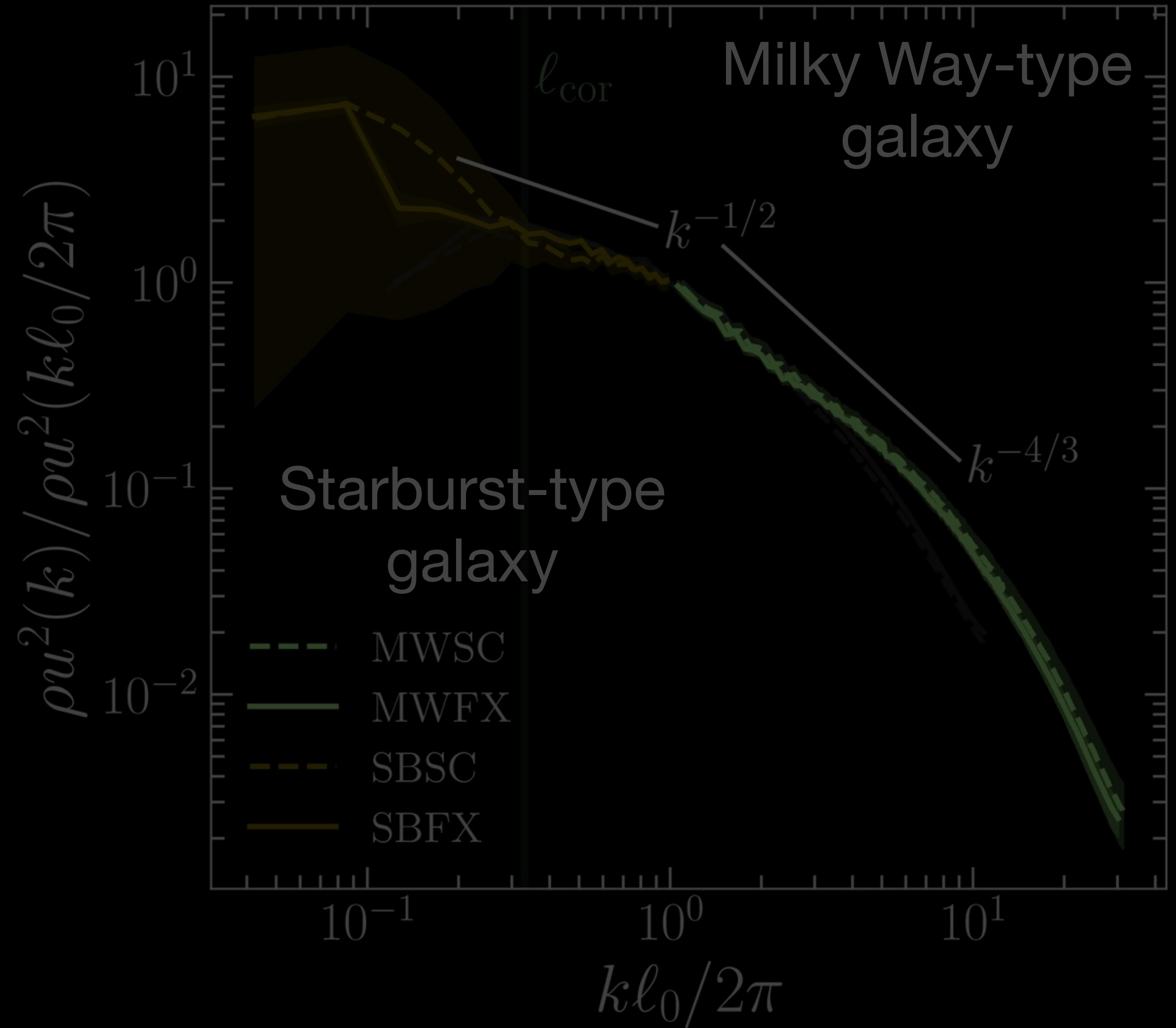
Universality between galactic models

$$\gamma_{\text{SN}_e,\text{SB}} = 10 \gamma_{\text{SN}_e,\text{MW}}$$

Velocity spectrum



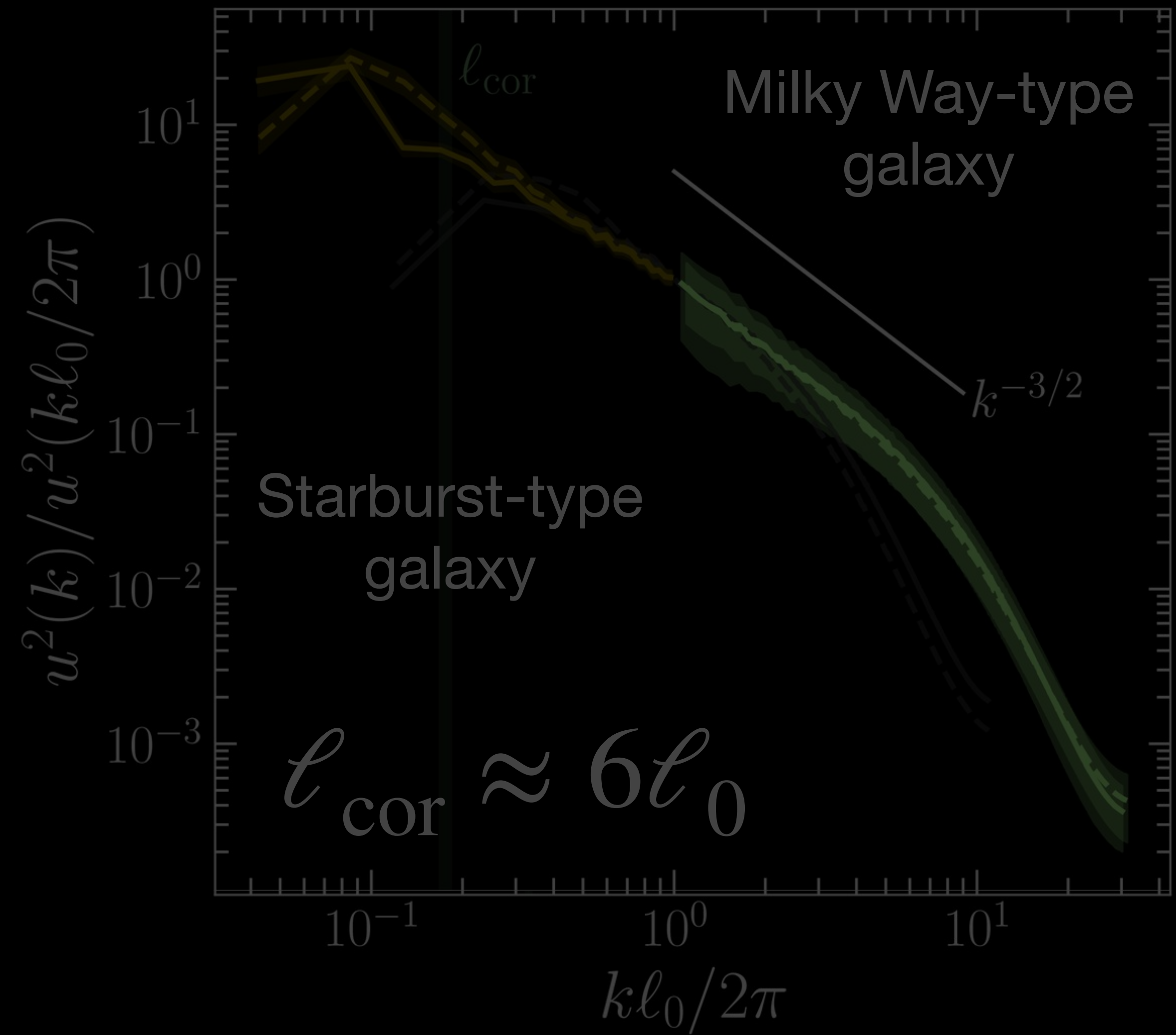
Kinetic energy spectrum



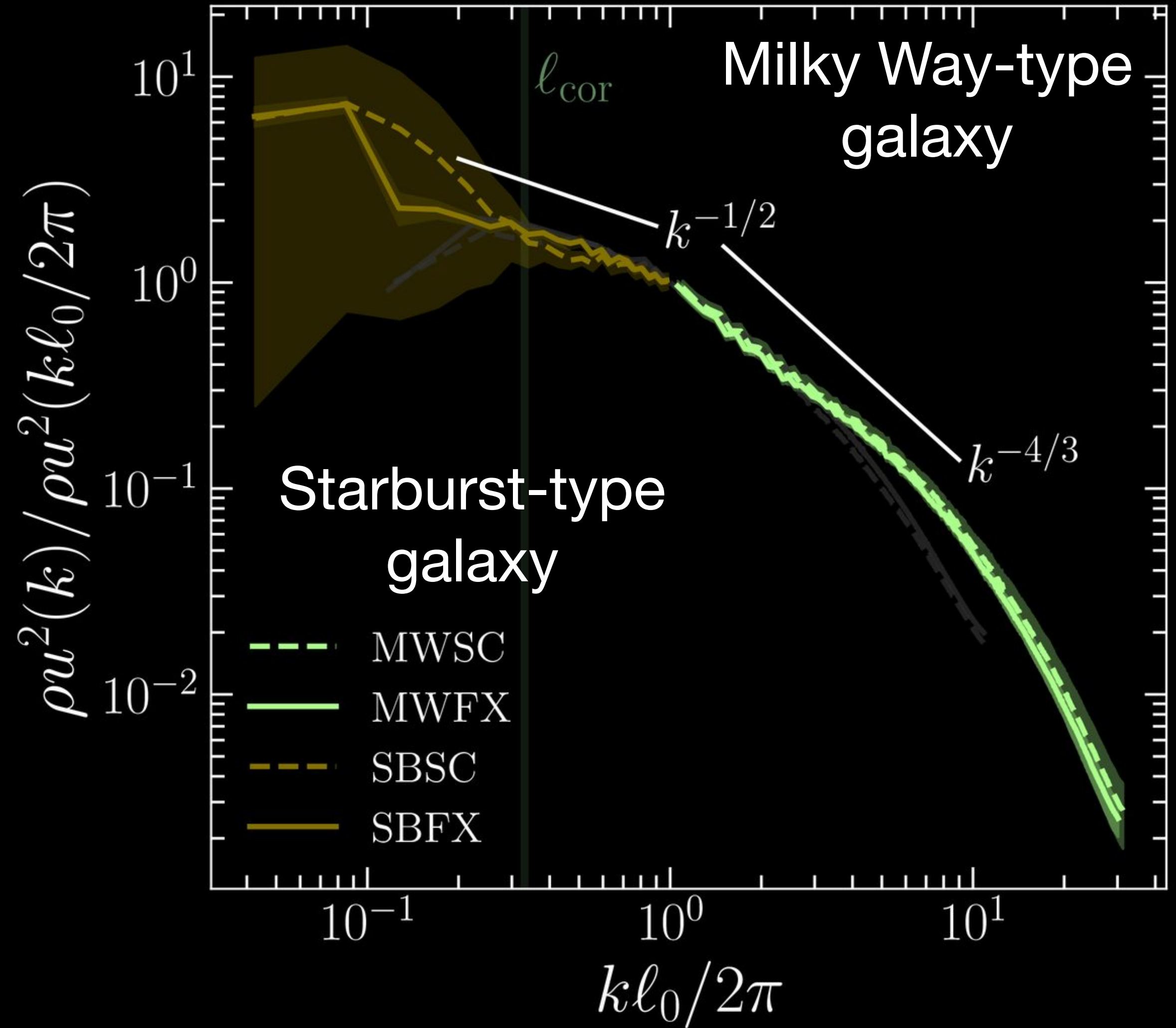
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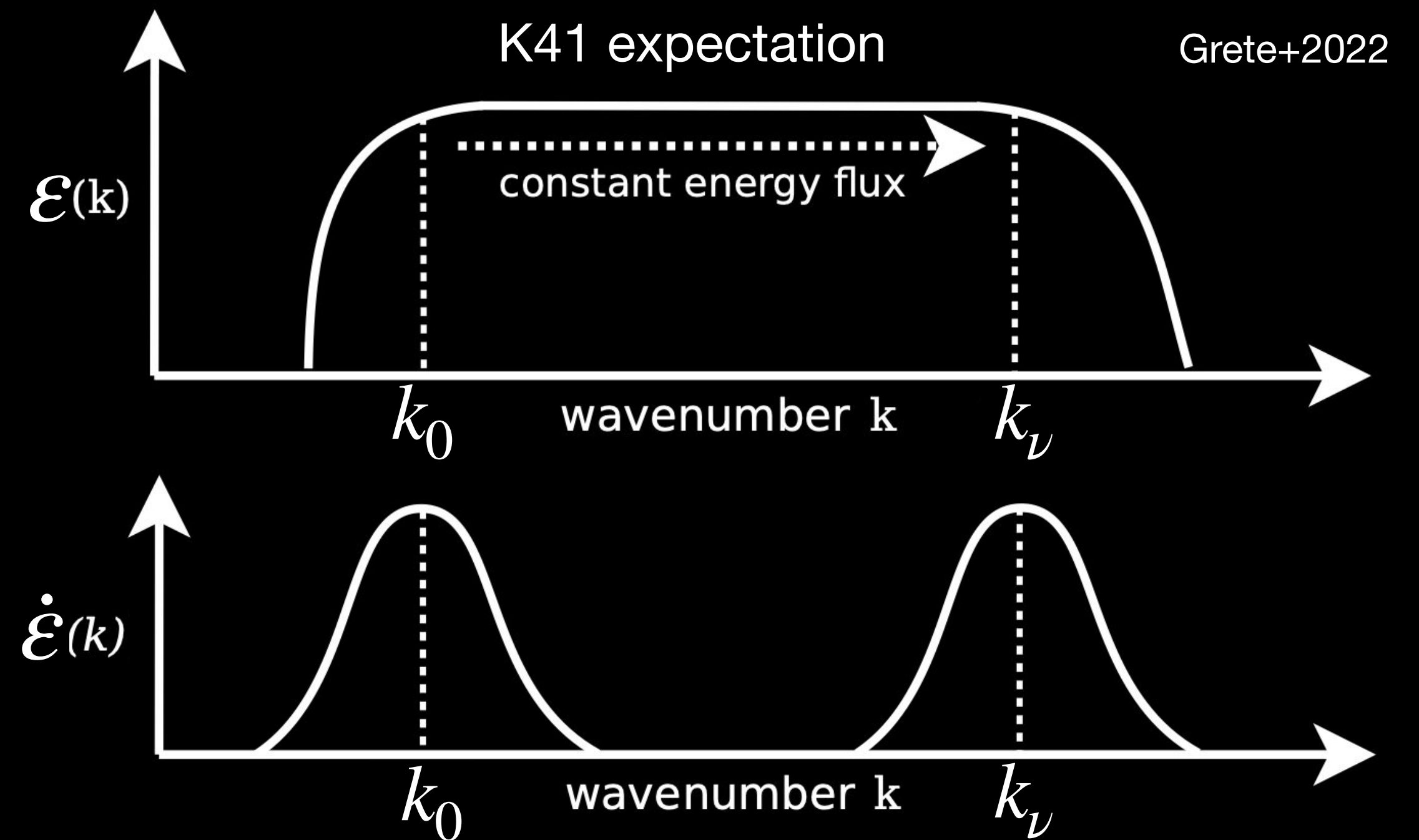
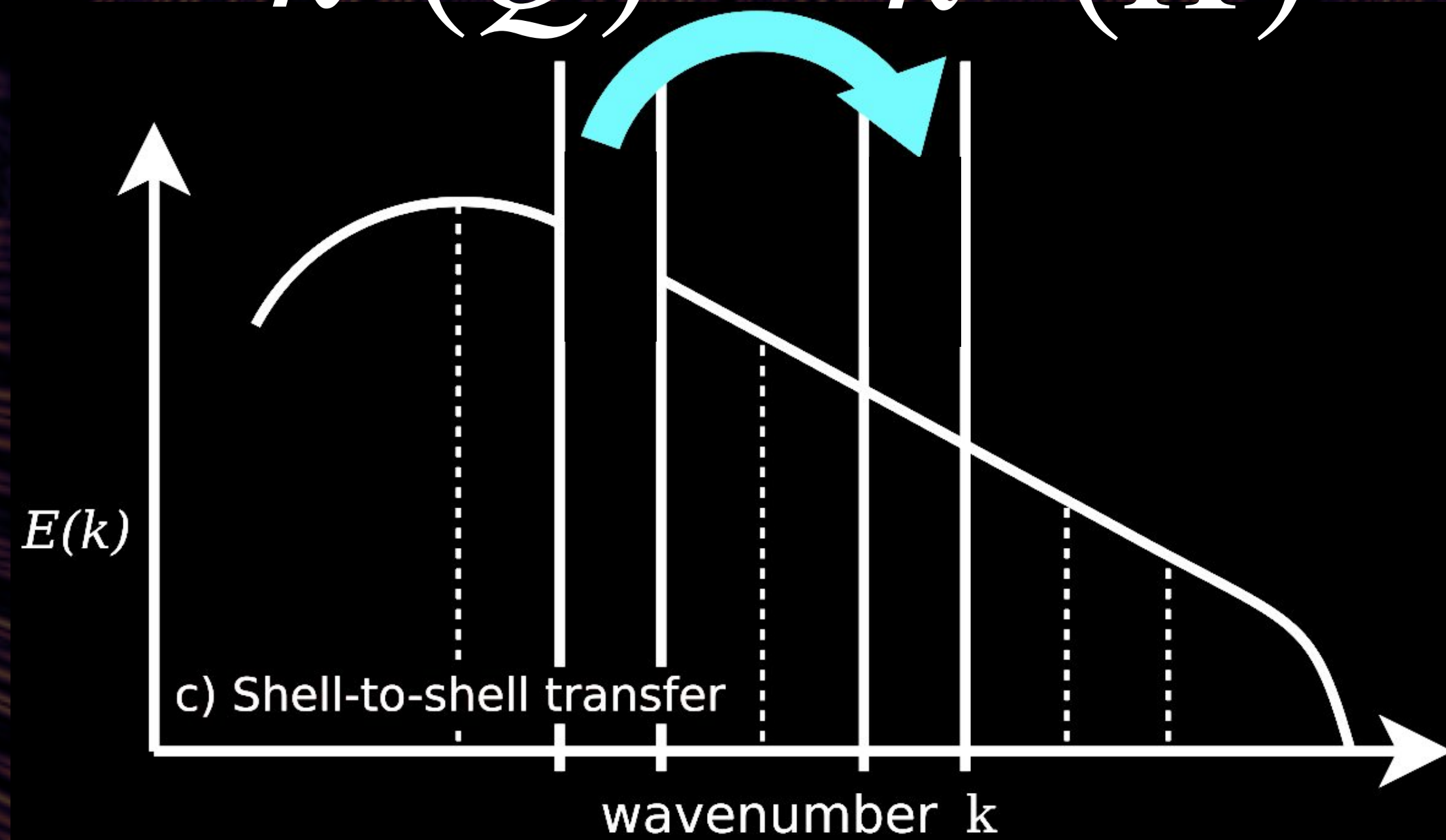


Kinetic energy spectrum



But to understand the turbulence we must understand the flux rates from mode to mode!

$$k''(Q) \quad \varepsilon \quad k'''(K)$$



Grete, O'Shea +2022a,b,23

Dar+2001; Verma+2004; Mininni+2005; Alexakis+2005

But to understand the turbulence we must understand the flux rates from mode to mode!

Momentum conservation:

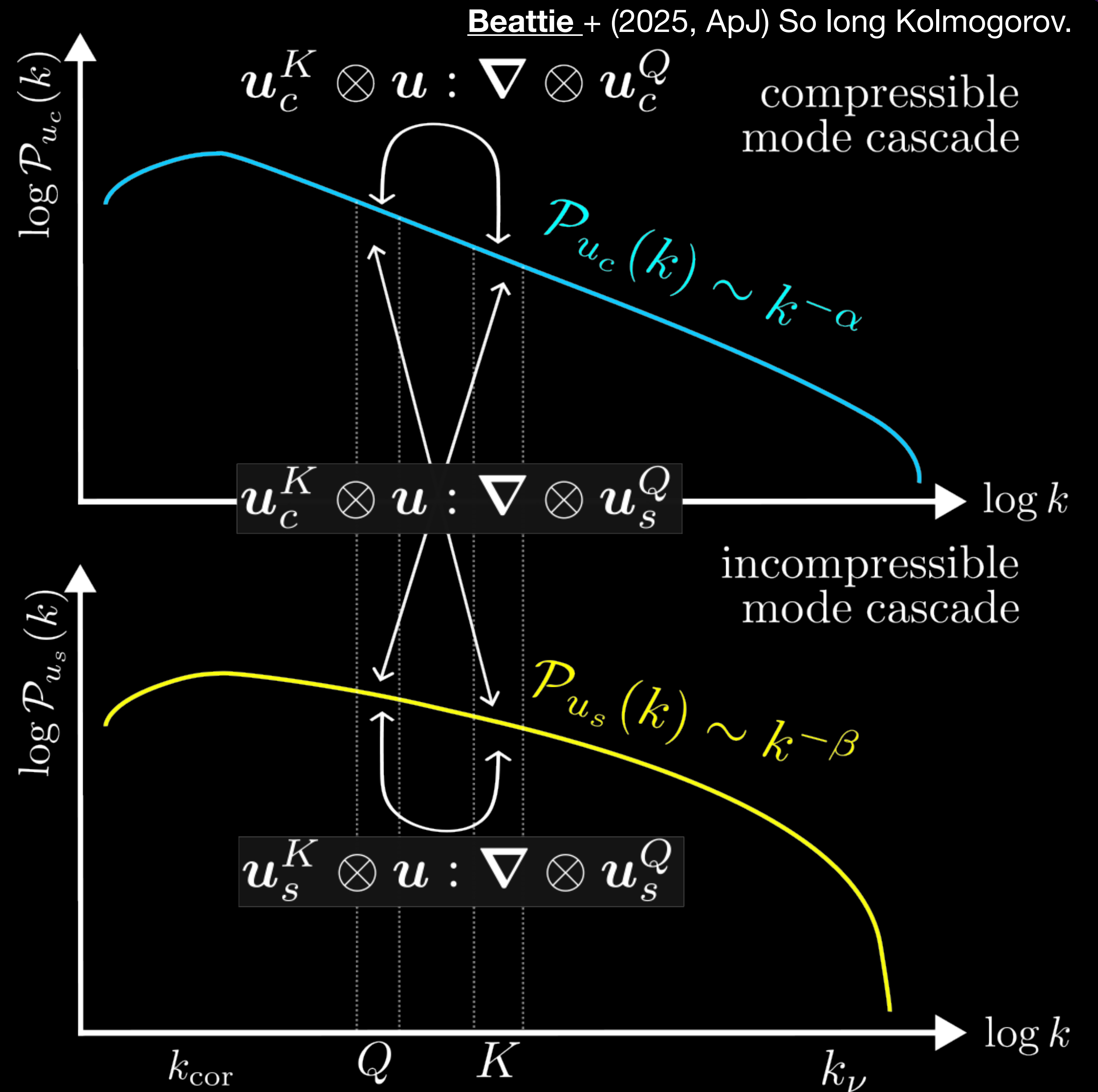
$$\overset{\text{doner}}{Q} + \overset{\text{receiver}}{P} + \underset{\text{mediator}}{K} = 0$$

$$\overset{\text{doner}}{Q} \xrightarrow{\overset{\text{receiver}}{P}} \underset{\text{mediator}}{K} = -K \xrightarrow{\overset{\text{receiver}}{P}} \underset{\text{mediator}}{Q}$$

$$\mathbf{u}^Q = \mathbf{u}(\mathbf{r}^Q) = \int \delta^3(\mathbf{k} - Q) \mathbf{u}(\mathbf{k}) \exp \{2\pi i \mathbf{k} \cdot \mathbf{r}\}$$

Compressible
modes and
solenoidal modes
live very different
lives!

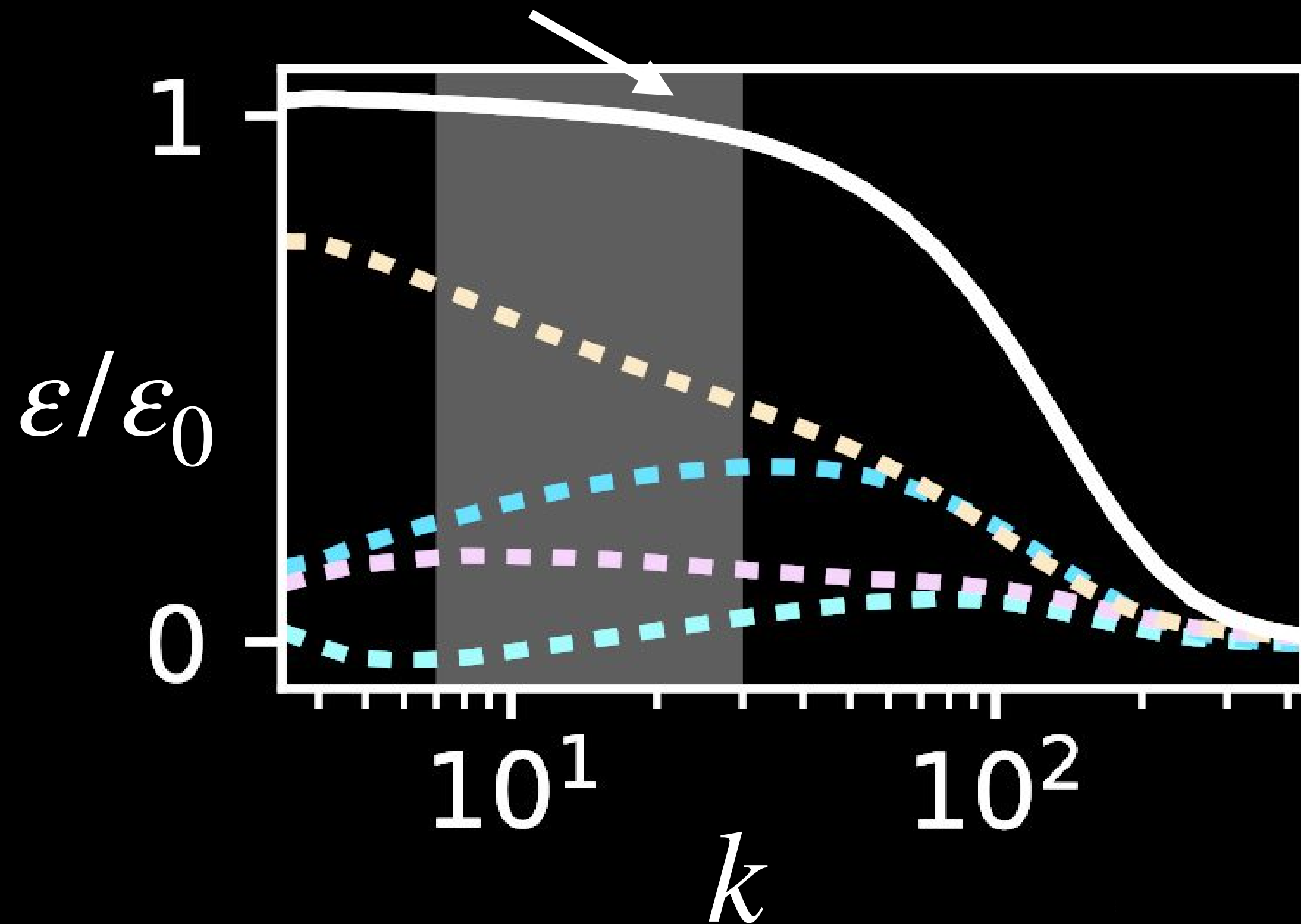
We should study
them separately!



Expectations from compressible MHD turbulence in a box

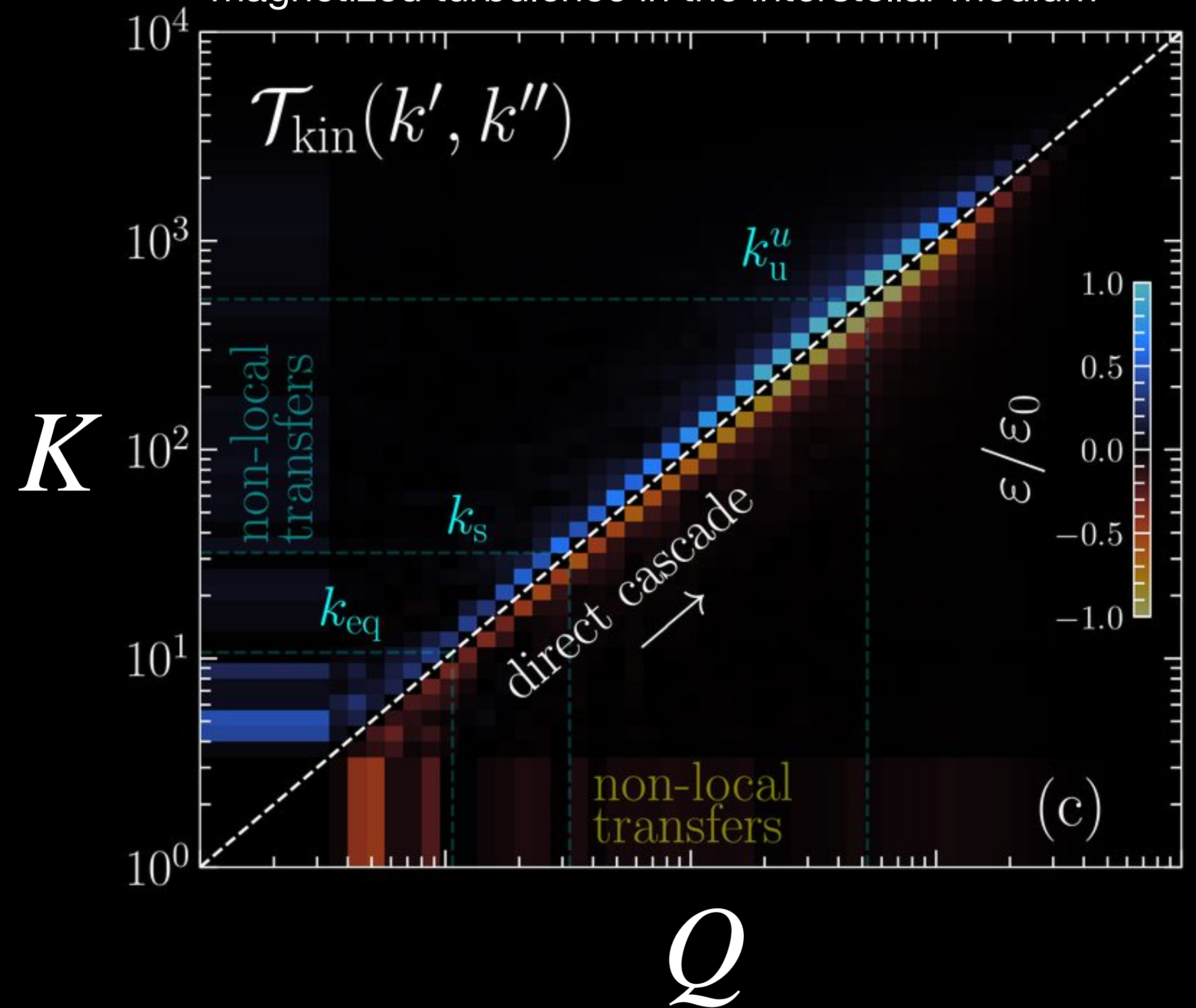
$$Q \xrightarrow{P} K$$

$$\varepsilon \sim u^3/\ell \approx \text{const.}$$

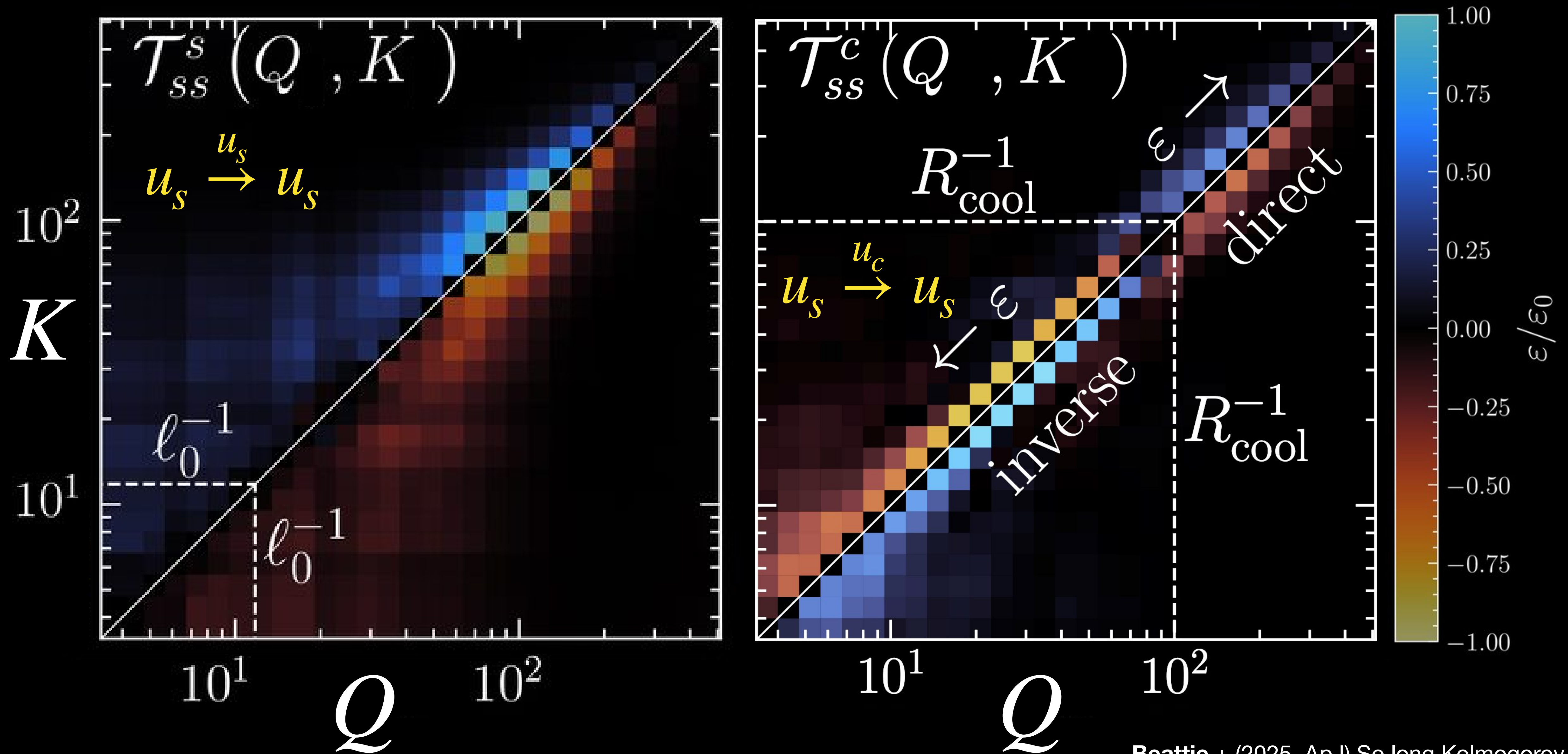


Grete, O'shea + 2022

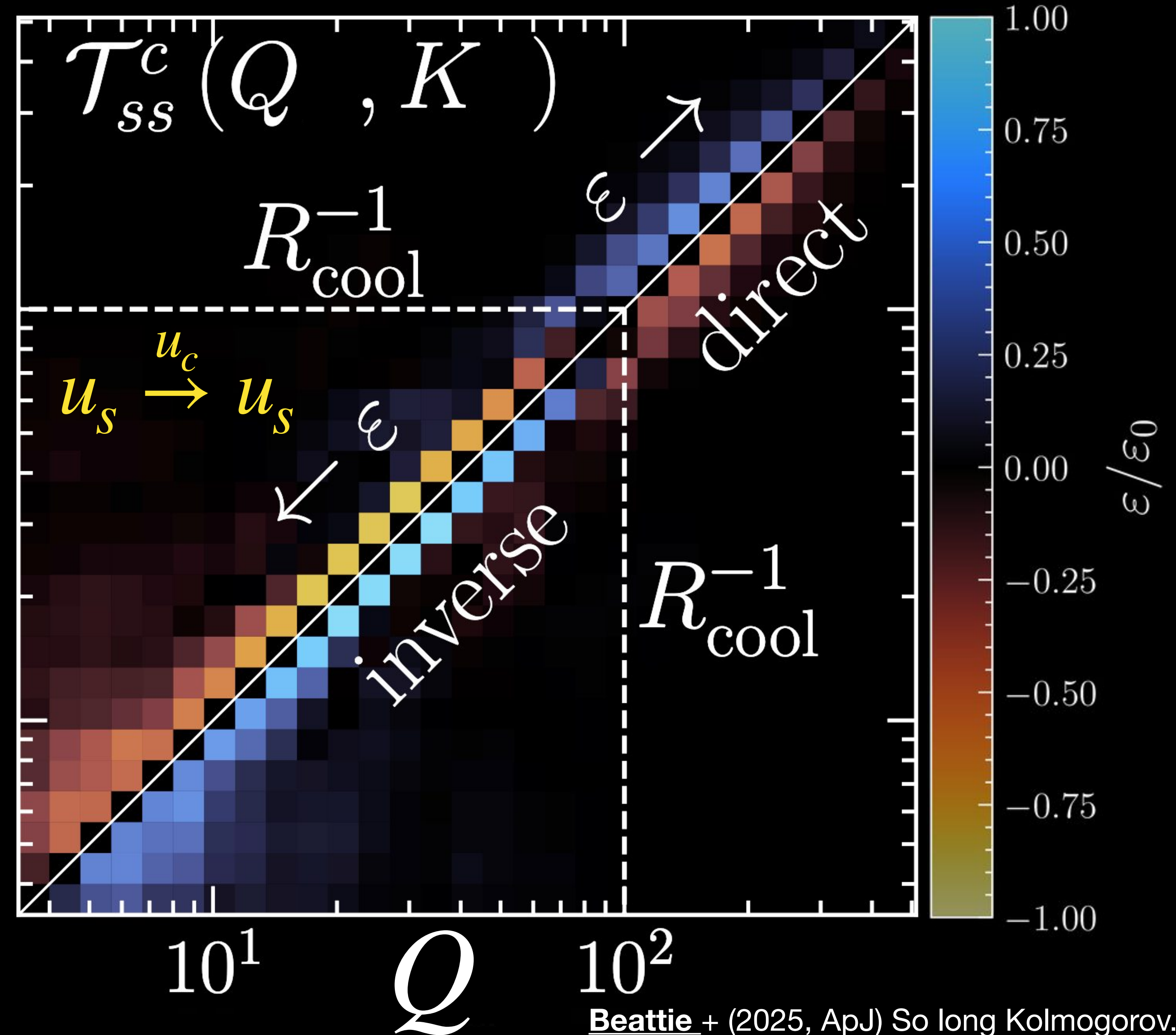
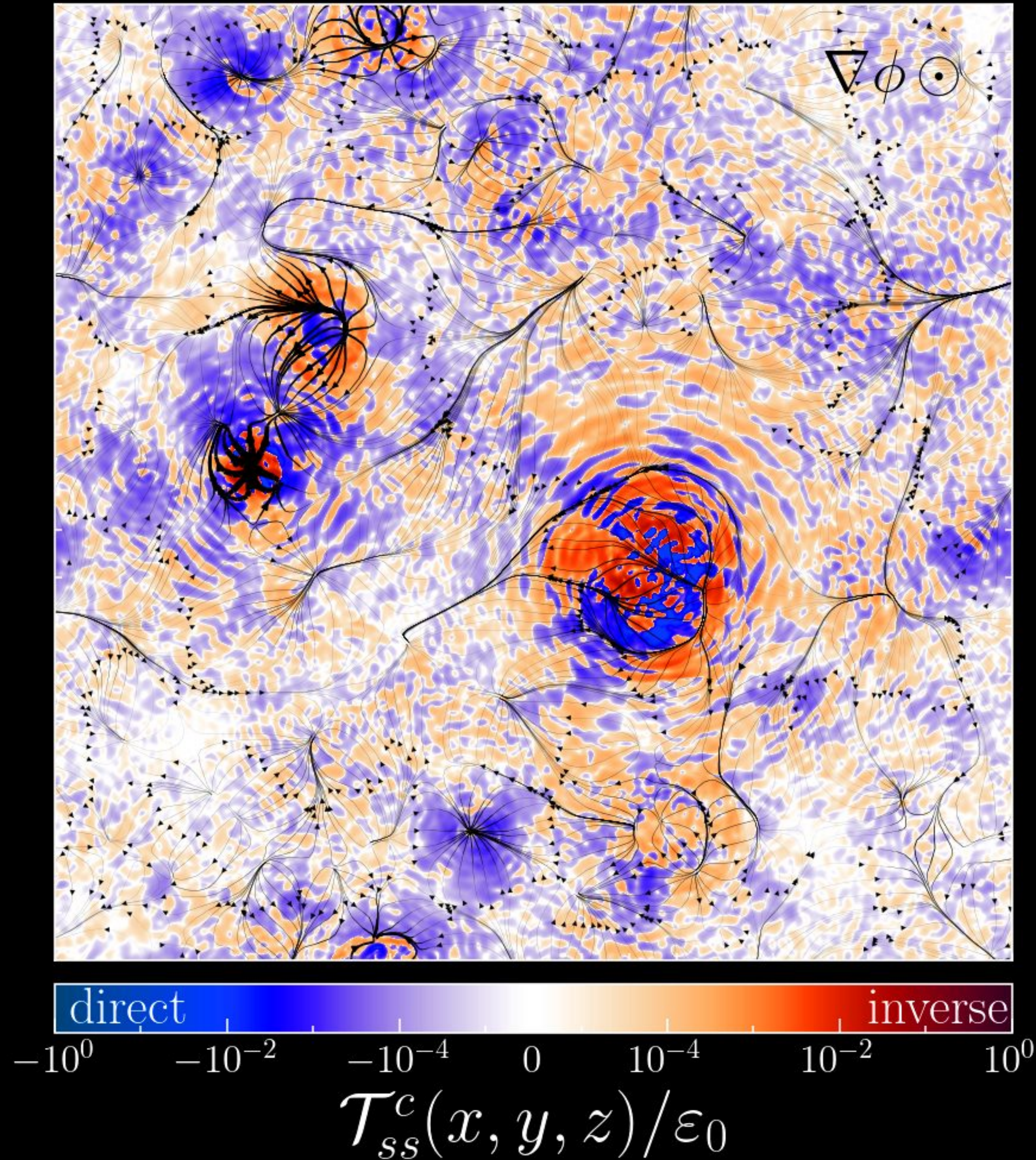
Beattie+2025 (*Nature Astronomy*). The spectrum of magnetized turbulence in the interstellar medium



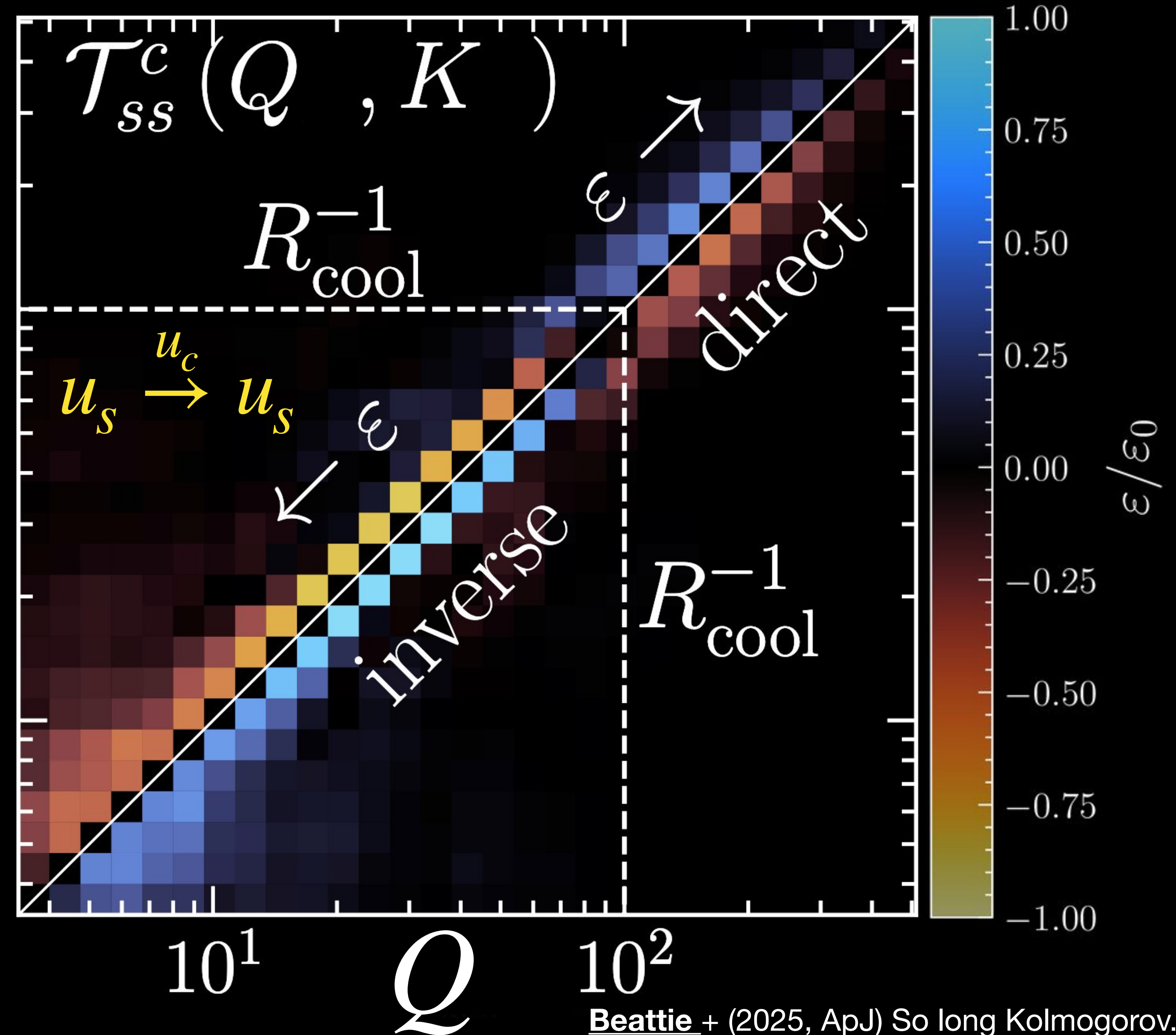
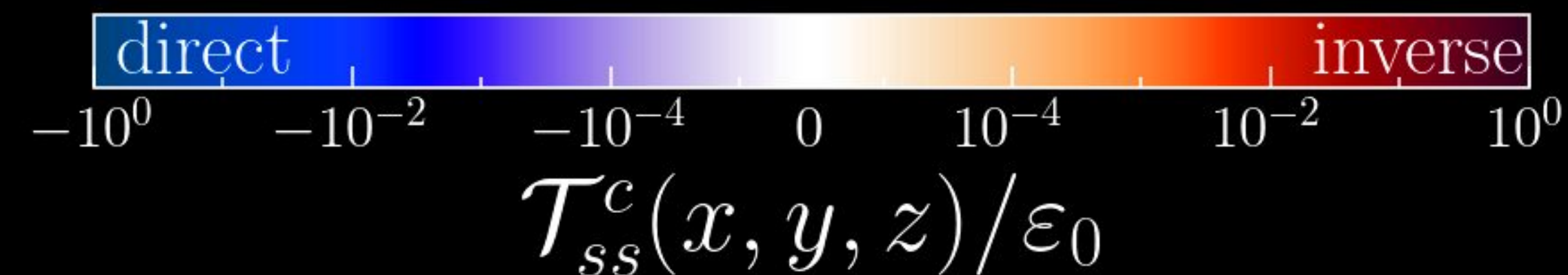
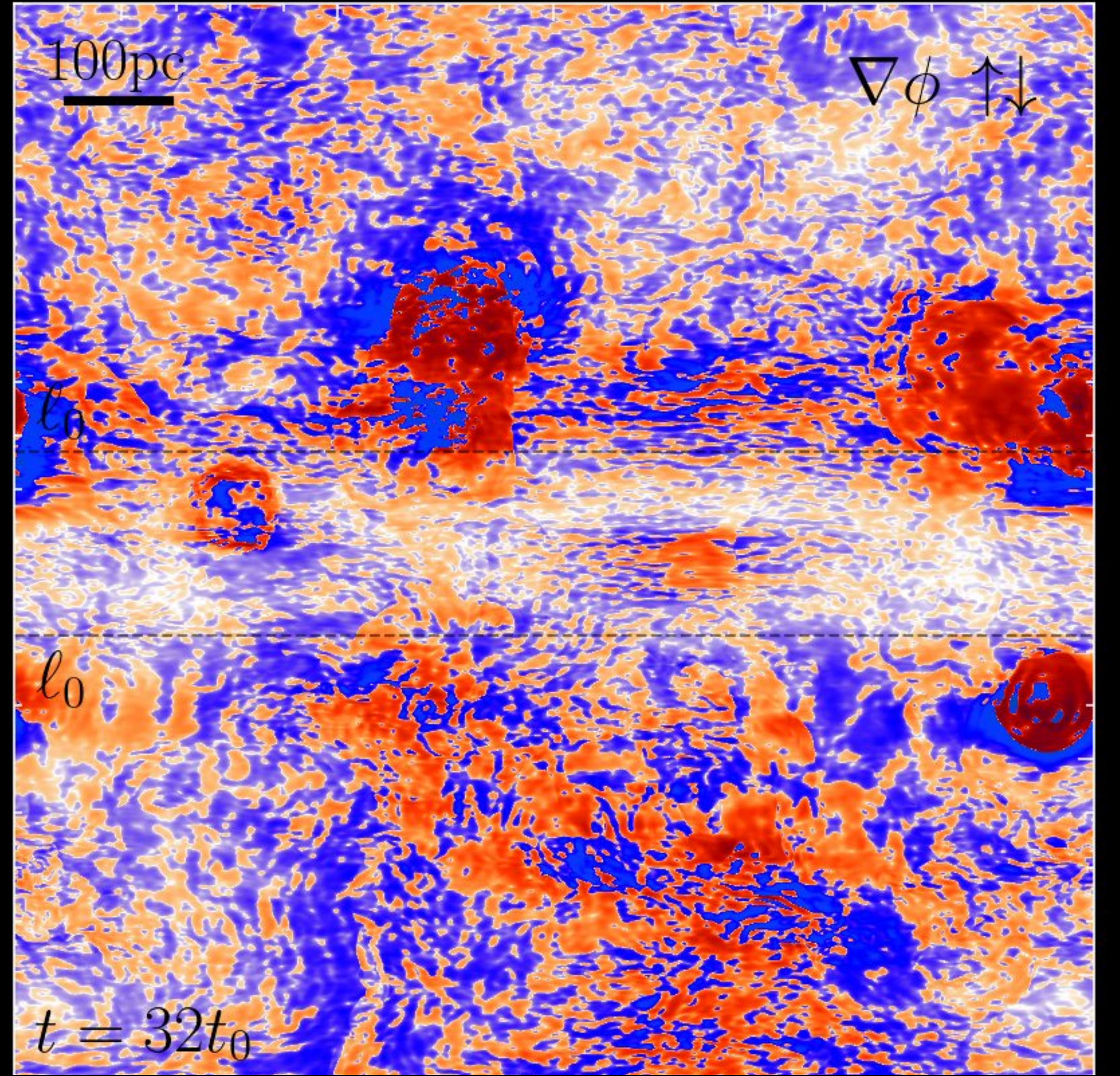
The incompressible turbulence ($u_s \rightarrow u_s$ cascade)



The incompressible turbulence ($u_s \xrightarrow{u_c} u_s$ cascade)



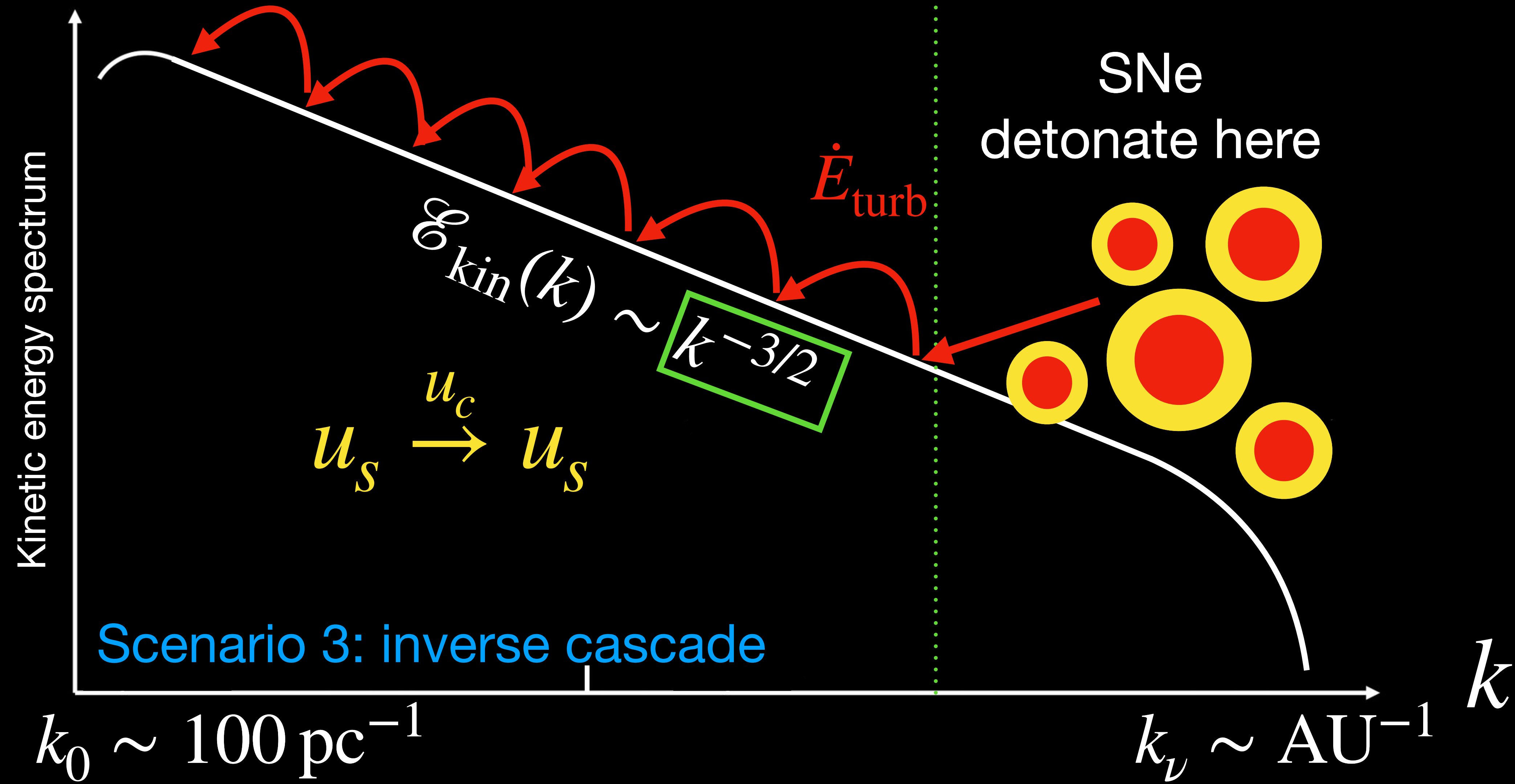
The incompressible turbulence ($u_s \xrightarrow{u_c} u_s$ cascade)



The naive ($\mathcal{M} \approx 2$) warm ionized medium spectrum
(the volume-filling spectrum)

$$R_{\text{cool}}^{-1} \approx 14 \text{ pc}^{-1}$$

Cioffi + 1998
Blondin + 1998



$$\mathcal{E}_{\text{kin}}(k) \sim k^{-3/2}$$

$$\dot{E}_{\text{turb}}$$

SNe
detonate here

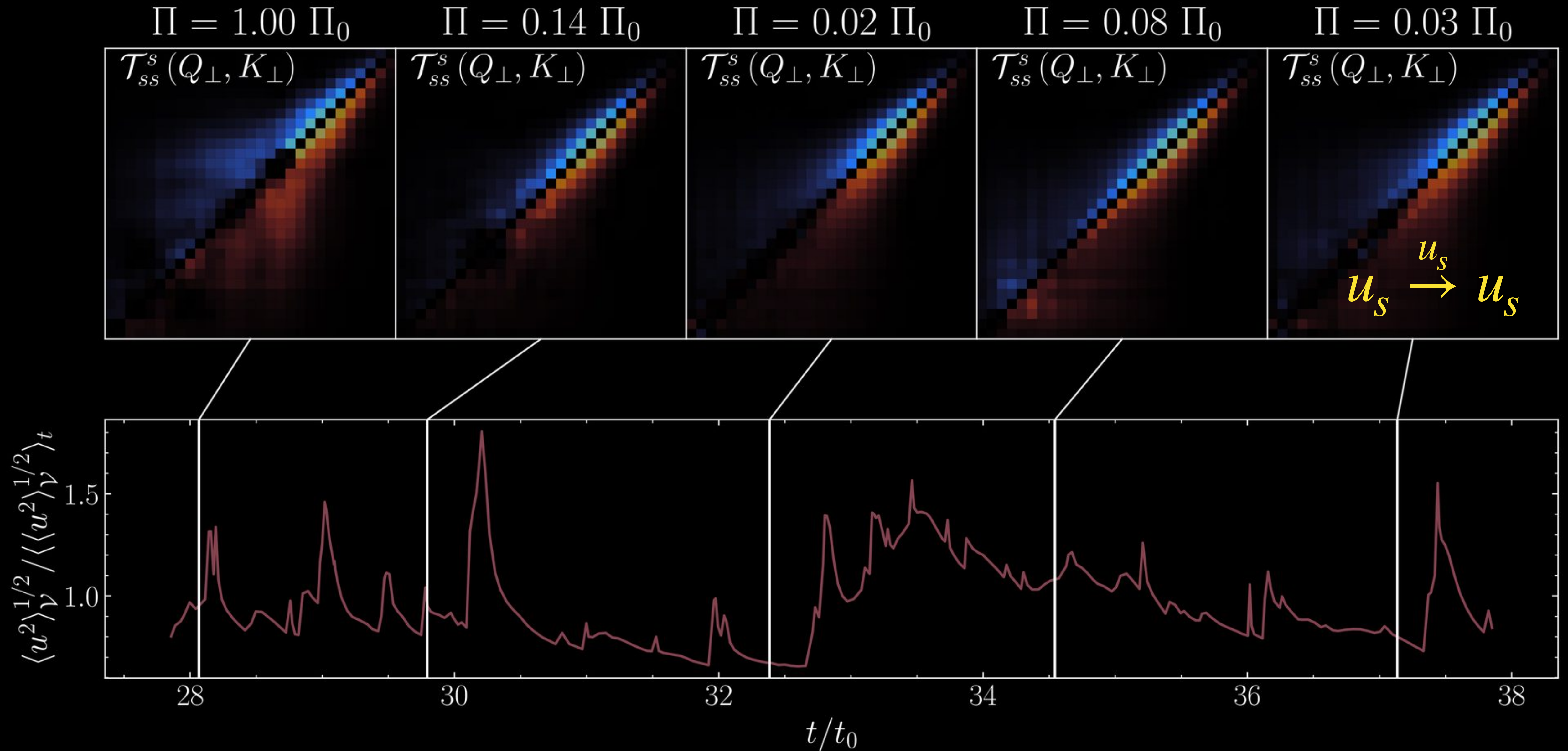
Scenario 3: inverse cascade

$$k_0 \sim 100 \text{ pc}^{-1}$$

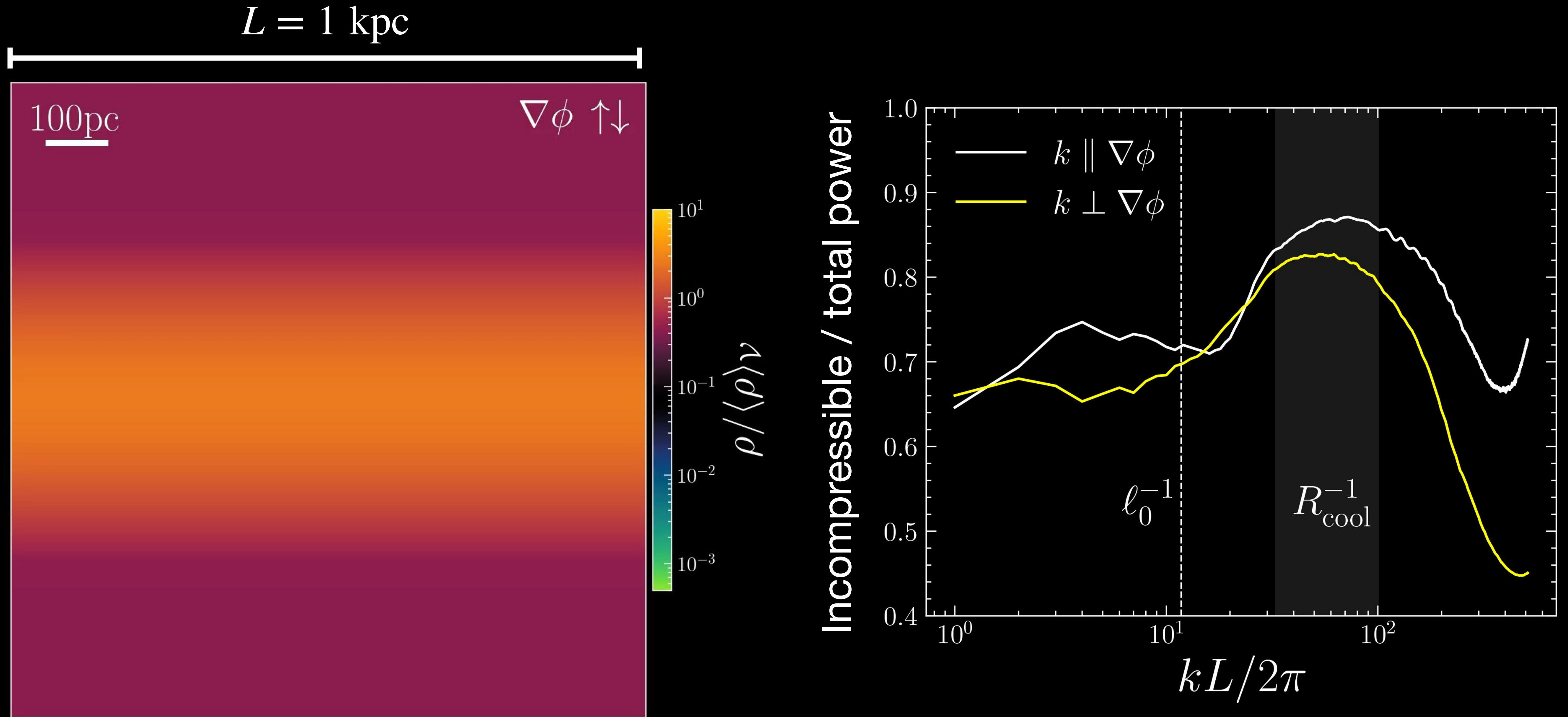
$$k_v \sim \text{AU}^{-1}$$

$$u_s \xrightarrow{u_c} u_s$$

The entire incompressible cascade comes and goes



Where does the incompressible turbulence come from?



Where does the incompressible turbulence come from?

No vorticity!

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u} + P \mathbb{I}) = - \boxed{\rho \nabla \phi + \dot{n}_{\text{SNe}} \mathbf{p}_{\text{SNe}}(Z, n_{\text{H}})},$$

Where does the incompressible turbulence come from?

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$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u} + P \mathbb{I}) = - \boxed{\rho \nabla \phi + \dot{n}_{\text{SNe}} \mathbf{p}_{\text{SNe}}(Z, n_{\text{H}})},$$

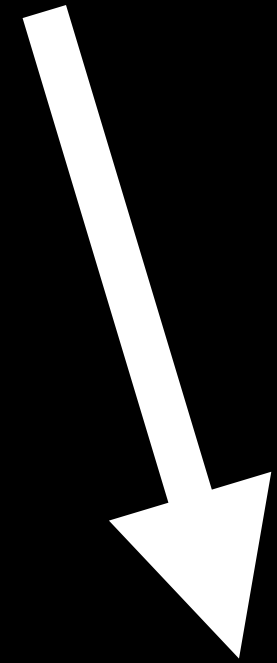
take the curl, Fourier transform and square

$$\frac{1}{2} \frac{\partial \omega(\mathbf{k}) \cdot \omega^\dagger(\mathbf{k})}{\partial t}$$

Where does the incompressible turbulence come from?

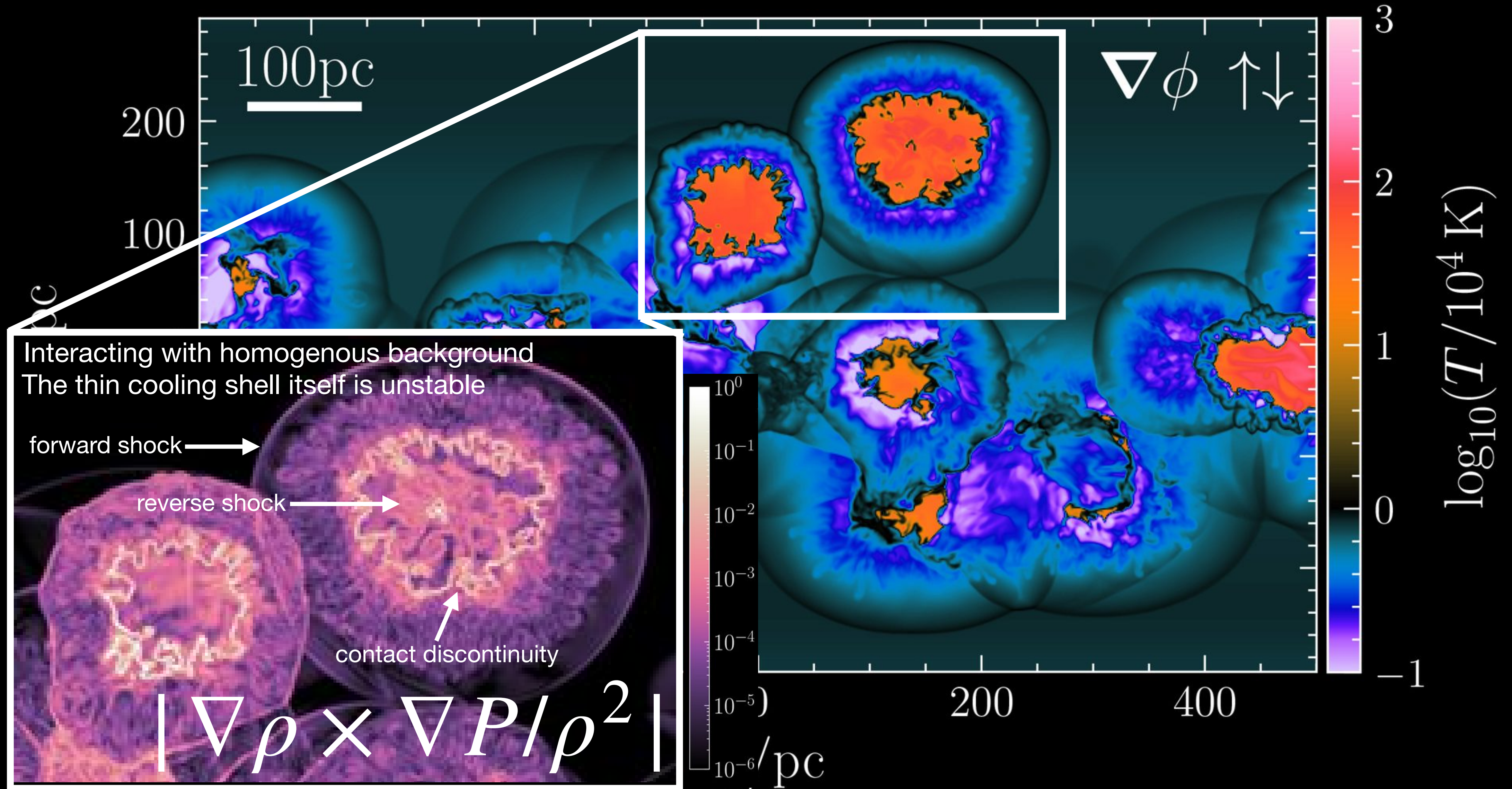
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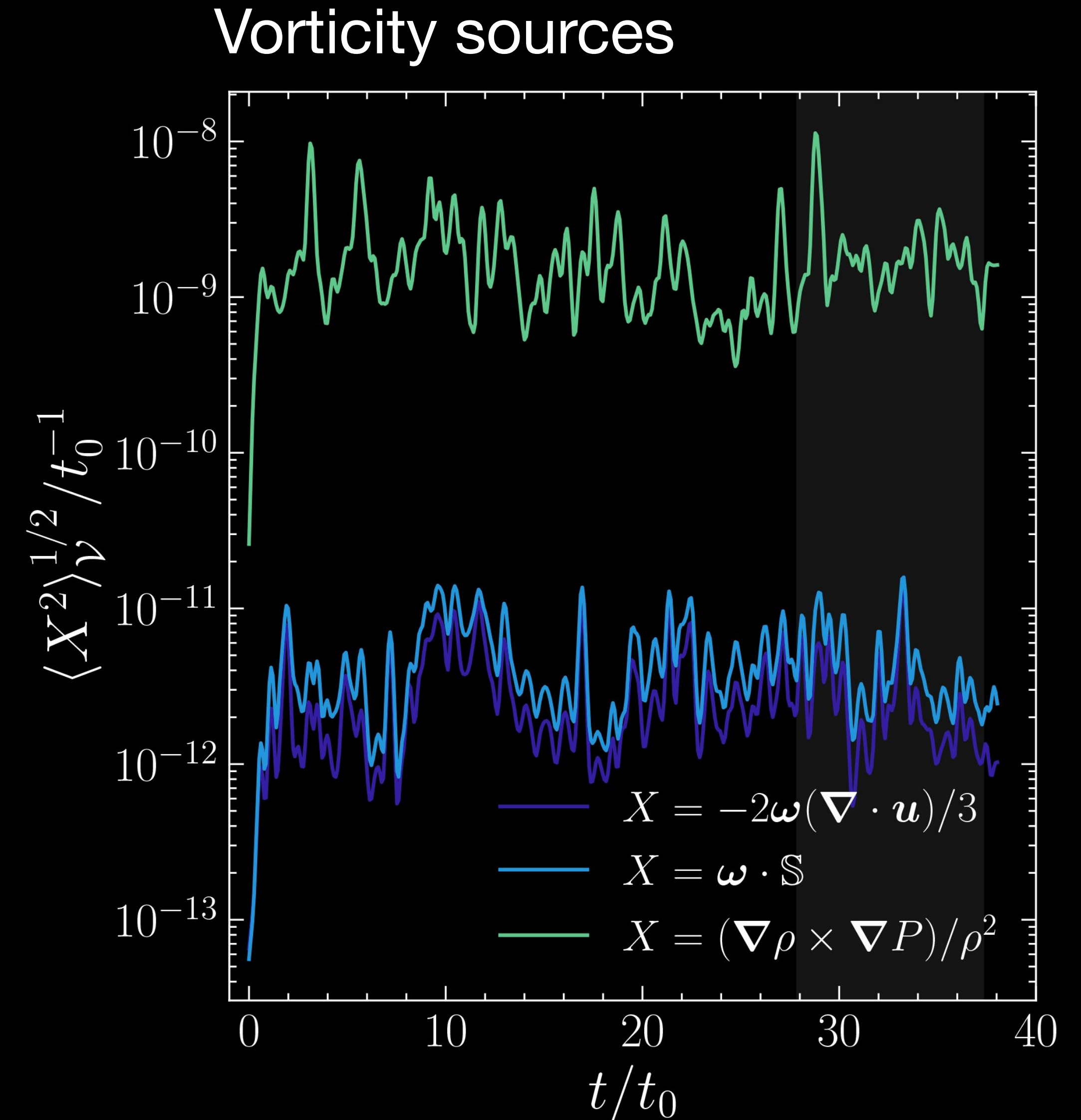
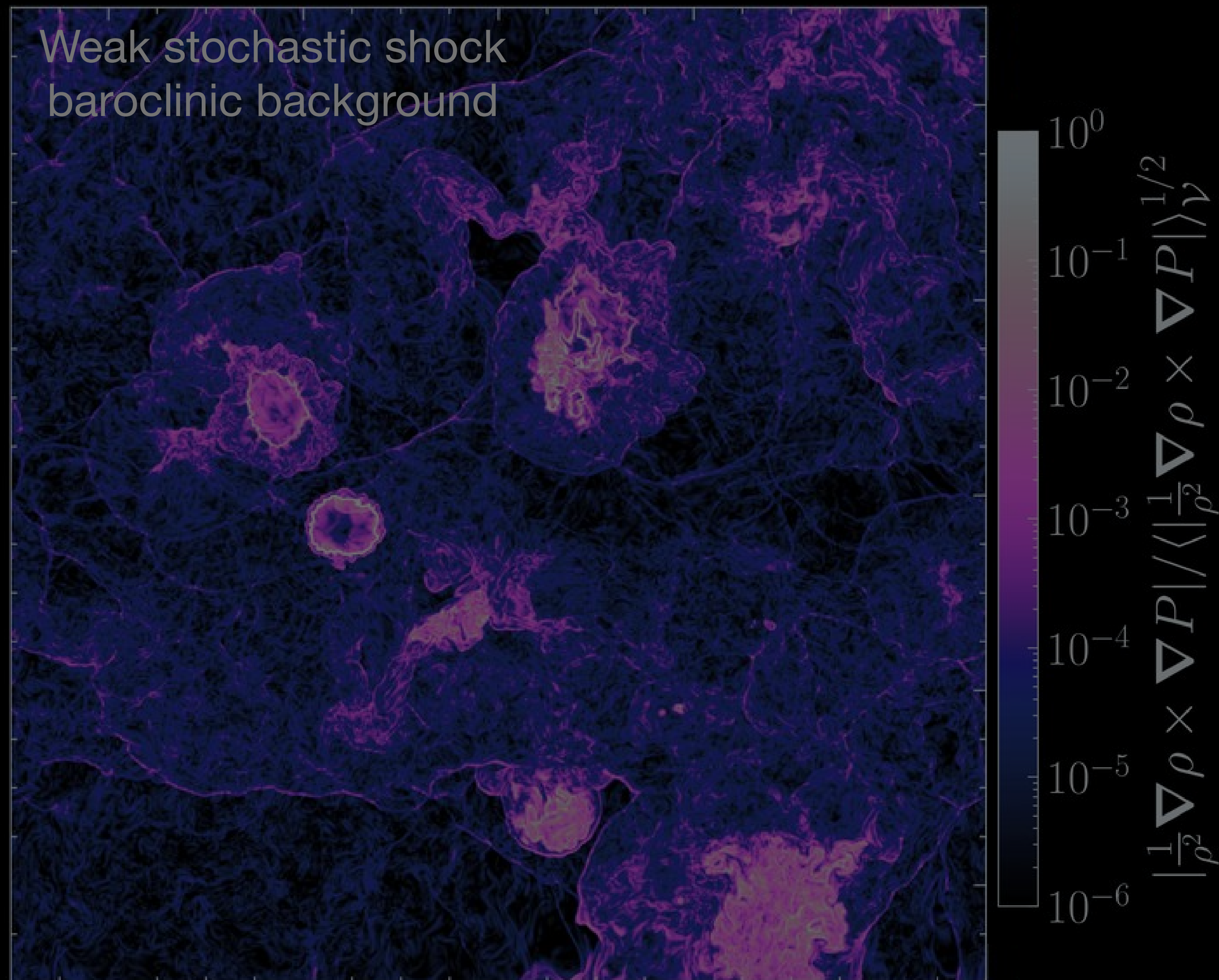


$$\frac{1}{2} \frac{\partial \omega(\mathbf{k}) \cdot \omega^\dagger(\mathbf{k})}{\partial t} = \boxed{\text{Cascade}} \left[- \frac{d\Pi_\omega(\mathbf{k})}{d\mathbf{k}} \right] + \boxed{\text{Source}} \left[\text{Re} \left\{ \omega(\mathbf{k}) \cdot (\nabla \rho \times \nabla P / \rho^2)^\dagger(\mathbf{k}) \right\} \right] - \boxed{\text{Sink}} \left[\mathcal{D}(\mathbf{k}) \right],$$

Where does the incompressible turbulence come from?

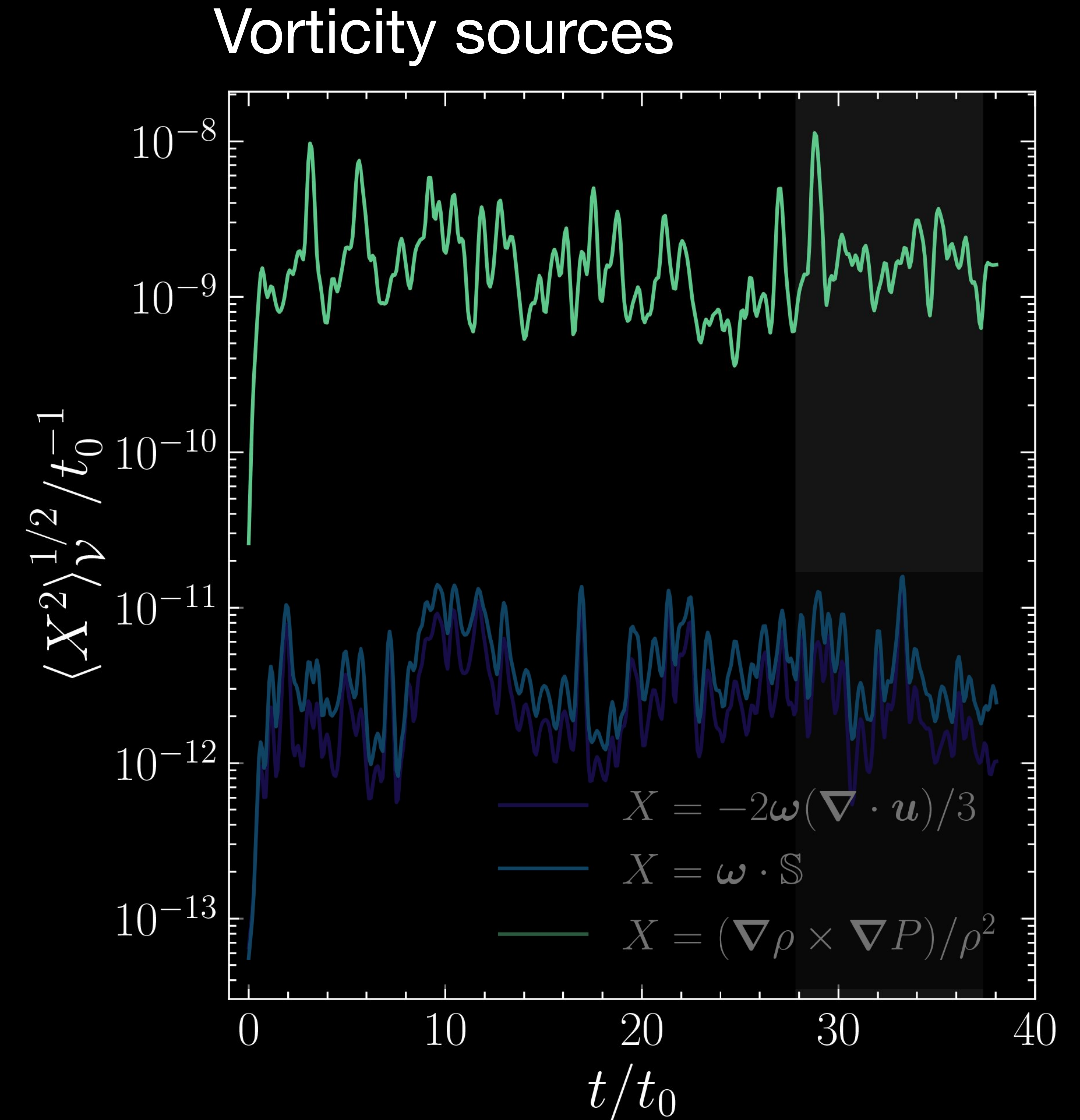
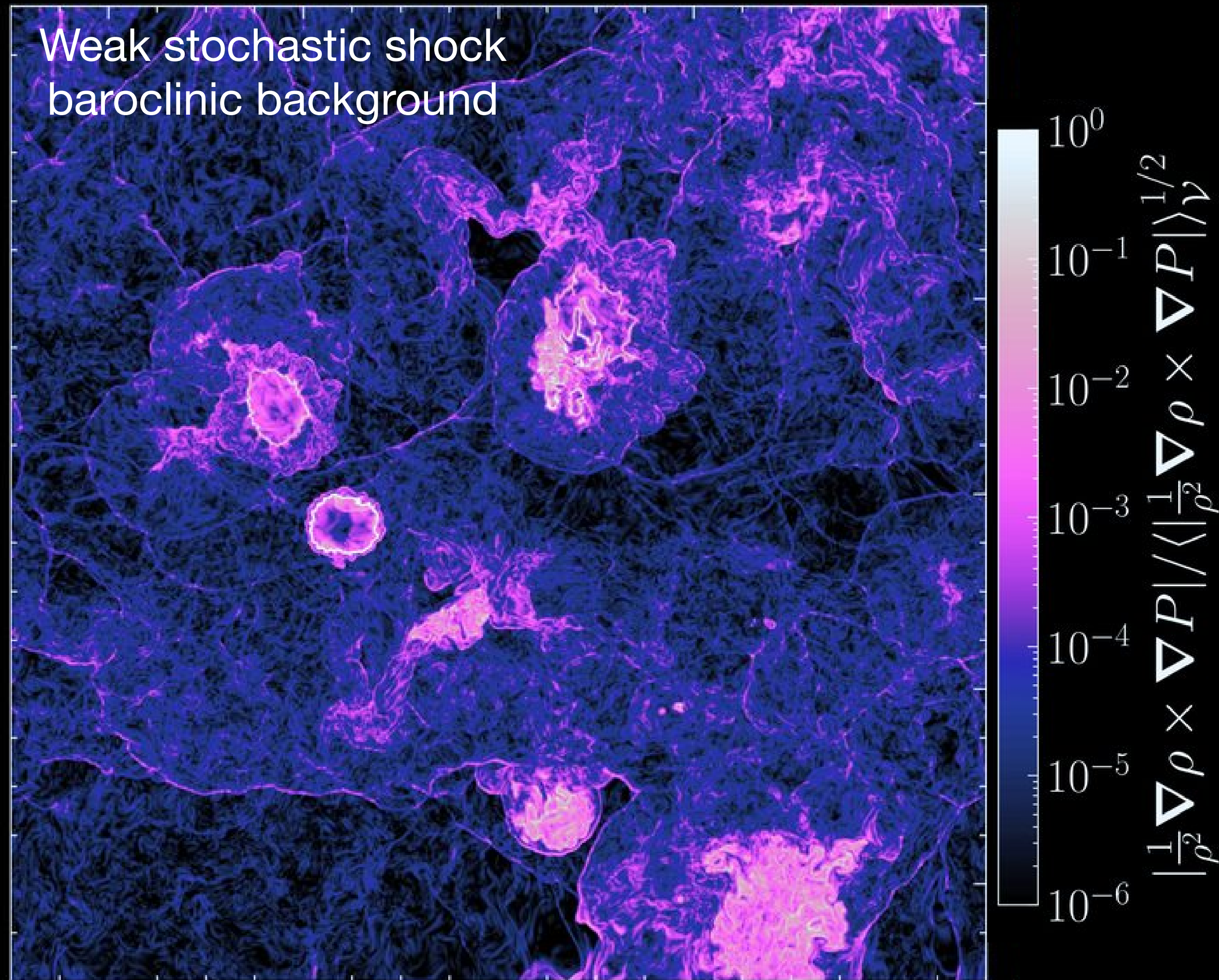


Where does the incompressible turbulence come from?



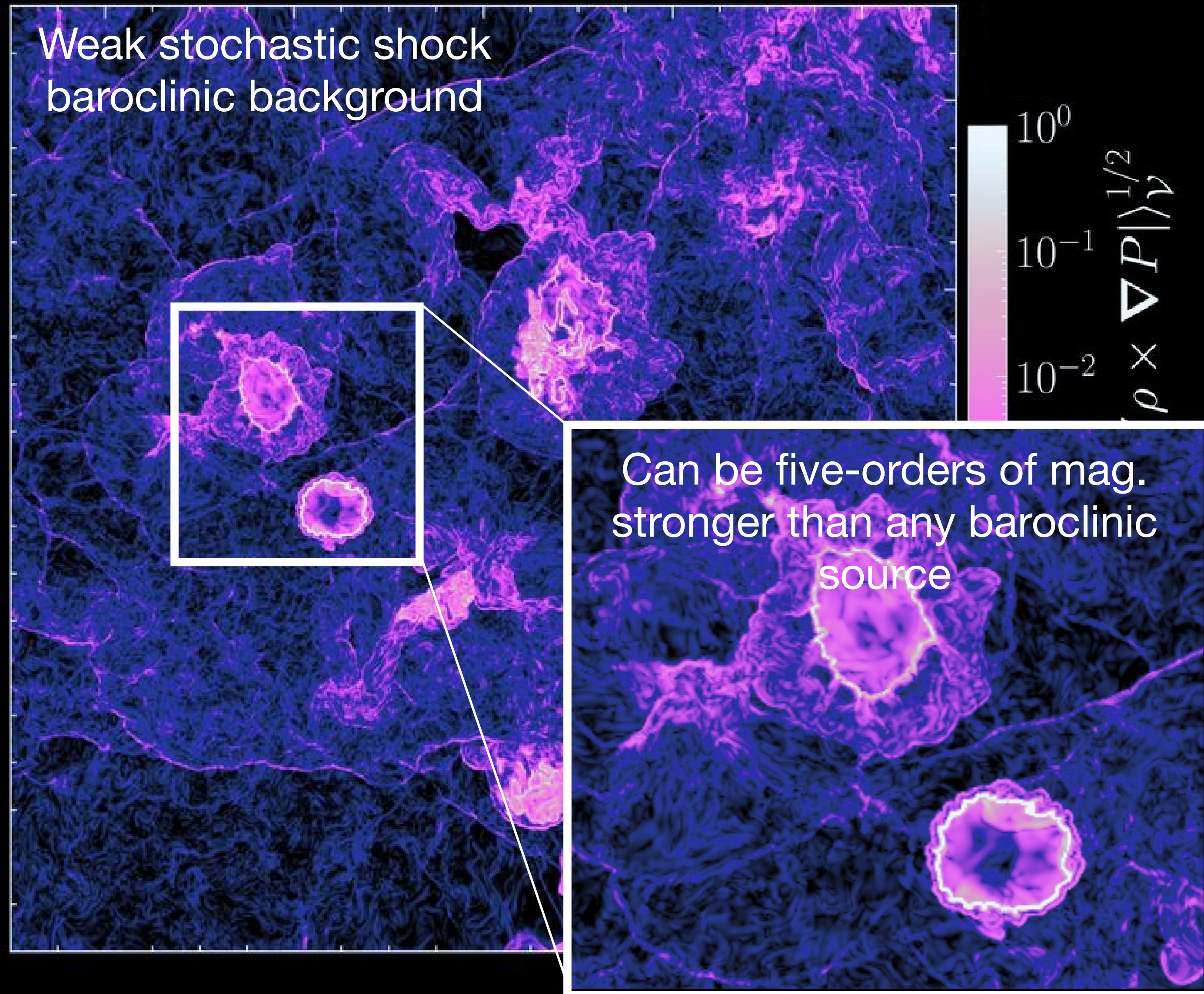
Completed dominated by baroclinic
battery at all times.

Where does the incompressible turbulence come from?

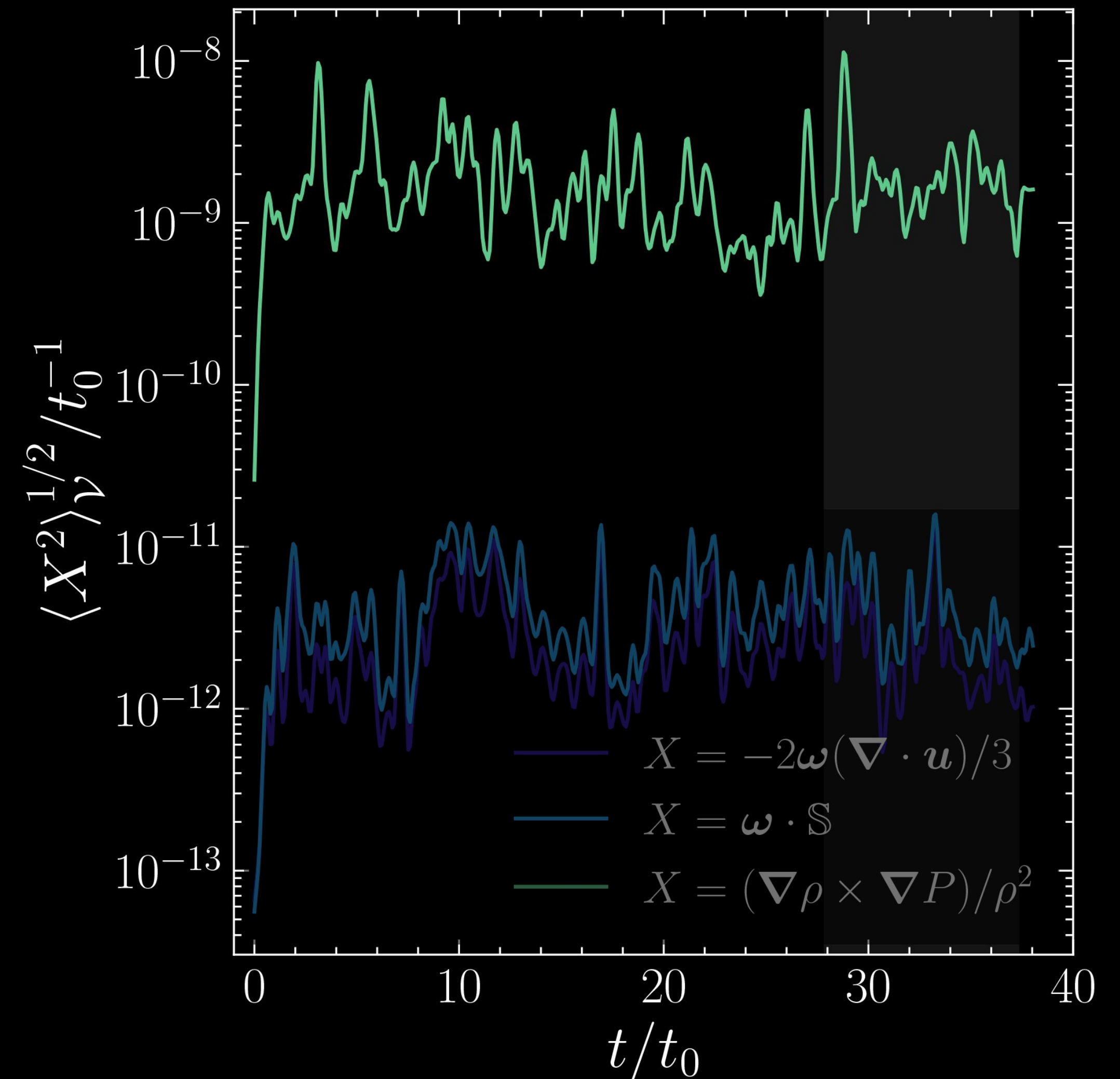


Completed dominated by baroclinic
battery at all times.

Where does the incompressible turbulence come from?



Vorticity sources



Completed dominated by baroclinic battery at all times.

Where does the incompressible turbulence come from?

$$\frac{1}{2} \frac{\partial \omega(\mathbf{k}) \cdot \omega^\dagger(\mathbf{k})}{\partial t} = \overset{\text{Cascade}}{\boxed{-\frac{d\Pi_\omega(\mathbf{k})}{d\mathbf{k}}}} + \overset{\text{Source}}{\boxed{\text{Re} \left\{ \omega(\mathbf{k}) \cdot (\nabla \rho \times \nabla P / \rho^2)^\dagger(\mathbf{k}) \right\}}} - \overset{\text{Sink}}{\boxed{\mathcal{D}(\mathbf{k})}}$$

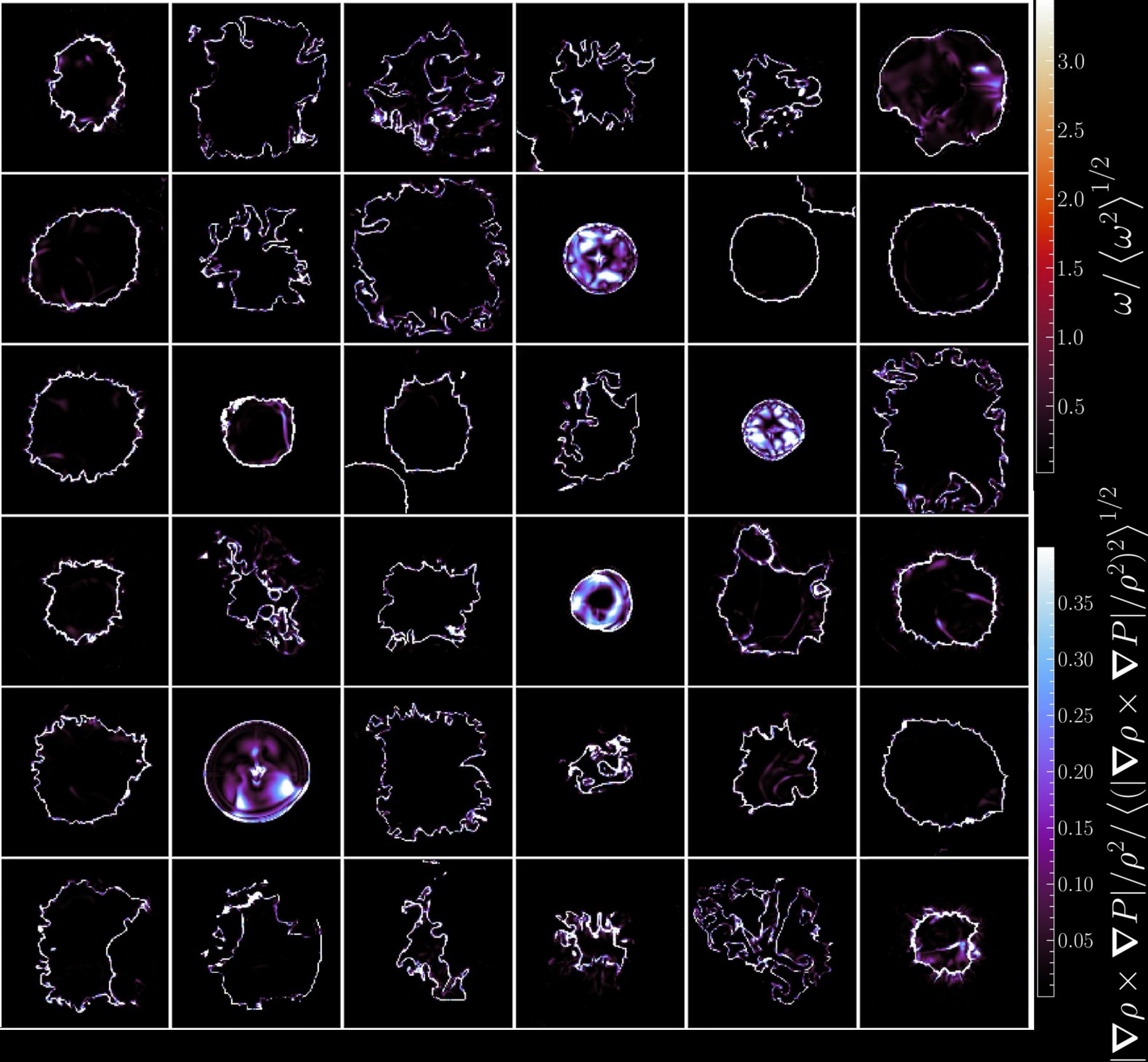
Assuming isotropic, stationary, high-Re

$$\overset{\text{Cascade}}{\boxed{\frac{d\Pi_\omega(\mathbf{k})}{d\mathbf{k}}}} \approx \overset{\text{Source}}{\boxed{\text{Re} \left\{ \omega(\mathbf{k}) \cdot (\nabla \rho \times \nabla P / \rho^2)^\dagger(\mathbf{k}) \right\}}}$$

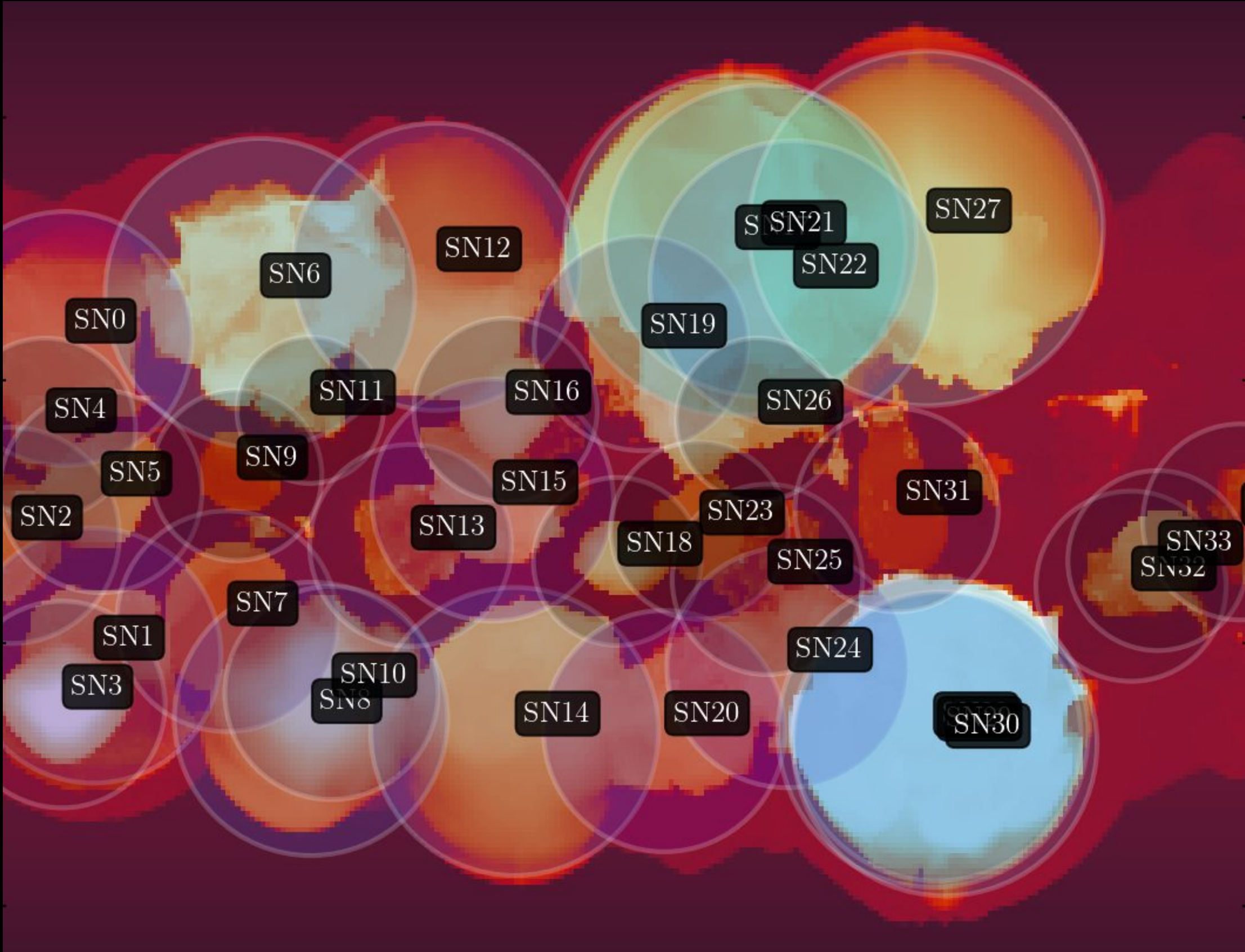
$$\mathcal{P}_{\omega\text{B}}(k)$$

Where does the incompressible turbulence come from?

Beattie (2026; submitted ApJL) Supernovae drive large-scale incompressible turbulence from small-scale instabilities



Cluster all SNR in 3D to extract local statistics



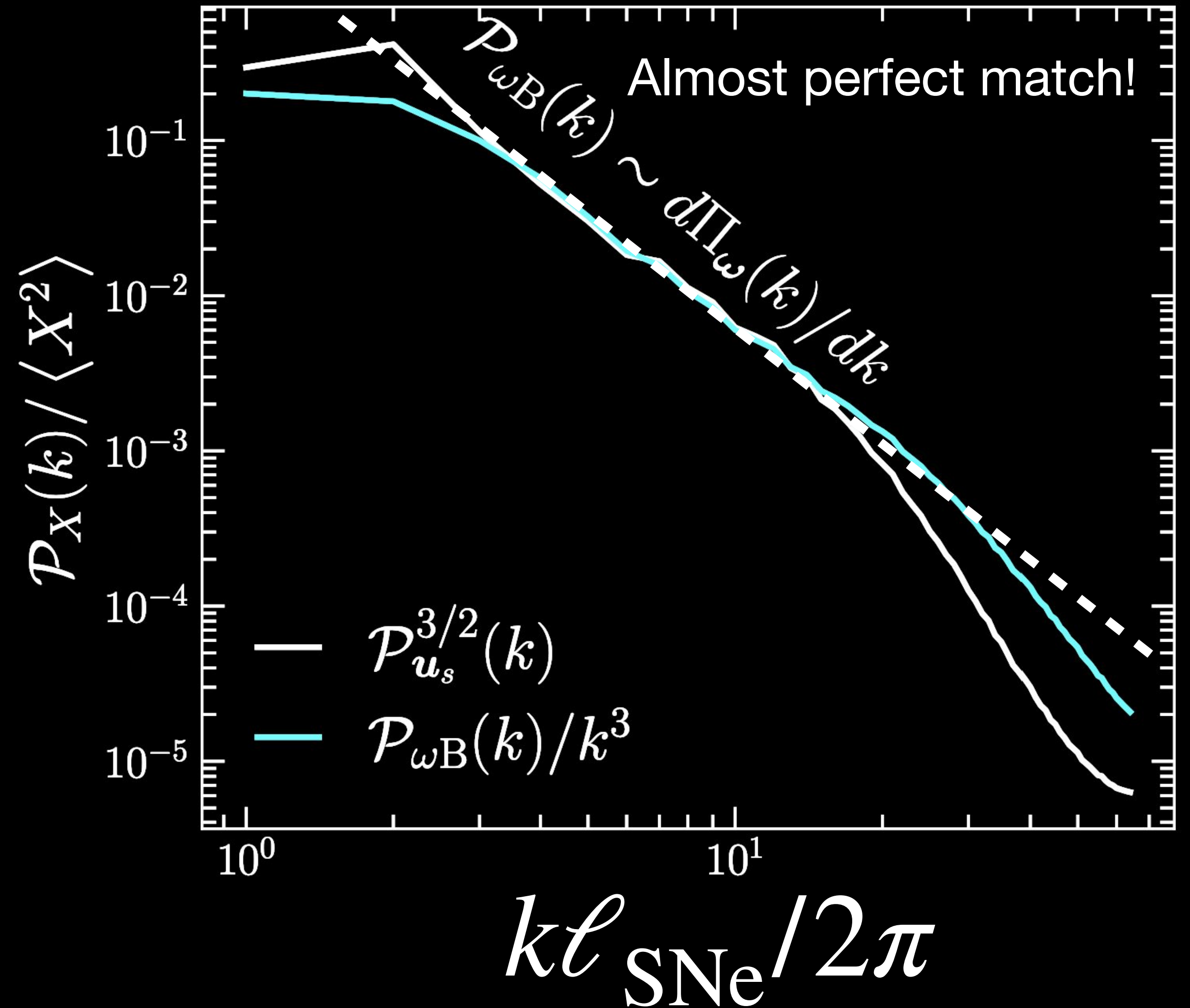
Where does the incompressible turbulence come from?

Assuming isotropic, stationary, high-Re

$$\mathcal{P}_{\omega\text{B}}(k) \approx \frac{d\Pi_{\omega}(k)}{dk}$$

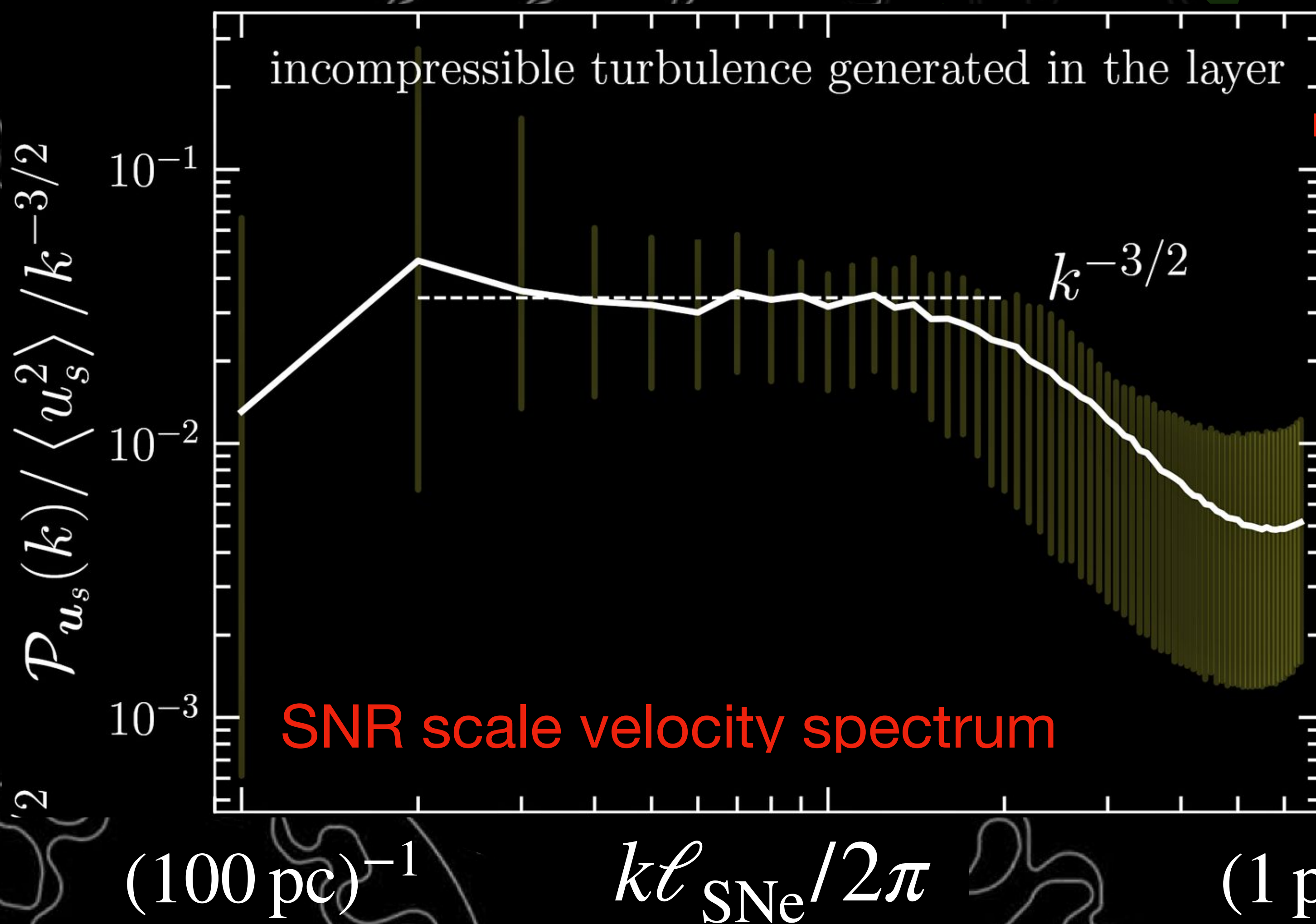
relation between incompressible mode spectrum and baroclinic source spectrum

$$\mathcal{P}_{\omega\text{B}}(k) \sim k^3 \mathcal{P}_{u_s}^{3/2}(k)$$

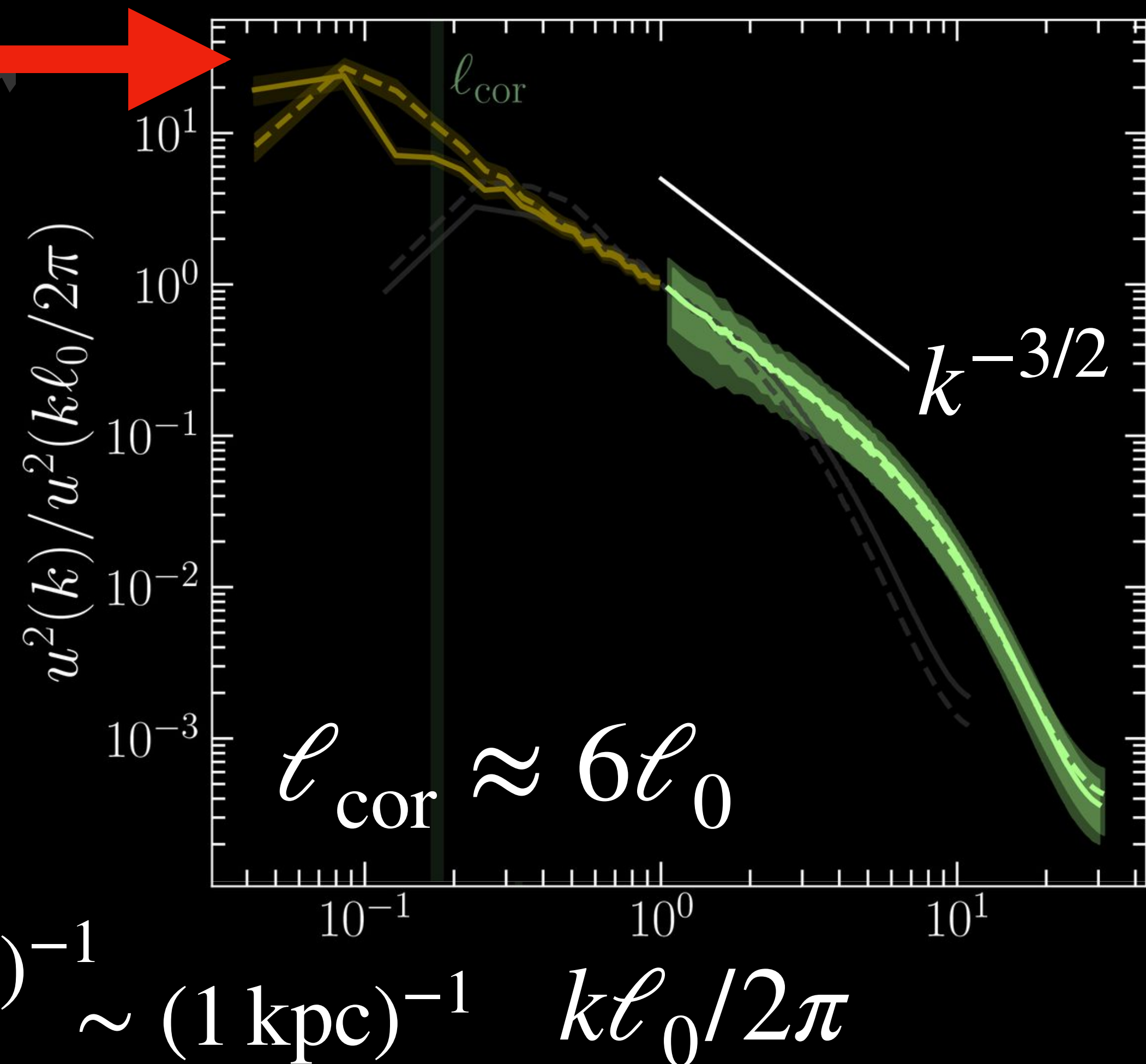


Incompressible modes generated on SNR scale imprinted on all scales

(scales larger than gaseous scale height!!!)



Global velocity spectrum



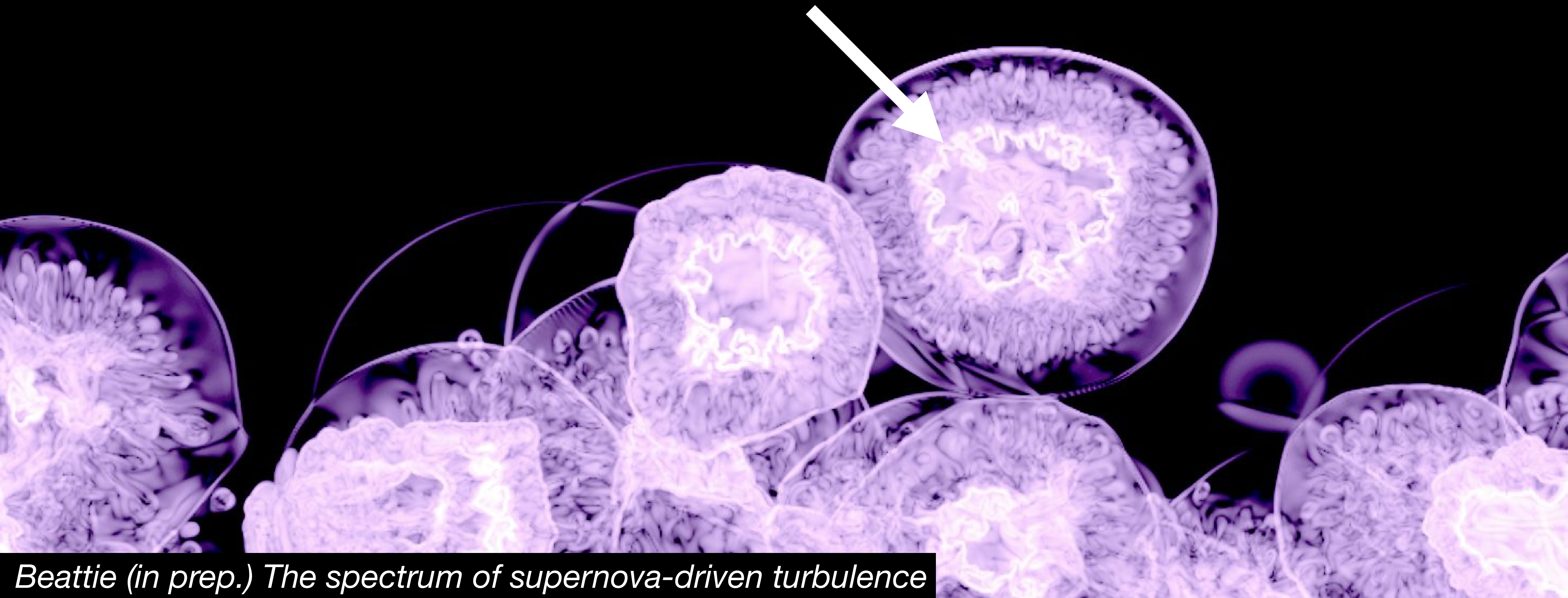
Can we predict the spectrum?

Baroclinicity is born from thin shell instabilities (RT & overstability)

$$\gamma \sim \sqrt{k}$$

Linear RTI

Vishniac (long wavelength limit; 1983)



Beattie (in prep.) The spectrum of supernova-driven turbulence

Can we predict the spectrum?

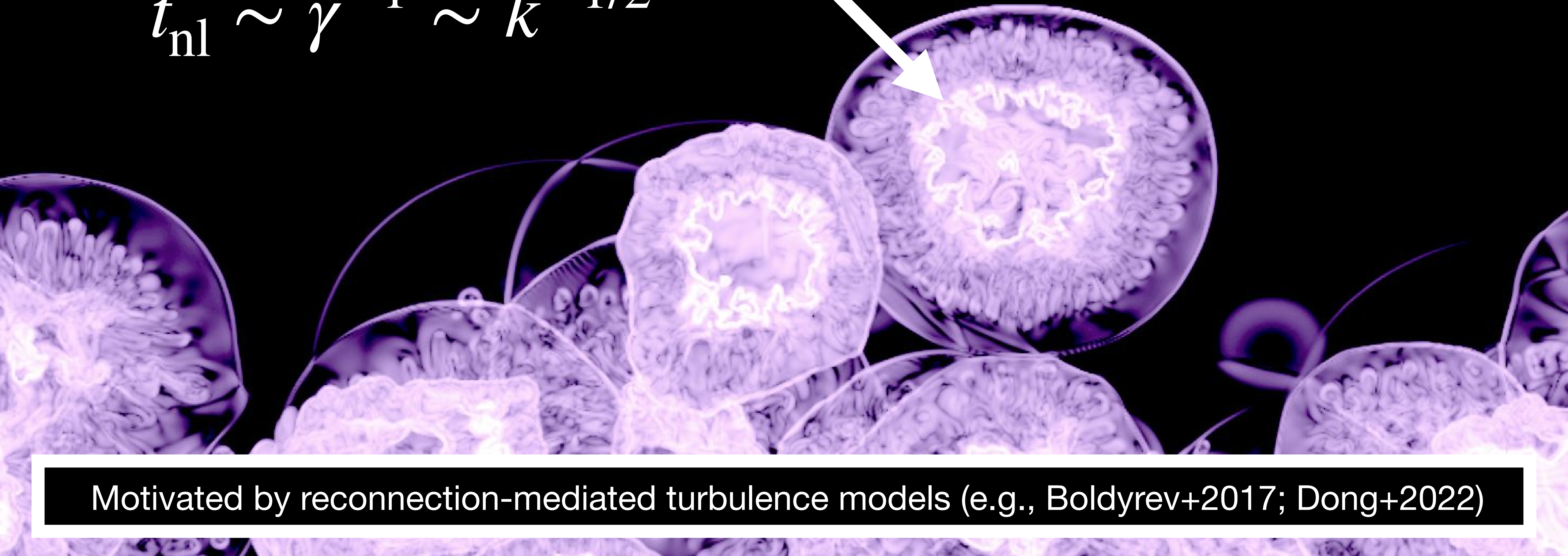
Baroclinicity is born from thin shell instabilities (RT & overstability)

$$\gamma \sim \sqrt{k}$$

Linear RTI

Vishniac (long wavelength limit; 1983)

$$t_{\text{nl}} \sim \gamma^{-1} \sim k^{-1/2}$$



Motivated by reconnection-mediated turbulence models (e.g., Boldyrev+2017; Dong+2022)

Can we predict the spectrum?

Baroclinicity is born from thin shell instabilities (RT & overstability)

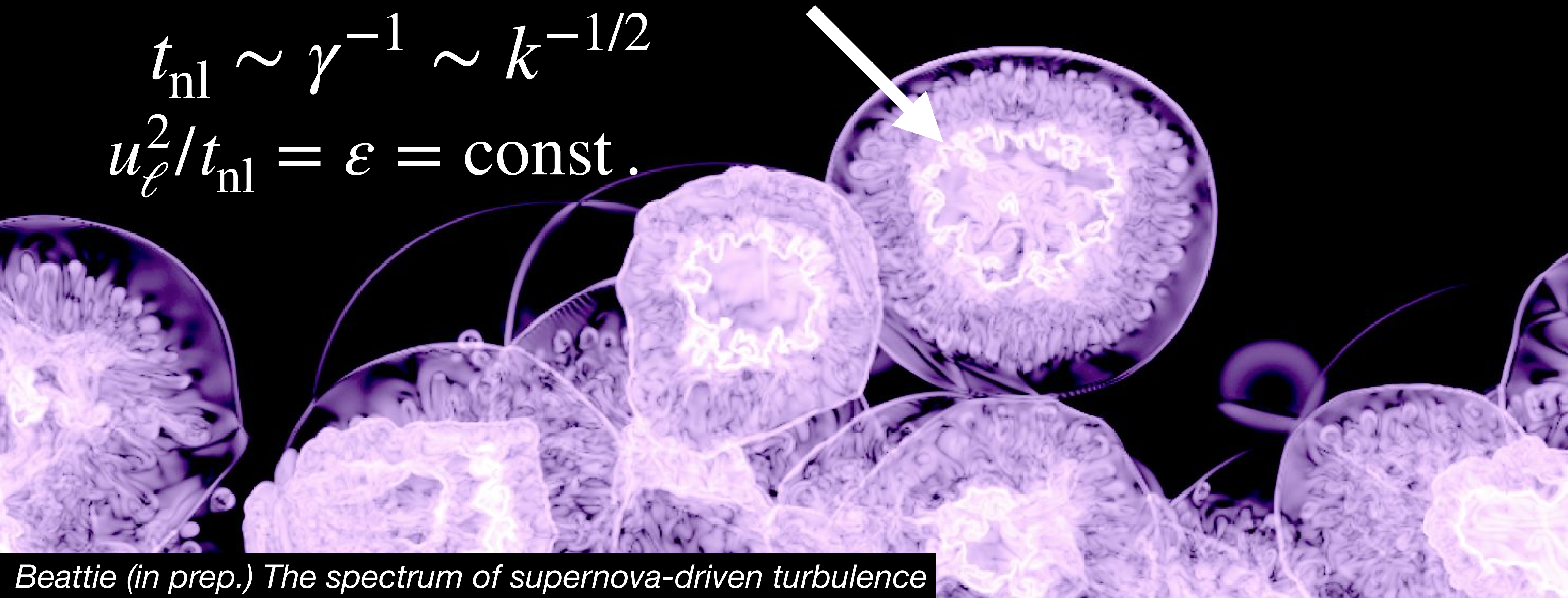
$$\gamma \sim \sqrt{k}$$

Linear RTI

Vishniac (long wavelength limit; 1983)

$$t_{n1} \sim \gamma^{-1} \sim k^{-1/2}$$

$$u_{\ell}^2 / t_{n1} = \varepsilon = \text{const.}$$



Can we predict the spectrum?

Baroclinicity is born from thin shell instabilities (RT & overstability)

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Linear RTI

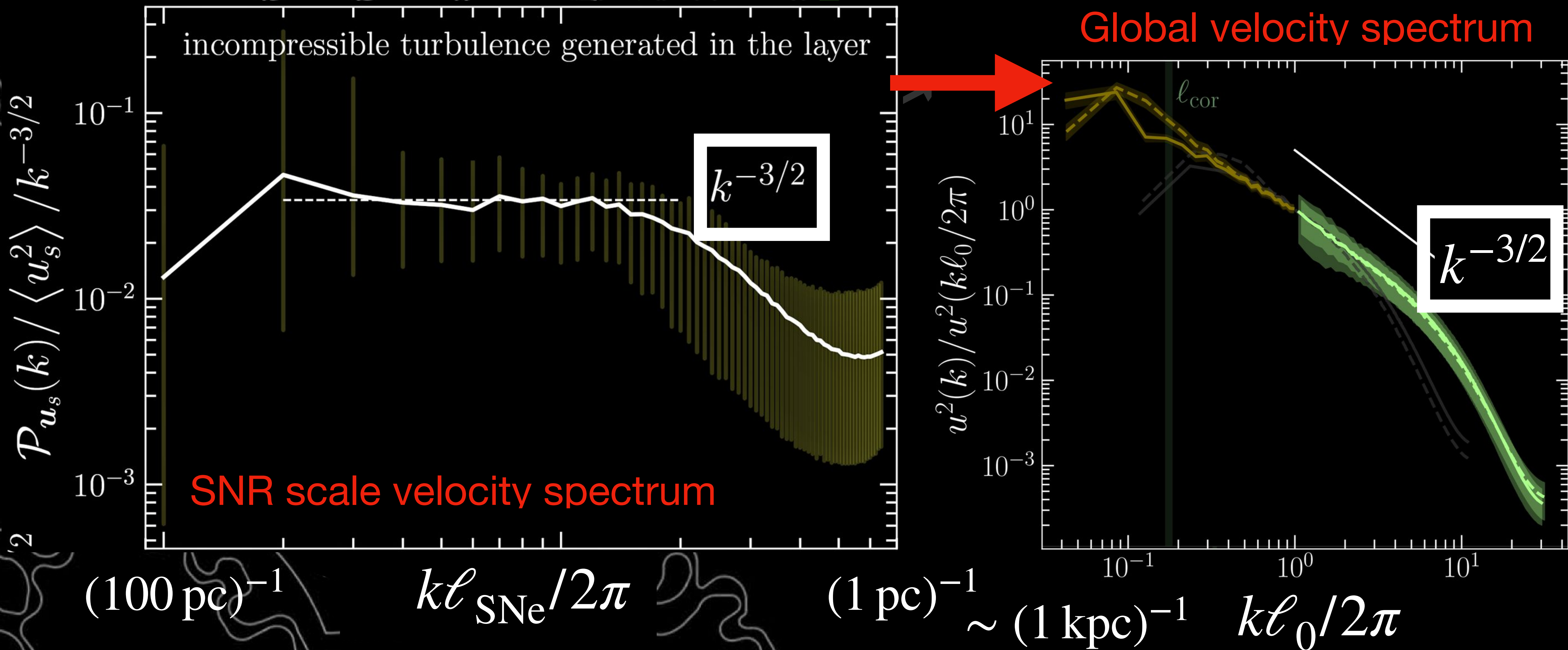
Vishniac (long wavelength limit; 1983)

$$t_{n1} \sim \gamma^{-1} \sim k^{-1/2}$$

$$u_\ell^2 / t_{n1} = \varepsilon = \text{const.}$$

$$u_\ell \sim \ell^{1/4} \iff u^2(k) \sim k^{-3/2} dk$$

Shell instabilities drive and imprint themselves on the spectrum!



What's the driving scale of SN-driven turbulence?

From baroclinic feeding

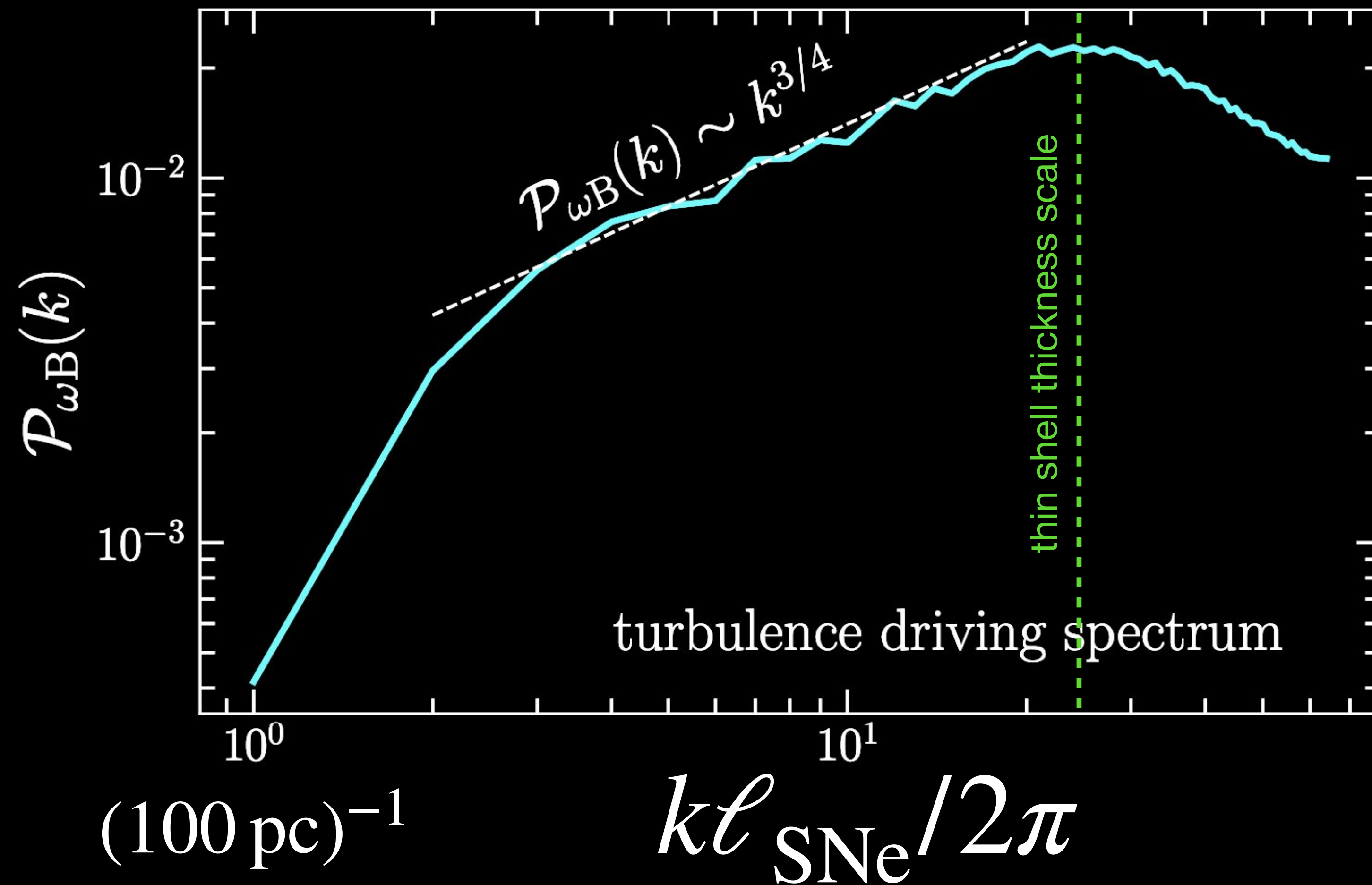
$$\mathcal{P}_{\omega B}(k) \sim k^3 \mathcal{P}_{u_s}^{3/2}(k)$$

From RTI

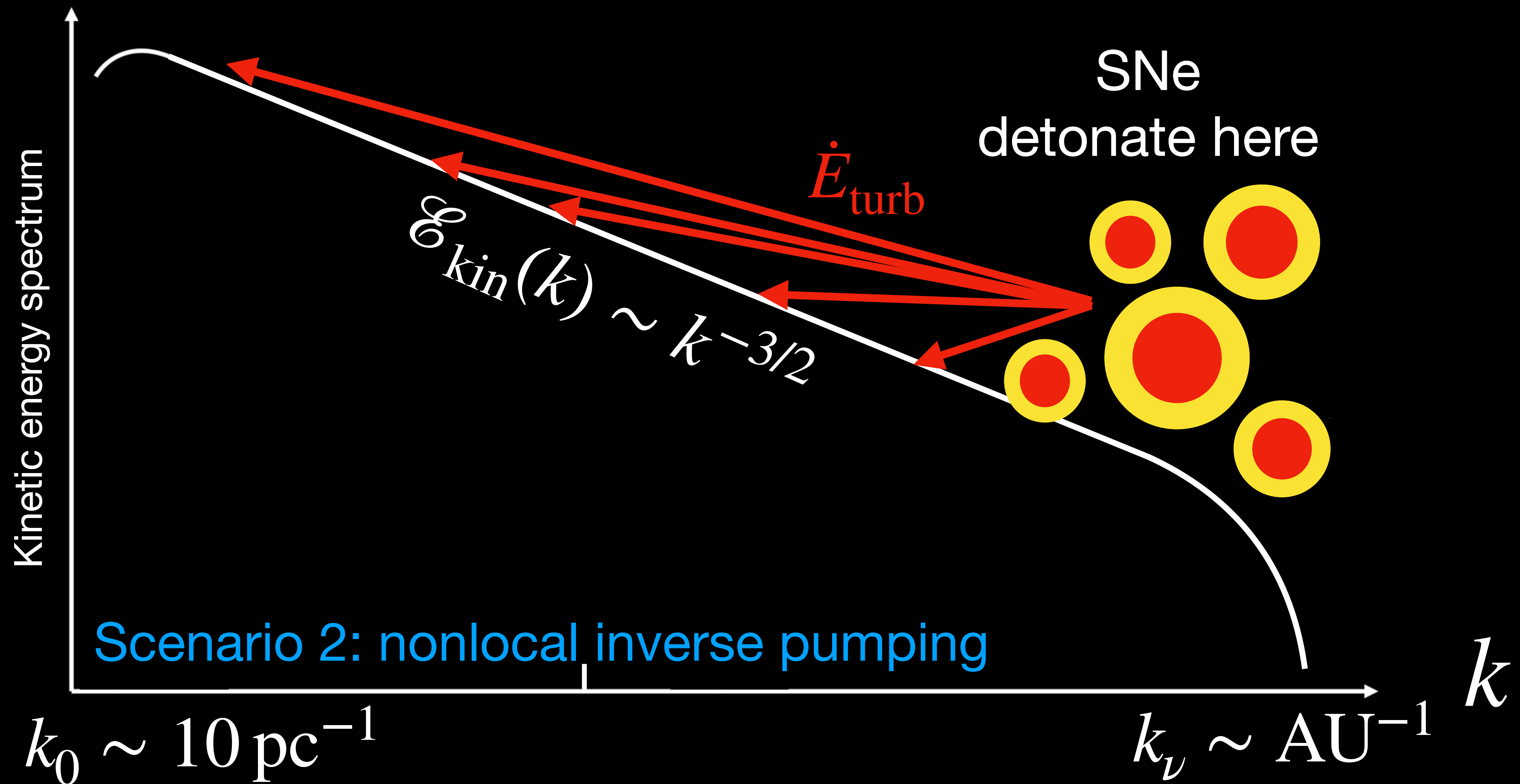
$$\mathcal{P}_{u_s}(k) \sim k^{-3/2}$$

Spectrum that drives the turbulence is therefore:

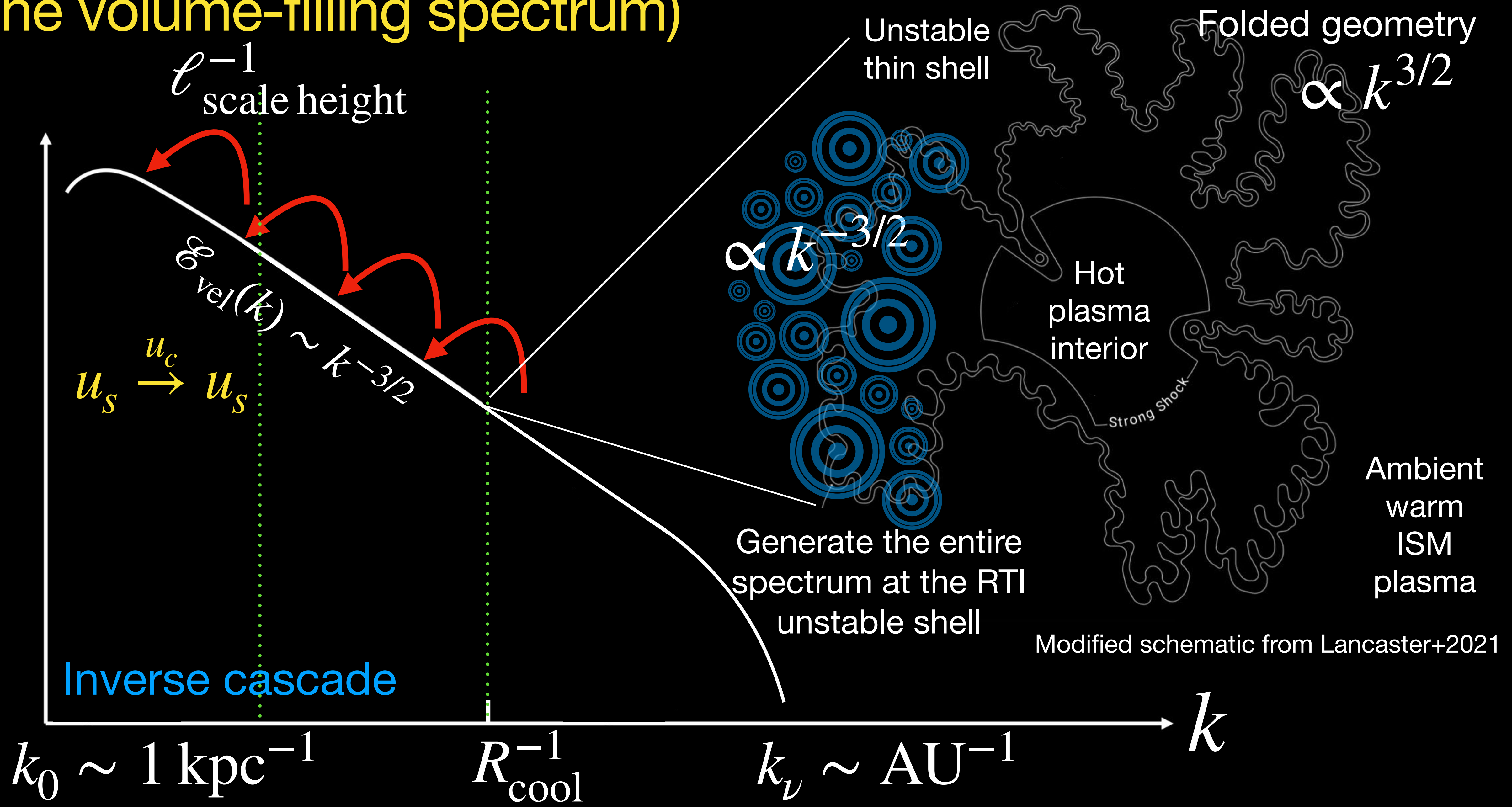
$$\mathcal{P}_{\omega B}(k) \sim k^{3/4}$$



The naive ($\mathcal{M} \approx 2$) warm ionized medium spectrum (the volume-filling spectrum)



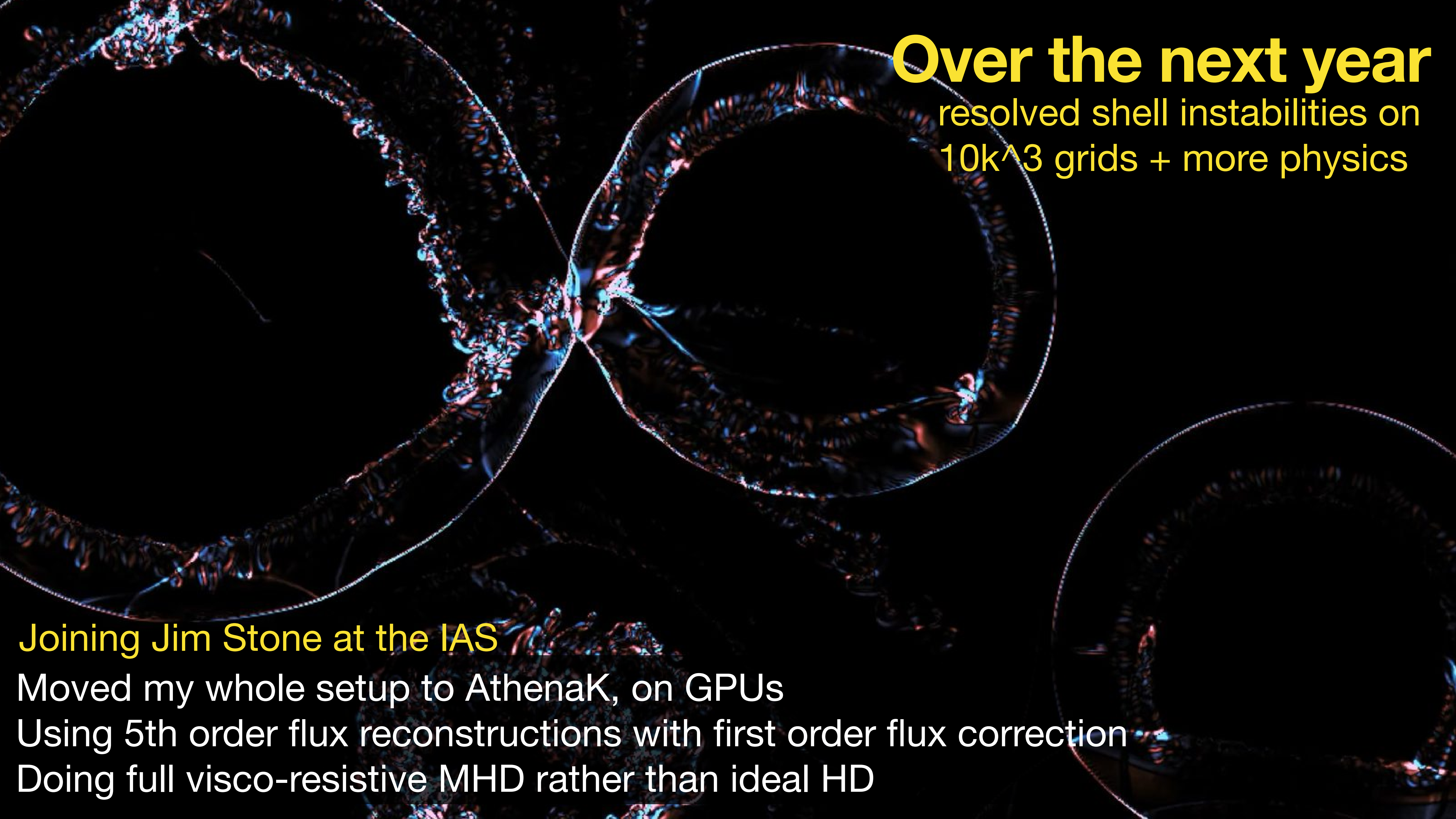
The SN-driven warm ionized medium spectrum (the volume-filling spectrum)





Summary for this seminar

Supernova-driven turbulence (and hence $\sim$ galactic turbulence) resembles nothing of the standard Kolmogorov model!



Over the next year
resolved shell instabilities on
 $10k^3$ grids + more physics

Joining Jim Stone at the IAS

Moved my whole setup to Athenak, on GPUs

Using 5th order flux reconstructions with first order flux correction

Doing full visco-resistive MHD rather than ideal HD

$$\partial_t B = -\nabla \times E$$

$$E = -u \times B + \eta J - \frac{\nabla P_e}{en_e}$$

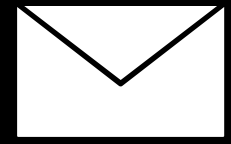
Over the next year
the sparks of magnetogenesis

$$\partial_t B \propto \frac{\nabla P \times \nabla \rho}{\rho^2}$$

That's our
vorticity
(turbulence)
source term!!!

Question:
Can we spark the
first magnetic fields
in the death of the
first stars?

Thanks, questions?



james.beattie@princeton.edu



[@astro_magnetism](https://twitter.com/astro_magnetism)



Compressible modes and solenoidal modes live very different lives!

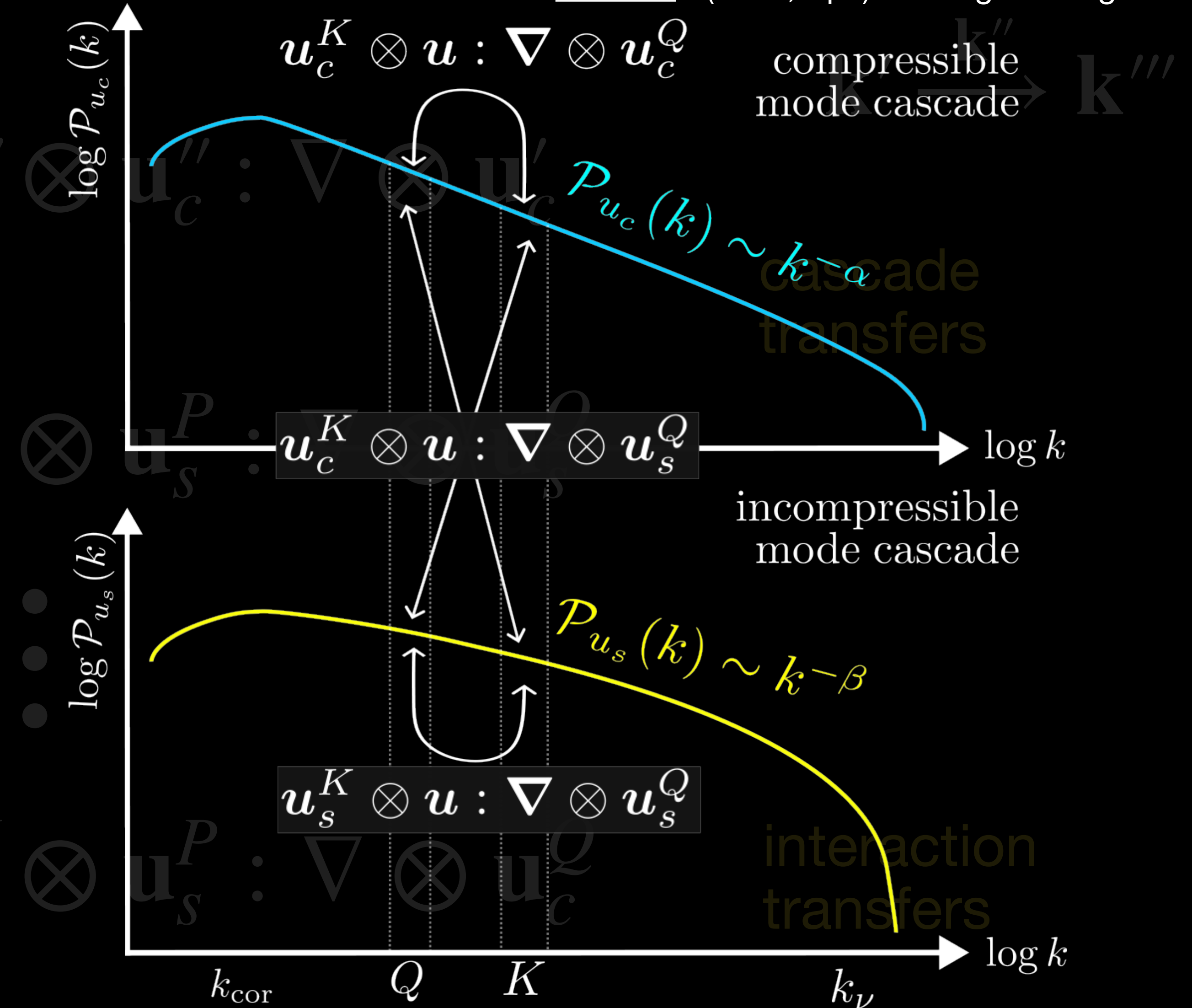
We should treat them differently!

Beattie + (2025, ApJ) So long Kolmogorov.

$$\mathcal{T}_{cc}^c(Q, K) = - \int dV \mathbf{u}_c'^{''}$$

$$\mathcal{T}_{ss}^s(Q, K) = - \int dV \mathbf{u}_s^K$$

$$\mathcal{T}_{cs}^s(Q, K) = - \int dV \mathbf{u}_s^K$$



Compressible modes and solenoidal modes live very different lives!
We should treat them differently!

$$Q \xrightarrow{P} K$$

$$\mathcal{T}_{cc}^c(Q, K) = - \int dV \mathbf{u}_c^K \otimes \mathbf{u}_c^P : \nabla \otimes \mathbf{u}_c^Q$$

cascade
transfers

$$\mathcal{T}_{ss}^s(Q, K) = - \int dV \mathbf{u}_s^K \otimes \mathbf{u}_s^P : \nabla \otimes \mathbf{u}_s^Q$$

$$\begin{aligned} u_s \cdot u_c &= 0 \\ \nabla \cdot u_s &= 0 \\ |\nabla \times u_c| &= 0 \\ u &= u_c + u_s \end{aligned}$$

$$\mathcal{T}_{cs}^s(Q, K) = - \int dV \mathbf{u}_s^K \otimes \mathbf{u}_s^P : \nabla \otimes \mathbf{u}_c^Q$$

interaction
transfers

But to understand the turbulence we must
understand the flux from mode to mode!

kinetic energy density

$$\frac{1}{2} \overbrace{\partial_t \rho \mathbf{u}^Q \cdot \mathbf{u}^K} + \underbrace{\mathbf{u}^K \cdot \nabla \cdot \mathbb{F}_{\rho \mathbf{u}}(\mathbf{u}^P, \mathbf{u}^Q)} = 0 \quad Q \xrightarrow{P} K$$

energy flux rate density
from transport between $\mathbf{u}^Q$ and $\mathbf{u}^K$

$$\mathbf{u}^K \cdot \nabla \cdot \mathbb{F}(\mathbf{u}^P, \mathbf{u}^Q)_{\rho \mathbf{u}} = \mathbf{u}^K \otimes \mathbf{u}^P : \nabla \otimes \mathbf{u}^Q + \dots$$

$$\varepsilon \sim u^3 / \ell$$

But to understand the turbulence we must
understand the flux from mode to mode!

$$\mathbf{u}^K \cdot \nabla \cdot \mathbb{F}(\mathbf{u}^P, \mathbf{u}^Q)_{\rho \mathbf{u}} = \mathbf{u}^K \otimes \mathbf{u}^P : \nabla \otimes \mathbf{u}^Q + \dots$$

$$\mathcal{T}_{uu}(Q, K | P) = - \int dV \mathbf{u}^K \otimes \mathbf{u}^P : \nabla \otimes \mathbf{u}^Q$$
$$\varepsilon \sim u^3 / \ell$$